

Mobile Assisted Language Learning

The increased use of sophisticated mobile devices opens up new possibilities and challenges for language teachers and learners, which has led to an increasing need to consider issues relating to mobile technologies specifically. To date, there is no comprehensive book-length treatment of issues relating to mobile-assisted language learning (MALL). This book fills that gap, providing a resource for present and future language teachers, and for graduate students of applied linguistics and TESOL, to understand how mobile devices can best be used for language teaching. It is founded on existing research, practice and theory, and offers a balanced perspective, based on the author's own experiences with mobile learning – considering the limitations of such an approach, as well as the benefits. Written in a practical and approachable tone, it provides a much-needed guide to MALL, and its fascinating insights promote further debate within the field.

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For Takana

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Abbreviations

ADDIE	Analyse, Design, Development, Implementation, and Evaluation
ANT	actor-network theory
BYOD	bring your own device
CALL	computer-assisted language learning
CST	complex systems theory
DST	dynamic systems theory
DVD	digital video disc or digital versatile disk
ENIAC	Electronic Numerical Integrator And Computer
GPS	Global Positioning Service
GSM	Global System for Mobile Communication
HCI	human–computer interaction
ICT	information and communication technologies
iPAC	personalisation, authenticity, collaboration
LMS	learning management system
MALL	mobile-assisted language learning
MOOC	massive open online course
MP	megapixel (an indicator of the quality of the resolution of a picture)
MP3	MPEG (Motion Picture Experts Group) layer-3 (usually an audio file)
MP4	MPEG (Motion Picture Experts Group) layer-4 (usually a video file)
MRI	magnetic resonance imaging
NFC	near-field communication
PDA	personal digital assistant
PMI	physical mobile interaction
QR code	quick response code
RAM	random-access memory
SD card	secure digital card
SLA	second-language acquisition
SMS	short message service
SNS	social networking service

TALL	technology-assisted language learning
TAM	technology acceptance model
TELL	technology-enhanced language learning
TPACK	technical, pedagogical, and content knowledge
USB	Universal Serial Bus
UTAUT	unified theory of acceptance and use of technology

1 *Introduction*

1.1 Introduction

The use of technology in education has always been somewhat controversial. This may seem like an unusual statement to make at the start of a book that deals with the use of mobile technologies in teaching and learning contexts, but pointing this observation out from the outset helps to frame several of the relevant issues pertaining to the acceptance – and resistance – of technology, including its position in discussions of theoretical, empirical, and practical issues surrounding its use. Although technology has featured more prominently in education than could have possibly been imagined since the spread of the COVID-19 virus at the beginning of 2020, there still remain strongly divided opinions as to its long-term use as a viable option to quality education rather than a stopgap until the world recovers from the disaster. The controversy surrounding technology usage in education is caused by a complex net of interrelated factors that are difficult to explain in isolation of one another, and yet in some ways have shaped the way that technology has come to be viewed in the larger educational context. This includes, to some degree, how it has been viewed as an academic discipline. Attitudes towards technology have ranged from enthusiastic or overly optimistic at one end of the spectrum to critical or doubtful at the other, and these attitudes have both given rise to and resulted from the controversies surrounding technology use in education. Looking at these controversies and the reasons behind them may lead to a more balanced view of technology – including, of course, mobile technologies – in language teaching and learning to form a more solid foundation on which to understand the concepts and contexts, and to see how best to anticipate and deal with the potential challenges.

Among the many controversies, perhaps the most obvious has centred around *pedagogical* aspects. Since the beginning of the field

of computer-assisted language learning (CALL), discussions about the effects of learning through technological versus non-technological means have held prominence. Some practitioners have embraced new technologies as they appear, while others have been more hesitant to accept them. For some, technology is viewed as an integral part of keeping up with the times (Hanna, Brown, Dede, Olcott, Poley, Schmidt & Tallman, 2000; O'Flaherty & Phillips, 2015), where it is seen as an indispensable tool that provides significant benefits for teaching and learning. For others, however, such technologies are little more than a gimmick, something that can be used to perk student interest for a time but with little or no added real educational value, or even detracting from valuable class time (Reid, 2014; Rogers-Estable, 2014). Depending on the ways in which technology is used, however, both of these perspectives may actually be correct. Technology most certainly does have the potential to add elements to a teaching and learning environment that can enhance learning, but at the same time, if technology is simply used for the sake of the technology itself without careful planning and implementation, then the benefits for learning can be so greatly diminished that non-use can be a more effective option.

A second controversy is related to *socioeconomic* aspects. The digital divide (i.e., the disparity that exists between those with access to technologies and those without) has been a topic of discussion since the 1990s. Widespread access to information and communications technologies (ICT) was seen as being closely linked to socioeconomic development, and the setting up of infrastructure to allow stable and affordable Internet connections has been an ongoing challenge. Mobile devices such as mobile phones and tablets have been seen as potentially having an equalising effect, where mobile broadband has made Internet access more available to users in less affluent regions such as in Africa (Gillwald, 2017) and South America (Galperin, 2016). At the same time, however, debates have also taken place surrounding the dangers of accentuating the digital divide, where users around the world are spending considerably more money on communications than is stipulated in the statistics set out by the Broadband Commission for Digital Development (2015, cited in Gillwald, 2017). Although the digital divide has most widely been discussed at a national or regional level, the discrepancy is also relevant at an institutional or even an individual level. Institutionally, such divisions can result in a type of technological eliteness, where institutions that can afford expensive technologies are somehow seen as providing better services than those that with less advanced resources. It is not difficult to see how this links to pedagogical concerns, with many institutions

feeling real or perceived pressure to provide similar levels of technological resource infrastructure in order to provide an image of a better learning experience for learners (O’Callaghan, Neumann, Jones & Creed, 2017). While it would be difficult to draw a direct link to confirm whether wider access to technologies will necessarily result in better learning outcomes, it is also difficult to argue that there is *no* relationship either, and having greater access to technology does seem to provide greater opportunities for learning if it is used appropriately. That is to say, if learners have access to technology, there is at least the chance for learning to take place, but this is based heavily on how the technology is used. It is individually, however, that we may see the greatest impact of mobile learning with regard to the digital divide. Requiring learners to use their own mobile devices for education can impart burdens upon those in less advantaged socioeconomic circumstances than their peers, which can cause stress and/or embarrassment to them, feelings of inferiority, and potentially even detrimental impacts on motivation to engage in learning through their mobile devices at all.

Thirdly, there is *academic* controversy, one that somehow views CALL as a lesser field to the broader parent fields of second-language acquisition and information technology. CALL has often been branded as lacking in theoretical foundation and academic rigour, and while there may have been some evidence of this in the early days of CALL research, there is also an extremely solid foundation of well-conceived and well-conducted research that has made a significant contribution to our understanding of other fields as well. A seminal article by Coleman (2005) drew attention to this issue, indicating that CALL has often lacked the “mutual respect” (p. 20) of other fields, evidenced by publications in CALL journals citing research from respected SLA journals but very little evidence of the reverse. More than a decade after this observation, the trend still seems to stand largely true, as seen by the lack of references to CALL-related journals in articles that have a similar focus but do not use technology. Technology can provide relevant data on language teaching and learning and insights that are made possible only through the adoption of technology (Blake, 2000). Despite the fact that disseminating research in CALL journals has become increasingly competitive and publishing in high-ranking journals in the field is now considered extremely difficult, the image clearly persists of CALL research as being somehow less rigorous than other, more “established” fields (Leakey, 2011), and it is difficult to predict when or if it will be put on a similar standing with research in SLA or other educational fields.

Finally, technologies have been a part of *administrative* controversy, where pressures have been placed on teachers – and ultimately learners – to use new technologies, frequently with little explanation or support provided, and input in the selection of technological resources is often not sought from the teachers who will actually be expected to use them. The underlying reasons for technology adoption by administrators are, no doubt, complex and have ranged from actual or anticipated cost-cutting, promotion of institutional image, and betterment of the teaching and learning environment, although the real benefits in each of these regards have been somewhat questionable (Bowen, Chingos, Lack & Nygren, 2014; McPherson & Bacow, 2015). There have been, of course, multiple unanticipated outcomes from the introduction of technology by administration, some of which are more positive and others more negative. Positively, aside from the benefits associated with support for learning itself through technology, in some ways, it has made the exchange of information among administrators, teachers, and learners more transparent, where the channels of communication are somewhat more open than in the past. Negatively, the relative ease with which technology makes collecting and analysing data also means that teachers may be subjected to more frequent centrally administered online evaluations. While evaluations in themselves may not necessarily be problematic, they do have the potential to place greater pressure on teachers to strive towards higher evaluation scores (Lejonberg, Elstad & Christophersen, 2018), which may or may not be an accurate picture of better teaching. Moreover, evaluations may even contribute to less willingness to experiment or to be innovative in order to avoid potential failures (e.g., Bennett, Dawson, Bearman, Molloy & Boud, 2017; Carless, 2009).

The cost issue has always been a contentious one, and attempts to use technology to save money inevitably result in shifts towards other expenses such as maintenance of the technologies and hiring sufficient support staff to ensure that these technologies run smoothly (Reid, 2014). The quality of education that is provided by technologies designed to replace the teacher has consistently drawn debate from many stakeholders – administrators, teachers, students, and even parents – with claims by many commercial providers that their products are comparable with human teachers that are difficult to substantiate in actual practice. Apart from the oversimplification of the role of the teacher as little more than a provider of content and feedback, claiming that technology can completely replace human teachers largely ignores the myriad human interactions that are an integral part of learning in virtually all aspects of life. This argument itself brings us

back to the pedagogical controversy, which in turn clearly illustrates the interrelatedness of the various factors at play regarding technology in education.

Early CALL practitioners lamented the lack of appropriate teaching materials, software, and trained staff, likening these problems to those of the language laboratories which preceded them (Higgins & Johns, 1984). Indeed, the lessons that were to be learned from language laboratories were still painfully evident in much of the literature written at the time about using computers in language learning. Claims from CALL research also closely paralleled these concerns, and researchers were often in one of two camps: on the one hand, a lack of computers where learners competed with one another to use the limited machines available to them (Fitzgerald, Hattie & Hughes, 1985), and on the other, an over-prevalence of computers which remain underused due to insufficient skilled teachers and the paucity of appropriate teaching resources (Cuban, 2002; Dunkel, 1987). In recent years, we have an abundance of materials and technologies – particularly with most learners having their own devices – but a lack of infrastructure to ensure that these are used properly, meaning that these materials are often not being used in a time- and cost-effective manner.

These examples are far from exhaustive, but they do serve to give us some insights into the controversies that are involved in the adoption and integration of technology in language teaching and learning, of which technology itself is just one factor, and possibly even the factor which is most easily controlled. With the wider use of mobile devices such as mobile phones, smartphones, and wearable technologies appearing in language teaching and learning, these controversies still exist in many shapes and forms. Pedagogical factors remain central, with some believing that mobile learning is the answer to problems that occurred beforehand. This is a concern that was expressed by Bax (2003) about virtually any new technology in language teaching contexts, well before mobile learning started to enter the mainstream (see Stockwell & Reinders, 2019, for a discussion). Mobile learning has long attracted the interest of teachers and administrators, but pedagogy has generally lagged behind the prospects of what it might become. Even now, we see people who are considering using mobile learning ask what app to use, devoid of any contextual information. This question shows a lack of appreciation for the complexity of the field and is akin to asking what language textbook should be used without specifying the skills to be targeted, the level of the students, or the relationship with other elements of a course of study. MALL – like CALL – really does seem bound in expectations that it will make

teaching and learning easier, provided the appropriate app, software, or website can be located. Of course, this view is not universal, but from my experiences with talking about MALL around the world, this very is indicative of the type of questions that I am frequently asked.

From the Field: The Digital Divide

I recall that several years ago, all of the students in one of my classes had smartphones, apart from one. I was not aware of this initially, as all students had responded that they owned smartphones in an informal survey about the technologies that they owned in the first class of the semester. I asked students to try to use materials that they could access through their mobile phones in class, but this one student declined, looking only at his textbook. After class, he came to me and said that he did not have a smartphone as it was too expensive for him to afford the initial contract cost and the monthly charges, and he only owned a GSM phone so that he could keep in contact with their parents as necessary. I assured him that the materials functioned quite well on GSM phones, but the student said that he felt embarrassed to be seen using his older phone in front of the other students. Eventually, he did engage in a small proportion of the activities on his mobile phone, but I learned a valuable lesson as a teacher that day about the dangers of making assumptions regarding the technologies that our learners possess and their feelings about feeling inferior because they can't afford the technologies owned by their peers.

1.2 The Nature of MALL

The spread of mobile devices has taken place at an enormous rate, with contracts for Internet connections through mobile phones surpassing those of desktop computers from as early 2012 (see Pegrum, 2014, for a detailed overview). Mobile devices have become an everyday part of the lives of many people in their social, work, consumerist, and entertainment agendas (Castells, Fernández-Ardèvol, Qiu & Sey, 2007), to the extent that many people – particularly young adults – would find it difficult to survive without them (Burnell & Kuther, 2016). Mobile devices have in many ways become an extension of our bodies. We carry mobile devices – now most commonly smartphones, but also tablets or even wearable technologies – with us at nearly every waking moment (and as an increasingly common problem and to the detriment of the quality of our sleep, many people have them near their bedside even while sleeping). The fact that they are almost always close at hand is obviously one point that has made them

a target for educators, but if we are to consider how they might be used effectively in learning contexts, we also need to think about what it is about these mobile technologies that makes them such a central part of our everyday lives. Carr (2011) suggests that technologies may be roughly divided up into four main categories: (1) an extension of our physical strength, dexterity, or resilience; (2) an extension of the range or sensitivity of our senses; (3) a way of enabling us to reshape nature to better serve our needs or desires; and (4) a way to extend or support our mental powers. Mobile devices may take on any one of these roles in some way, but the most obvious links to language learning would be their ability to extend the range of our senses (such as enabling us to communicate with others at a distance) and extending or supporting our mental powers (through acting as a notebook, a camera, a dictionary, or a search engine, to name but a few). This ready access enables learners to “exploit small amounts of time and space for learning” (Traxler, 2007, p. 8), but exploiting these times and spaces requires learners to make learning a part of their everyday schedule, where they can take advantage of times that may previously have been wasted. In other words, if learners carry their mobile devices with them to both learning and non-learning locations, they will have greater opportunity for engaging in learning activities, if only they decide to make the most of them.

The portability of mobile devices makes possible another potential benefit helping to contextualise learning – that is, to make learning relevant to the specific situations that learners find themselves on a day-to-day basis (Stockwell, 2014). In other words, the attractiveness of mobile learning is that it not only allows learners to spend more time engaging in learning tasks, but also that these tasks can be made to relate to actual experiences to make them more meaningful to each individual learner. Having access to mobile devices that can provide information means that unexpected or unplanned learning situations, such as needing to explain something in the target language to someone on the street, can be taken advantage of by seeking and immediately using this information in authentic contexts. In addition to portability, mobile devices also allow improved opportunities for communication. The fact that mobile devices are typically associated with various social activities of users throughout the day also makes them attractive to attempt to exploit this social element of learning (see Ushioda, 2011). Furthermore, the flexibility and multimodal and non-linear possibilities of mobile devices make them ideal for learners to adjust them to their own particular learning times, spaces, preferences, and goals (Kress & Pachler, 2007). In all, mobile devices, theoretically at the very least, seem to be an ideal tool in which to make language

learning more accessible and relevant to learners. However, the problem is that this has also led to expectations that have often preceded actual empirical outcomes.

This brings us to ask what mobile-assisted language learning really is, how it is perceived, and what these expectations that are held about it actually are. Pegrum (2019) proposes that the “mobile” part of mobile learning may relate not only to mobile devices but also to mobile learners and mobile learning experiences. Although the general perception of mobile learning is typically bound to the use of mobile devices, those devices can, of course, be linked to the mobility of learners and their experiences (mobility is discussed in more detail in Chapter 9). Thus, in the context of this book, MALL refers to learning a second or foreign language¹ through the use of one or more of various mobile devices including, but not restricted to, mobile phones (including smartphones), tablets, personal digital assistants (PDAs), MP3/MP4 players, electronic dictionaries, and gaming consoles. The definition of what is actually included in the list of mobile devices has been surprisingly difficult to determine. Some have contended that the list might include laptop computers (Kukulska-Hulme & Shield, 2008), while others have argued against this (van’t Hoof & Vahey, 2007). On this issue, Puente’s description (cited in Pegrum, 2014) provides a useful distinction between *mobile* and *portable* devices, where portable devices are typically used at Point A, closed down, and then used again at Point B, whereas mobile devices can be used at Point A, Point B, and anywhere in between if so desired.

A commonly held view of MALL by laypersons is that it refers exclusively to the use of these mobile devices in “outside” locations when the user is in transit or, using the previous example, when learners are at somewhere between Point A and Point B. This is, of course, a common use of mobile devices, but research has shown that many learners opt to use them at home, even when other technologies are available (e.g., Stockwell & Liu, 2015). MALL can also be used to refer to the use of these devices inside the classroom, where learners use mobile devices to carry out certain learning tasks or activities. These devices may be provided by the teacher for the duration of the task or activity, or learners may use their own devices – such as using their own phones, tablets, other similar devices. Thus, I would argue that learning through mobile devices does not necessarily need to refer

¹ It could be argued that MALL, like CALL, could also include learning of the first language, but this type of inclusion is extremely rare in the literature. For this reason, MALL has been limited to the learning of a second language in both second and foreign language contexts only in this book.

to learning on the move, and that using mobile devices such as smartphones or tablets at home is still very much a part of mobile learning in that the users feel the devices they are using are a part of their toolkit of resources that they may choose from for learning. The distinction between the mobility or portability of devices may end up being a moot point. We are starting to see a merge between different devices that were once considered to be separate entities, such as laptops and tablets, where the functionalities are overlapping. Laptops are exhibiting the features that were once associated with tablets, for example a touchscreen; and tablets and even smartphones are becoming used more widely for functions that might have been in the realm of laptop and desktop computers – such as word processing, creating spreadsheets, or other office-related uses.

Defining specific devices for mobile learning is becoming increasingly more difficult. Emerging wearable technologies, most notably watches and other devices like Google GlassTM, would also be classified as mobile devices, and although there is only a limited amount of research on wearable technologies for language learning at the time of writing (de la Guía, Lopez Camacho, Orozco-Barbosa, Brea Luján, Penichet & Lozano Pérez, 2016), the potential is certainly evident (Bower & Sturman, 2015; Sykes, 2018). These devices typically require an interface with another mobile device such as a smartphone or tablet (although there are some devices that can operate with an independent Internet connection), so the correlation with or dependence on other technologies would need to remain in the consideration of the factors in their use. Furthermore, implanted technologies would be considered as mobile in that they must naturally be carried inside the body with the user at all times, but at this stage, research is limited to assistive technologies such as for people suffering from hearing disorders (e.g., Beeres-Scheenstra, Ohnsorg, Candraia, Heinzmann, Castellanos, De Min & Linder, 2017). These are areas where mobile learning is likely to continue to develop in the future, and they are discussed in more depth in Chapter 9.

The ways in which mobile devices are selected and used will vary considerably depending on the functionality and availability of technology – as well as the experiences, skills, goals, attitudes, and preferences of the multiple participants in the individual context such as the teachers, learners, and administrators. This is obviously an enormous issue, and it takes up a large portion of this book, but specific examples of designing for MALL are included in Chapter 8. As already described, one of the goals of MALL activities is to take learning outside of the classroom and into reality, where learners can not only take advantage of those gaps in time and space but also take

their learning into the world; other goals of MALL activities include personalising learning for 'learners' own needs (Kukulska-Hulme, 2016); interacting with the environment using wireless, GPS, or QR code functions (e.g., Chen, Liu and Hwang, 2015); providing information suited to specific situations through context awareness (e.g., Santos, Saneiro, Boticario and Rodriguez-Sanchez (2016); and expanding upon computer-based activities to keep content fresh in learners' minds (e.g., Sharples, 2014). At the same time, MALL also strives to enrich activities inside the classroom. Learners can have access to learning resources (de la Fuente, 2014) and authentic materials (Ducate & Lomicka, 2013), or teachers can augment existing paper-based materials by providing links to multimedia that can enable a more interactive experience (Solak & Cakır, 2015), to name a few of the potential in-class uses. While these are just a sample of the types of activities that might be included in MALL, it is evident that MALL should encompass more than just delivering simplified and somewhat colourless content and activities on mobile devices as a substitute for computer- or paper-based versions (Squire, 2009). MALL can be highly dynamic, creative, and personalised if carefully planned and implemented, and it is this potential that should drive educators to explore how they can use it in their teaching and learning environments.

Needless to say, mobile learning does not mean that learning must be limited only to the device which is being used to engage in tasks or activities. The mobile devices may be used in conjunction with other non-mobile devices, and also with more traditional non-technological means, such as paper-based resources and materials. This can be seen through mobile devices being used to augment reality (see Godwin-Jones, 2016), such as enabling learners to interact with materials or even places around them, even with limited technological skills. This can even be achieved through using mobile resources that act as a supplement to paper-based or other materials, such as audio- or video-based resources that can also be used together with a textbook or other paper-based materials. Of course, mobile technologies can be used to support other activities through other devices like computers which have larger screens and keyboards that are easier for reading or typing, by acting as a resource such as a dictionary, reference tool, communication device, or an audio or video player. In this way, MALL is becoming a multimodal, multiplatform experience where the learner is interacting with multiple technological and non-technological options as a larger part of their learning experience.

1.3 Understanding MALL

Four years after the release of the first smartphones in 2007 (the LG Prada was followed by the iPhone in the same year), Traxler (2011) perceptively described mobile devices as “curiously both pervasive and ubiquitous, both conspicuous and unobtrusive, both noteworthy and taken-for-granted in the lives of most people” (p. 25). Although it has been more than a decade since this statement was made, it has become even more relevant now than it was when smartphones were just starting to find their place into our everyday lives. As educational applications of MALL increase, we are starting to see that these conflicting views still exist, and even though the mobile devices themselves are becoming more ubiquitous, there is both resistance and scepticism with regards to how they can be used effectively. The impact of technology is indeed unobtrusive in that we use devices these days with little thought of picking them up to find information or to communicate with others, but when educational purposes become the target, then there must necessarily be some degree of thought that goes into deciding what to use and how to go about doing it in order to achieve some learning outcome. For self-directed learning purposes, it may include making decisions about what app to use and how to use it (see Chapter 7), but even if tools have been assigned to learners, they are still faced with decisions and dilemmas about how, when, and where – and even why – they should use it, and how this fits into their daily or weekly schedule of private and educational uses of their mobile devices.

The impression that one often gets when talking about MALL is that it is viewed as being some kind of constant, as though MALL can be considered as a single teaching approach. As is described in Chapter 2, MALL is seen largely as being for self-study through apps, which does not take into account the complexities of what can be done through mobile devices based on the enormous range of technical functions that are now available. MALL can range from being a means for searching for information through a search engine on a mobile phone (Gu, 2016) or as an electronic dictionary (Levy & Steel, 2015) to using a dedicated vocabulary app like Quizlet (Tran, 2018), QR codes to promote interaction between students in class (Rivers, 2009), recording videos to develop presentation skills (Toland, Mills & Kohyama, 2016), to name a few. This small sample of the types of studies that can be carried out using mobile phones gives some insight into the complexities involved and the difficulties in referring to mobile learning as any kind of homogeneous type of learning. Surveys that ask learners about their attitudes towards so-called mobile learning have little to no meaning if

there is not some kind of supplementary information that can clarify *what* activities, as it is likely that most learners will possess very different attitudes towards different types of activities and tasks. MALL is a constantly evolving mixture of factors that capitalise on the various affordances inherent to each device and the new hardware and software that are developed over time. As a result, MALL refers to something that is fluid, and it is next to impossible to encapsulate it in a single term. In this way, when MALL is used in this book, it is describing a broad spectrum of activities and tasks that are tailored to different technologies, abilities, and contexts.

In understanding what MALL is, it also is helpful to consider also what it is not. As touched upon before, MALL is more than just a means for learners to engage in individual self-study, although it is often viewed that way by many who are contemplating using mobile technologies as a part of their teaching and learning environments. In one sense, this perspective is an example of a parallel that exists between CALL and MALL. Just as CALL often conjures images of self-study in front of a computer, MALL is envisaged as being predominantly dedicated language learning apps. In one sense, the image of mobile learning as equating with apps is probably not that surprising, given that apps are one of the most visible aspects of our current mobile devices. It is true that pretty much everything that happens through mobile devices is indeed done through apps, such as launching web browsers, social media, office tools, music, and even cameras. As described before, however, MALL is far more than this, and using mobile technologies opens up a whole world of interactive and social possibilities that can enrich the learning process qualitatively and quantitatively.

I would argue that MALL is not *fundamentally* different from CALL. There are some who might disagree with this statement, but thinking through what MALL is can help to make sense of this perspective. As with CALL, the ultimate aim of using technologies is to enhance the various language skills (namely, writing, listening, and speaking) and language areas (such as vocabulary, grammar, pronunciation, fluency, and so forth) that go into making a person into a proficient user of a language. The similarities lie not only in the ultimate aims of teaching and learning a language but also with the expectations for their impact on the learning process and the fact that they allow for instant access to a range of multimodal learning materials, social communities, and immediate and personalised feedback. As described earlier, they have also been through a similar path of evolution that makes them the target of comparisons with other methods, as will be described further in Chapter 4.

There are, however, a number of considerations that do need to be kept in mind in order to understand how to make the most of mobile technologies from the perspectives of both teaching and research. Firstly, the *affordances* of mobile devices need to be considered. Put simply, the term affordances is used to connote what something makes possible (see Norman, 2013) and is often used to refer to the technical functionality of the device, but it can also include environmental factors and the perceptions of how the device will be used (see Chapter 2 for a discussion). There are naturally going to be differences in the affordances of mobile devices from various other technologies such as computers (and, of course, variations between different mobile devices as well), so understanding the impact of these affordances can contribute to understanding what device or devices are most suited to a particular environment. Secondly, realising that mobile learning in recent years has relied heavily on people using their own devices means that there is a need to be aware of the differences between what learners will be carrying with them in terms of both hardware and software. This has, of course, happened with CALL to some degree now too, with many institutions opting to reduce self-access computer laboratories to cut maintenance and running costs as more students carry their own laptops, but many institutions still opt to keep computer rooms for class learning. Thirdly, tracking what learners do with mobile devices both inside and outside of the classroom is rather more difficult when learners are using their own devices. Computers in classrooms may have tracking software installed for research or learning purposes, but it becomes decidedly more difficult to track usage with learners' personal mobile devices. Finally, using mobile devices for assessment can be a challenge, particularly in classroom situations. A mobile may be used as a "replacement" for the computer in carrying out some activities, but their use in some forms of assessment may be difficult. The teacher may be able to keep a watchful eye on learners' interactions with computers either directly or through computer lab management software, but mobile devices may require a more vigilant approach. In saying this, ways of preventing cheating through mobile devices are also starting to emerge. The issues of using the tracking and assessment with mobile devices will be dealt with in more detail in Chapter 8, but there are obviously underlying issues regarding how learners use their devices that also need consideration. Tracking does make it possible to see not only *when* but also *how* students are doing required tasks or homework outside of class. Server access logs can allow teachers to see learner behaviour in doing tasks or homework, but that has also made it possible to see that many learners simply complete tasks without considering the reasons for doing so (Fischer, 2012).

Looking at both the product *and* the process can reveal rather undesirable engagement patterns, often because learners are just not aware of the reasons for doing these tasks and activities, which makes the value in requiring them to do so in the first place somewhat questionable (Palardy, 1988; Wallinger, 2000). Thus, an unexpected outcome of the affordances of technology is it can provide a window into learner behaviour that can lead teachers to reconsider their practice.

A term that is often misunderstood is *ubiquitous learning*. Ubiquitous learning is often used interchangeably with the term mobile learning, but these two terms refer to overlapping but quite different concepts. Looking at the meaning of these two terms individually sheds light on the difference; mobile means something that is on the move whereas ubiquitous means that something exists everywhere. Mobile learning, then, is most commonly used to refer to using mobile devices to engage in learning on the move, but as described earlier, it is also used to encompass using mobile devices in fixed locations as well. In contrast, ubiquitous learning means having access to learning technologies in whatever location the learner might be in, and this may include a combination of both mobile and non-mobile devices. In ubiquitous learning, the learner can use multiple devices that share data and information seamlessly between them, such as may be seen with cloud computing, while mobile learning is most commonly carried out on devices that are carried by the learner, which may or may not access this shared data or information. In this sense, mobile learning might be considered as a part – albeit an important one – of ubiquitous learning, where it makes up the range of tools that are available for the learner to remain connected to the learning environment. Where there may be limitations caused by the affordances of a mobile device, they can be compensated for by using other technologies, and at the same time, learners may be able to pick up where they left off using a less-mobile technology such as a desktop or laptop computer to continue engaging with learning content or others. It is evident that the affordances of mobile devices make it impractical to be the only electronic device to be used language learning in terms of screen size, text input, storage, and battery issues (Stockwell, 2016), so embedding mobile learning into the larger context where it complements other tools and devices would seem to be a preferable way to keep learners engaged in learning in various shapes and forms, which logically would point to the conclusion that MALL is an essential part of ubiquitous learning and vice versa.

The preceding discussion provides a very brief introduction to some of the key issues associated with mobile learning, gives some insights

into what mobile-assisted language learning is (and what it is not), and points out a number of key issues that will be discussed in more depth in later chapters. The remainder of this chapter provides an overview of each of the chapters in the book.

From the Field: Mobility in MALL

The image that is held of MALL, and indeed the main “selling point” of it, is that learners will take advantage of short moments in the middle of their busy schedule to engage in small snippets of learning (sometimes called “micro-learning” or “atomised learning,” as described in Chapter 8). Theoretically, this is a wonderful concept – and one that is thought to be suited to learners with shrinking attention spans – but it is based on assumptions that do not seem to hold true as often as teachers might hope. In my experience, the reasons for this appear to be that when learners are in positions that might be considered as mobile (in transit or even in a café or other location that they temporarily occupy), the distractions inherent to such locations make it difficult to concentrate sufficiently on the tasks they are trying to engage in. The problems of returning one’s attention to a learning activity after even just a few minutes away from it mean that there is inevitably going to be some lead-in time to prepare before any meaningful learning can take place. Looking at many of my own learners has shown that they also come to this realisation. Learners with longer commutes on public transport do, in fact, take advantage of this time, but for the most part, many learners seem to seek quieter locations where they can spend quality time on learning activities, even if they use their mobile devices. This does not necessarily mean that learners do not use the small snippets in time on occasion, but it seems that learners become aware of the limitations of trying to learn in short bursts and favour locations such as their own homes, where they can engage for longer periods without interruption.

1.4 Overview of the Book

Apart from this introductory chapter, there are eight chapters in this book, including the Conclusion – which brings together the various discussions raised throughout the book to address the title of this book, considering the concepts, the contexts, and challenges. Chapter 2 considers parallels between CALL and MALL and outlines how the technological affordances of various devices affects the ways in which they are used to achieve specific goals. It then describes the interrelationships among technology, research, practice, and theory. It outlines the evolution of mobile technologies and emphasises the

importance of moving from affordance-based practice through to pedagogy-based practice as technologies move through the hype cycle (cf. Gartner, 2016). The issue of push and pull modes associated with mobile technologies is outlined here, along with the ability of mobile devices to interact with other devices through a range of networking functions, the potential for mixed reality learning and context sensitive learning, and the use of other functions such as cameras and audio recorders for language learning. The chapter continues by pointing out some of the limitations of research on technology use in language education and then considers some of the complexities of both theory and practice in MALL in order to lay the foundations for the later chapters.

The main thrust of Chapter 3 is to discuss how dramatically teaching and learning are changing, largely as a result of developments in technology. These changes have brought about shifts in the roles of the teachers, of learners, and even of the technologies themselves. Not only do teachers have to manage their teaching environments, but they also need to manage their technological skills and the emotional load that goes along with the pressures of maintaining digital literacy. Learners are faced with having greater expectations to use technology, while at the same time it is expected that they are already skilled in using technologies for learning purposes. It explores the possible future directions of education, where teachers and learners need to consider not only what information needs to be learned but also what information is acceptable to be referenced. They also need to develop skills in evaluating information from the enormous amount of available resources. The chapter also explores the view of teaching and learning in formal and informal contexts and looks at how mobile technologies have impacted both of these learning situations. It deals with the emerging concept of mobile literacy, outlining what learners should be aware of when engaging in learning and non-learning activities on their mobile devices. Finally, given that mobile devices are constantly “switched on” and generally carried almost constantly by learners, the chapter discusses the issue of motivating learners to take advantage of their devices to engage in activities that are associated with learning languages.

Research into MALL has proven to be somewhat complex when compared with much of the earlier research in the field of CALL, which is the focus of Chapter 4. The reasons for this are twofold. Firstly, by the nature of the device, there is an expectation that a significant proportion of learning through mobile devices will take place out of class. While this does not preclude using devices in classroom settings (and there have been several very interesting studies

that have looked at the use of mobile devices in class), it is exponentially more difficult to determine how learners are using their mobile devices in uncontrolled and unsupervised settings. Secondly, learners are typically using their own devices, although early studies into mobile devices typically involved using loaned PDAs or even iPads, whereas similar data for mobile devices owned by learners is not available because it is not logistically possible to install tracking software on them. Learner usage data then will need to take the form of data that can be stored on a central server or reported data, each of which face potential limitations. Thus, the purpose of this chapter is to outline the difficulties involved with undertaking research in MALL as well as describing various innovative approaches that have been undertaken, illustrated through appropriate recent examples from the literature.

Chapter 5 deals with theory in MALL. Discussions of the role of theory in CALL have been rather limited to date and have typically relied heavily on second-language acquisition (SLA) theories or other theories related to learning and pedagogy. This in itself is not particularly surprising or inappropriate, but there has been very little discussion of the role of technology in the learning process. The majority of the work that considers theory relating to technology to date has revolved around the multimodality that modern technologies make possible. This has included looking at the cognitive load associated with each mode and how access to information through different channels can facilitate learning. These aspects are equally relevant in MALL, but there are added elements with using mobile devices that must also be taken into consideration. These include the way in which technologies can be used as an ongoing reference tool, where users can look up information quickly and easily rather than committing them to memory. These aspects were also relevant to a far lesser degree through non-mobile devices, but the more frequent the access we have to mobile technologies, the more likely we are to pick them up to seek out information rather than attempting to retrieve information already stored in our memories. This has an impact not only on theories of learning but most certainly on theories of pedagogy.

Chapter 6 explores the physical, psychosocial, and pedagogical issues associated with MALL. The physical characteristics of mobile devices such as the size of the screen and the input methods have long been an issue when considering their applicability for learning, but these have often been considered as a necessary trade-off in order to maintain their portability. In addition to this, however, there are also psychosocial issues considering the position of mobile devices in the minds of the learners and teachers, such as a personal tool for private

uses or a tool that can be applied to any use as required. This perception of mobile devices is often a product of the social context in which they are used, and this will likely vary depending on the region, the socioeconomic status, and the age group of the users. Furthermore, the dangers in distractions both inside and outside of the device are described here along the psychological impact of mobile devices on learners' abilities to concentrate on multiple tasks. Finally, the chapter discusses the issue of pedagogy when learning through mobile devices and the factors that may be thought to contribute to successfully achieving learning goals and sustaining task engagement. The details of research studies that look at the impact of each of these elements are described.

The ultimate user of MALL is the learner, which is the main focus of Chapter 7. Learner agency is an issue that helps us to gain a better understanding of how and why learners are or are not able to make appropriate choices applicable to their learning. Agency may be related to individual characteristics of the learner, but it may also be supported by proxy or collective agency from the teacher or larger environment. Even learners who exhibit agency experience difficulties in making appropriate choices about their learning, which is why learner training is essential. In MALL, learner training is key in ensuring that learners understand the reasons for engaging in mobile activities as well as what is expected of them from teachers. Stockwell and Hubbard (2014, 2015) provided evidence that ongoing training in technological, strategic, and pedagogical aspects can have a very powerful effect on the ways in which learners view mobile and engage in learning activities. Thus, the chapter argues for ongoing training that guides learners to develop autonomy through evaluation of the strategies they employ and sharing their learning experience with others to help them reflect on their learning.

Chapter 8 explores the concept of design in MALL. Design can take place at multiple levels, starting with the larger learning environment, through to the digital artefacts that are used by learners and the design of tasks for using the artefacts within the given environment. Some of the key models of general design of the learning environment, artefacts, and tasks are provided with examples of how they relate directly to MALL. The complexities of designing assessment are also discussed. The chapter concludes with basic guidelines to bear in mind when designing a MALL and outlines a list of principles for the successful implementation of mobile devices in language teaching and learning contexts.

Finally, Chapter 9 brings together the arguments covered in the previous eight chapters and returns to the title of the book: concepts,

contexts, and challenges. The concepts that need to be kept in mind for the future of mobile learning are explored, along with the impact of the context on language teaching and learning through mobile technologies. Along with these, the current and prospective challenges are also investigated, with a view to seeing how these challenges can be overcome to make the most of what MALL can be. The potential future directions in which mobile learning may be considered to evolve will also be discussed here, not in terms of evolving technologies, but ways that the field seems to be headed and how these can relate to meaningful research and practice that is needed in both the short and longer term.

There are an enormous number of issues associated with mobile learning that could have been covered in this book, and while I have attempted to give attention to what I consider to be the main issues, there are obviously other issues that are emerging at the time of writing and different viewpoints and perspectives that others may feel strongly about from their own individual experiences. Mobile-assisted language learning varies in so many ways, depending on the attitudes, skills, and preferences of learners, teachers, administrators, and other stakeholders in the learning context to the various technologies that may be used, the networking environment, availability of resources, the language being studied, social expectations, and the list continues. This diversity contributes to the ever-changing nature of the field and has also brought about some extremely creative and even ingenious uses of different mobile technologies. It is hoped that this book will spark an interest in what mobile-assisted language learning can be while, at the same time, allowing potential adopters of mobile technologies to embark on their journey with their eyes open to the benefits and dangers that might be associated with MALL. It is becoming quite undeniable that mobile-assisted language learning will be a part of most language learning environments in some shape or form as time goes by, so having an idea of what has been done and what we still need to investigate further can help to lead towards meaningful research and practice.

1.5 Discussion Questions

1. Do you see the digital divide as being problematic within your own teaching or learning context? Why or why not? If so, how might this be dealt with specifically in your context?
2. How would describe what MALL is in a sentence? How would you respond to questions about the best apps for language learning?

3. In what way would you see MALL as being different from CALL? Do you think that this difference is important? Why or why not?
4. What would you include in a list of devices that would be appropriate for MALL? Give a rationale for each of these. Are there any mobile devices that you would not include? Why?

2 *Fundamental Considerations of Teaching with Mobile Technologies*

2.1 Introduction

MALL has already been in existence for nearly two decades at the time of writing, and mobile devices have undergone an enormous shift in capabilities since they first came onto the market in the early 1990s. The primary function of the first generation of mobile phones (often referred to as “bricks” due to their large, blocky shapes) was to enable telephone calls where people were not constrained to a single location, and for the first time, they made it possible to send text-based messages to others without the need for a fixed landline with a fax or telegraph. This provided people, particularly those in small businesses, with an extremely liberating tool, and marked the beginning of communication in “any place” and at “any time.” It was not until mobile phones reduced in size and included a texting feature that the potential of using them for a wider range of function – including learning functions – became more of a logistical reality. Ownership of mobile phones with these functions did increase, but during the late 1990s, they were still only slowly entering the market for university-level students. SMS texting was an increasingly common way of communicating, although in some countries such as Japan, texting was rather late to appear, with mobile-to-mobile emails being the primary text-based communication tool.

Of course, it should also be remembered that mobile phones were not the only mobile devices to develop through the 1990s and early 2000s; portable MP3 players, electronic dictionaries, PDAs, small netbook computers and laptop computers also became more commonplace. From around this time, research that looked at how these devices, particularly MP3 players and electronic dictionaries, could be used appeared, and a view of how they could be applied to certain aspects of language teaching and learning became apparent. The primary functions that became possible as a result of these technological

developments were the ability to listen to audio, both authentic materials and materials that were prepared for learning purposes, and the ability to stop, pause and listen again to these resources became possible. Electronic dictionaries also allowed learners to add an audio component to hear the pronunciation of words that they looked up, as well as to check words across several dictionaries to see them used in various contexts. This audio component was indeed an important aspect which contributed to mobile devices being considered as having wider potential as a tool in language teaching and learning. Another important function of electronic dictionaries was to make it possible for learners to handwrite words that they were looking up, adding a more kinetic element to the learning process as well.

The emergence of the smartphone and tablets with touchscreens in the late 2000s meant that functions that were once distributed across several devices came to appear together in a single device. Smartphones and tablets became equipped with high-quality MP3 players, dictionaries became available as apps, and other functions as described in the previous section were also added, such as cameras and other interactive capabilities. This led to an explosion of developments over the past decade with the hard-fought competition between various developers of mobile devices such as Apple, Google, Huawei, Oppo, Samsung and Xiaomi. In some ways the users have been the ultimate winners in some ways of this competition in that processing power, battery life, memory, storage, operating systems, camera quality and screen size have improved exponentially over the past several years as developers strive to outdo one another to attract customers to purchase their products. Ignoring the other advances from GSM phones (which were also surprisingly sophisticated devices), the first iPhone had a 3.5" screen with a 2.0 MP camera, 620 MHz processor, 128 MB of RAM and up to 16 GB storage when it was released in 2007. The latest version of the iPhone at the time of writing has a 5.5" screen (far from the largest available) with a 12 MP camera, 2.49 GHz dual core processor, 4 GB of RAM and up to 512 GB of storage. There are, of course, far more powerful devices available on the market at present, but the continuity of the iPhone from the earliest models makes comparisons of the developments immediately obvious. Although there is vast variation in the sophistication of mobile devices owned around the world, now more than three-quarters of the population own a mobile device with internet access (GSMA Intelligence, 2019). The impact on teaching and learning is that the vast majority of learners will already own their own devices at the time of commencing study, be it formal or informal, and their mobile devices are likely to already feature in some way in the learners' mind in their toolbox of learning resources.

It is not surprising that the path followed by MALL as a field has had some similarities to that of CALL over this period, but at the same time, there have been variations that have appeared as a result of the differences in ownership of technologies and the functionality of the devices themselves. Furthermore, the foundation that has already been set in place by the twenty-odd years of CALL research has also played a role in shaping MALL research and practice, and at least some of the lessons learned from CALL have been applied to MALL. In saying this, there has definitely been some degree of “wheel reinventing” taking place as well, and it is important to reflect on how both MALL and CALL have come to be where they are today in order to consider how to seek out the best teaching and learning practices with mobile technologies.

2.2 Using the Terms CALL and MALL

As touched upon in Chapter 1, CALL and MALL are in some ways misunderstood, and the terminology itself can be a cause of confusion. Both the terms CALL and MALL are often used in a somewhat nebulous manner. They may sometimes be used to refer to a specific technology (a computer, a mobile device, etc.), sometimes a teaching approach (i.e., teaching with a computer or mobile device), and sometimes both. “CALL” was used in a sense that CALL was somehow homogenous, and there appeared to be some shared understanding of what “CALL” was. Discussions between teachers would frequently include, “Do you use CALL?” and this was largely understood to mean some kind of courseware or other commercial package for use in the classroom. This is, of course, very much a misnomer, in that CALL refers to using technologies to achieve different learning goals in different ways, as can be seen by the enormous amount of work carried out over the years, of which courseware is a relatively minor technology, with research using various generic technologies appearing most frequently (see Stockwell, 2012). In the same way, MALL has also been used in a somewhat confusing manner. The word “mobile” in MALL is sometimes used to refer to the types of device used (i.e., mobile devices) and sometimes used to describe the nature of the learning as being something that happens while the user is in transit between locations (i.e., learning on the move).

The frequently tacit view of the meaning of “computer” in CALL or “mobile” in MALL can have a strong impact on the way in which these interrelated fields are viewed, and the naming of them can lead to assumptions that may not necessarily accurately reflect what the fields represent. For example, there has been continued argument for many

years about whether the use of the term CALL is appropriate to represent the broad range of technologies that are being seen in teaching and learning environments, and there have been multiple alternatives proposed over the years, most notably TELL (technology-enhanced language learning) and TALL (technology-assisted language learning). While the arguments in favour of using these alternatives are understandable to a certain extent, the question of the actual necessity to do so still remains. There have been a number of discussions that have supported the continued use of the term CALL (Hubbard & Levy, 2016; Levy & Hubbard, 2005; Stockwell, 2012b), with one of the primary reasons being the fact that the term has been in widespread use for at least three decades now, and it continues to be used as the title of various books and journals. CALL has included more than just desktop and laptop computers for many years now, and it is not in any way different from or inferior to some emerging field with a different name like “TELL” that looks at the broader range of technologies. The research that has appeared in books and journals using the term “CALL” has been extremely broad and is not limited to a single type of technology. Over the years, it has been used to refer to mainframe computers, personal computers, laptop computers, laser discs, e-book readers, PDAs, tablets, mobile phones, MP3 players and electronic dictionaries, among others.

Depending on the view of what a “computer” is, the term CALL is most certainly sufficient to encompass all types of technologies that would be considered as being relevant to discussions on their use in second-language teaching and learning. Definitions of what a computer actually is may vary, but in its simplest form, a computer must allow for some means of input, have a processor to perform the required operations with this input and then provide the result as output. The ways in which this input can take place will differ depending on the device, and input methods have undergone an enormous number of changes over the decades since the general public has had access to personal computers. Input for these early computers consisted largely of just a keyboard and some kind of storage device such as cassette tapes, 5 ¼" floppy disks and small hard drives (if we take it for granted that ticker tape machines were not exactly the mainstream for personal use), but over time, storage capabilities increased dramatically with more durable floppy disks and hard drives, CD-ROMs, laser discs, DVDs and even voice input appeared. More recently, USB flash memory sticks; input methods such as mice, trackpads, styluses and touchscreens; and alternative ways of linking machines to other sources and devices such as Wi-Fi, Bluetooth and NFC (Near Field Communication), to name a few, have been added to

the repertoire of potential forms of input (but some of these have been superseded and fallen out of usage completely).

As described at the beginning of this chapter, processors have also undergone enormous evolution over the years, and without going into specifics, not only are they much faster but they are also more capable of handling multiple tasks at the same time (although this remains a feature that does separate computer processors and mobile phone/tablet processors to a certain degree, where mobile processors are far less suited to running applications concurrently). Processed information can, for example, be made available to a screen, a speaker, a printer, as a file on internal or external storage or to the Internet or a different device or devices. If we use this definition of a computer as a device that has a means of receiving input, processing that input and then providing some kind of input based on the processed information, it is clear that many other devices – including electronic dictionaries, MP3 players, mobile phones and tablets – would share the vast majority of these features, with the primary difference simply being their portability, sometimes with limitations in their range of functions. If we look back at the ENIAC and other room-sized computers from the 1940s, we can see – physically at the very least – a closer relationship between smartphones and, say, laptop computers than we would see between computers from the 1940s and personal computers today, despite the same name being used. Thus, despite the fact that our modern mobile phones are indeed a computer of sorts, the way in which they were developed from initially being a portable replacement for fixed-point telephones means that they have come to be viewed as entirely different devices from computers. This is perhaps one of the biggest hurdles that practitioners with technology need to contend with, and it is also a view that extends to how mobile devices are perceived as being used by others, where assumptions can be made that learners are “playing” with their devices rather than using them for a study tool (see Chapter 6). Even though the term “computer” would seem to be quite applicable to the range of devices that would seem to be fit into the broader category of “technology,” overcoming the psychological barriers pertaining to what they are and how they are perceived as being used has the potential to contribute to this pluralistic view.

The term “mobile phone” (or “cell phone,” depending on geographical preferences) has for many years referred to devices that go far beyond the functionality of their namesake, the landline telephone. Even in the early 1990s, these devices allowed for sending of text messages, something that was not even thinkable for conventional fixed telephones (if we exclude sending faxes). The smartphones that

are the mainstream today obviously go even further beyond the description of a telephone, and research has shown that audio telephoning is now one of the lesser-used functions of these devices (Richmond, 2012). The operating systems and associated software of these devices are extremely sophisticated, allowing users to have access to various textual and audiovisual communication and social networking tools, games, internet browsers, calendars and office tools, most of which were very much in the domain of computers little more than a decade ago (cf. Kress & Pachler, 2007). Mobile devices have contributed to pushing the envelope further, as it were, with regards to touch technologies and wireless data transfer, and many of these features have been reverse-engineered into laptop and even desktop computers (although sales of the latter have dropped considerably in favour of the former, or hybrid devices that sit somewhere between a “laptop computer” and a “tablet”). While technically there remain differences between most mobile devices and computers in terms of processors, operating systems and other software, for the most part, the user experience makes it difficult to identify noticeable differences in performance between these different types of devices. It is in the portability, input methods and screen size, however, that we have the most salient differences, and this is largely the reason behind the differences in perceptions that many people hold, despite the fact that the similarities far outweigh the differences. In this way, the terms CALL and MALL have led to stereotyped views of what the fields are about – both for practitioners in the field and for people outside the field – that can lead to preconceptions about what technologies are used and the ways in which they are applied to language teaching and learning.

Another similarity that both CALL and MALL are subject to is comparison with other teaching methods. As described previously, the difficulties in relating either to any particular type of technology or approach make such a question quite meaningless. Since its early days, the question of effectiveness of CALL was frequently in the background behind decisions about whether institutions should be spending money on computer laboratories or other equipment. The problem is, however, that asking whether CALL is better than other methods is akin to asking if using textbooks is more effective than non-use of textbooks or if using a whiteboard is better than not using one. The variety of uses of technology make such comparisons questionable at the least, and possibly even counterproductive. We see a similar phenomenon these days with mobile technologies as well, where studies seek to ask questions about attitudes towards MALL and to determine whether or not MALL is effective compared to other methods,

including CALL or non-use of technology. Effectiveness of and attitudes towards any method – be it CALL, MALL, or otherwise – will depend on a great deal of factors that go beyond the technology itself.

Both CALL and MALL use sophisticated electronic devices in order to facilitate the teaching and learning of languages, but we must be aware of the physical, functional, social and perceived differences between computers and mobile devices to see how they have been – and subsequently could be – used effectively by learners to enhance their acquisition of their chosen target language. The different operating systems and software of various mobile and non-mobile devices may have slightly different functionalities, and this is also the case for many programs and applications that are used for language teaching and learning. Part of this is as a result of the physical and technological affordances of the device, which would certainly be thought to have an impact on what can and should be done through mobile devices. Needless to say, these affordances have also had an impact on the focus of research and the ways in which this research is carried out, as discussed in the following sections.

2.3 Emerging Affordances of Mobile Devices

The concept of “affordances” was originally coined by Gibson (1979, 2015), who used it to describe the perceived potential uses that an artefact (i.e., objects, devices or any natural or human-made things around us) may be used for. These may vary from the actual intended functions themselves, meaning that there will necessarily be individual differences that will impact the ways in which objects – or artefacts – will be used and viewed. Van Osch and Mendelson (2011) proposed three types of affordances: designed affordances, improvised affordances and emergent affordances. *Designed affordances* are those that have been built into the artefact by the developers and are created in such a way as to support the use of the artefact in a way that is expected by them. For example, a colouring book is designed for children to fill in colours of pictures that have been provided by the authors. The pictures in themselves are basically already complete, and the only thing that is required is to choose what colours the child wishes to use and to draw or paint them into the existing pictures. *Improvised affordances* are those that were not recognised or intended by the developers, but that users themselves find while using the artefact. Using the colouring book example, improvised affordances might be that children try to draw new pictures based on those that are in the picture book or add other things to the existing pictures such as people, furniture or a background. These may not have been intended

uses of the colouring book, but users improvise and add their own affordances over time. *Emergent affordances* are changes to the environment itself as a result of the artefact that were not consciously anticipated by the developers or the users. For instance, seeing their children enjoying a colouring book may prompt parents to create groups for them to play together with other children using their colouring books, thereby creating a sense of excitement and community that was not anticipated by the developers or the users themselves. This example suggests that affordances are a complex product of subsystems which include the characteristics of the artefacts and users (Maier & Fadel, 2009), and the broad spectrum of affordances that are ultimately attached to an artefact will often be far removed from the initial expectations of the developers, emerging over time through its use usually with repercussions that go beyond the actual artefact.

The term affordance has come into broad use in research associated with technology as well, with *technical affordances* describing the functions of a device and may encompass the designed affordances (but also include features such as the size and weight of the device). A modern smartphone, for example, was designed by developers for telephoning, using the Internet, taking pictures, GPS and so forth. Over years of use, improvised affordances have appeared as a result of users exploring the technical affordances in practice. The relationship between an artefact and the social context is known as *sociomateriality* (Leonardi, 2012; Orlikowski, 2007). As Suchman (2007) explains, it is essentially impossible to separate the social and the material in practice, and artefacts will be shaped by users in the same way that users will be shaped by artefacts. Individuals will interact with artefacts in different ways as a result of their individual characteristics and objectives, and these differences will ultimately impact upon the artefacts, the users and the larger environment (the issue of sociomateriality will be dealt with in more detail in Chapter 9). Thus, the affordances of different devices will depend not only in their functions (i.e., technical affordances) but also on the backgrounds, skills, experience and imaginations of the users adopting them in their individual environments.

The technical affordances of mobile technologies have undergone an enormous change over time since the late 2000s, in terms of greatly enhanced processing power, storage, operating systems and technical functionality. These changes have resulted in research that has focussed specifically on these emerging affordances and how they might be adopted or adapted to teaching and learning. From the beginning of MALL, research looking at how technical affordances

of new devices (including hardware and software) could be used for language learning has had a central position. For example, the SMS function of mobile phones was an early area of research, which was used for creation of mobile flash cards (Thornton & Houser, 2002) and for the sending and receiving of SMS messages, such as receiving information from the teacher (Kennedy & Levy, 2008), and for enabling learners to send necessary details to one another to complete information-gap activities (Kiernan & Aizawa, 2004). Web browsers made it possible to access the Internet for accessing learning tasks and activities (e.g., Stockwell, 2007) or conduct searches for information (Liaw, Hatala & Huang, 2010).

Other developments in mobile technologies have also been reflected in research pertaining to their usage. The camera function was one that attracted the attention of researchers, both in the form of still images and video recordings. Liu and Chen (2015), for example, had learners take photographs of scenes that reflected grammatical forms that they learned in class, and Gromik (2012) showed when learners uploaded videos of themselves speaking in English, it gave them confidence in speaking. Other features that have been used include QR (quick response) codes that enabled information in the real world to be quickly transferred to PDAs as they used a map to visit learning zones (Liu, Tan & Chu, 2010), or to provide information supplementary to pictures in materials to make them more interactive (Rivers, 2009). These interactive tools make it possible to link learning resources together with the real world and, hence, can give learners greater sense of reality with regards to their language learning.

Another feature that has been considered as having potential is the GPS function, where the location of the learner can be used as a means of triggering information. An example of this was used by Sandberg, Maris and de Geus (2011), who provided Dutch learners of English with phones with GPS capabilities. The learners then walked around a zoo, where they could retrieve information about the animals that they saw based on their location through the phones. Both of these studies are linked with the concept of *augmented reality*, which is when the real world is supplemented by digital resources. This can take the form of having a picture on a page in a book become a video file when a mobile device is held over it (Boonbrahm, Kaewrat & Boonbrahm, 2015), adding audio or annotations to items or places that learners might encounter in the real world using image matching (Liu & Tsai, 2013), or GPS that enables the device to obtain information either from the Internet or other nearby devices.

The emergence of apps through the various stores (App Store, Google Play, etc.) has made it possible for learners to have access to

an extremely large number of apps available either for free or commercially (or, in some cases, a free version which can be upgraded to a full version at cost). This broad range of apps has been empowering for learners, but at the same time, it also puts a burden on learners to try to make decisions about what apps are the most appropriate for them. While some have argued that choosing from the variety of apps that is available may be liberating to students (Mindog, 2016), for the most part, it is not surprising that many learners find appropriate app selection an extremely difficult task to achieve (Kim, 2013), and as such, they need guidance to choose which apps to use (see Chapter 3) and training in how to use them to benefit from them as much as possible (learner training is covered in more depth in Chapter 7). Whereas the literature into CALL has featured examples of development of dedicated software for language learning, MALL research has included the development of dedicated apps. In both cases, these apps are typically suited to a specific learning environment or context and, to that end, have also some limitations in their broader application, but these examples do provide insights into what is desirable design in apps. The affordances of mobile devices have remained a topic of investigation and, as Kukulska-Hulme and Viberg (2018) have proposed, include flexibility and continuity of use, socialisation, personalisation, active participation and self-evaluation, and links with real-world situations and culture. Knowledge of the affordances of mobile devices is certainly essential, but it is important to go beyond this and to see how this knowledge can be applied to using them in a pedagogically sound manner.

One problem that became apparent with the mobile devices owned by the learners themselves (BYOD = bring your own device) was that learners were bringing in tools not only running different operating systems on their smartphones (iOS, Android and, later, Windows Mobile) but also the hardware itself varied from learner to learner, with some owning the later models of mobile devices at one end of the scale and students using the most basic models at the other. This fact has implications for the choice of technologies by teachers to ensure that no learners are disadvantaged when compared with their peers because of financial reasons (see Chapter 1 for a discussion of the digital divide).

While the boundaries between modern mobile devices and more traditional technologies such as desktop and laptop computers are starting to become blurred, there are features and functions that are typically associated with devices such as mobile phones and, to a lesser degree, tablets. These include physical mobile interaction (PMI) using QR codes (Rivers, 2009), Global Positioning System (GPS) capabilities

(Cheng, Hwang, Wu, Shadiev & Xie, 2010), push notifications (Stockwell, 2013), and various communication tools that enable the user to have greater interaction with the surroundings, peers, interlocutors, and teachers (e.g., Mok, 2012; Tran, 2016). As alluded to earlier in this chapter, there are also often differences between the desktop and mobile versions of software and apps (and, indeed, sometimes between apps on different operating systems such as iOS or Android). At the same time, even the range of mobile devices carried by learners is decreasing, and where once learners might have carried MP3 players or e-readers in addition to their tablets or smartphones, these functions are being consolidated into the devices that learners already possess (Delcker, Honal & Ifenthaler, 2018).

2.4 Evolution of CALL and MALL Research

CALL research as we know it today extends across four decades, and there has been an enormous amount achieved over this period in terms of the quantity and quality of research which has been carried out. It is not surprising that much of what has appeared in CALL literature over the years has largely reflected developments of technologies described in the previous section. Research from the 1970s and 1980s focussed on standalone “microcomputers,” which were considered as being extremely small for the time and resembled many modern-day desktop PCs. Language learning tools included software that focussed mainly on grammar or vocabulary (Anderson, 1987), local networks where learners could interact with one another using text chat (Young, 1988), videodiscs which could allow for a limited degree of interactive multimedia (Peppard, 1989), and hyperlinks, where reading could diverge from the traditional linear model and allow learners freedom to move through a text or resources or access annotations with relatively little interruption (Tchudi, 1988). While there was a simplicity in the research from this era in some respects, the depth of thinking that went into the development and implementation of CALL materials should not be ignored.

It is easy to assume that CALL from this time was primitive, and although it may have been limited by the technologies to some extent, it is clearly evident that the view and expectations (both positive and negative) of using technology in language teaching and learning concepts were insightful and have done much to inform practice of today. Themes that appeared during this time have reoccurred in later decades once the technology caught up with the imagination and creativity of researchers from this era. For example, intelligent syntax checkers (Galletly, Butcher & Lim How, 1989; Paramskas, 1986),

natural language parsers (Imlah & du Boulay, 1985), intelligent tutoring systems (see Chapelle, 1989), and computer-mediated communication such as computer-based messaging, mailing, and discussion forums (Murray, 1988) have continued to be the focus of CALL research some twenty to thirty years after they first appeared in the literature. Though, of course, advances in technology have meant that there is a greater degree of sophistication in the software and applications that are used, we can see that there was a lot of early work that provided a very solid foundation for later development.

One area where we have seen significant development, however, is in the type of research that is carried out using technology. Early CALL research consisted predominantly of comparative research to determine the effectiveness of CALL compared with non-CALL means (see Burstson, 2003), often as a means of proving that the time, effort, and expense that went into the development and implementation of CALL resources was worthwhile. Failure to prove that technology was somehow more “effective” than more traditional means was to admit that the benefits of introducing technology did not exceed (or even reach) the personnel and financial costs associated with it. As a result of this, there was a tendency to see research that showed “unreasonable effectiveness” (Felix, 2005), in that when considering the methods in which a technology was used, the reported outcomes were highly unlikely to be possible and, indeed, failed to be replicated in later studies. The CALL literature of the 1980s and 1990s contained quite a significant number of studies that compared use and non-use of technology, but considering that the technology was often an addition to the existing method used, it was not surprising to see more favourable results when technology was introduced into the equation. A further problem that Hubbard (2005) pointed out was much of the pre-2000 research was typically carried out with a small number of subjects for a short period of time, and often termed “pilot studies.” There were rarely any follow-up studies performed to confirm the validity of results that may well have been – at least in part – a result of the novelty of using newer or more “exciting” tools rather than as an outcome of innovative pedagogies which capitalised on the technology. Indeed, despite the arguments in favour of CALL – that it was motivating for learners in terms of its individualised nature, opportunities for learner control, and “rapid, frequent, and non-judgemental feedback” (Warschauer, 1995, p. 30) – research has consistently demonstrated that while there may be motivational effects for learners in the short term, learner engagement is not maintained after this initial novelty effect has worn off (see Stockwell, 2013, for a discussion), suggesting that it is the pedagogy adopted for using the

technology rather than the technology itself which is likely to contribute to longer-term task engagement (task engagement is discussed in more depth in Chapter 7).

In fact, later studies that were carried out over longer periods of time which controlled for the novelty effect in the early stages have more often than not tended to show “no significant difference,” but it could also be argued that the pendulum of proving the effectiveness of technology had swung a little too far back the other way, where researchers became overly cautious of making any claims in favour of the technology. While studies may not show any immediate favourable results of technology, it does not necessarily mean that there have not been other impacts that might manifest themselves quite some time after the data have been collected and analysed, so to dismiss the role that technology plays on the basis of a single study over a limited period of time is just as problematic as holding an overly positive view. Comparisons of CALL and non-CALL are, in many ways, comparing “apples with oranges,” where, despite some areas where comparisons might be possible, there are too many differing features that render direct comparisons somewhat uninformative.

Though we have seen that there were many extremely inventive and innovating uses of technology, some of the early literature into technology use in language teaching and learning contexts showed a lack of creativity or attempt to take advantages of the features of the technology, and many CALL activities were essentially digital equivalents of paper-based versions. Of these, many allowed for automatic grading but often included relatively superficial interaction or feedback in the shape of hollow compliments such as “Well done!” for correct answers and encouragement like “Good try” when learners failed to produce the expected responses to activities. This type of research was more concerned with what could logistically be achieved through the technology and did not take into account the body of non-CALL literature that was already making headway in showing when, what, and how feedback could benefit learners (e.g., Sheppard, 1992; Truscott, 1996). In saying this, feedback did feature in the CALL literature as well, with research that explored automatically generated feedback in intelligent grammar checkers (van der Linden, 1993) and human feedback from the teacher through communication tools such as email (Hoffman, 1994). Studies of this type paved the way for later research that considered the complexities of the relationship of task, feedback, and learner responses to feedback.

There is no doubt that the emergence of the Internet in the early 1990s had an impact on the face of what CALL meant. Prior to this, CALL predominantly consisted of standalone machines running

software or CD-ROMs (Cummins, 1998) or machines that were networked locally to allow for some degree of interaction amongst users through dedicated software (Beauvois, 1995). While these tools came to allow for multimedia as the technology developed, it was not until the emergence of the Internet that learners were able to interact more freely with other learners outside of their own classes; classes of target-language speakers for language exchange; or other groups such as chat groups or even newsgroups, bulletin board systems, or discussion sites intended for native speakers to exchange opinions on their areas of interest. In fact, the Internet (or World Wide Web) was the impetus for a boom in research in computer-mediated communication, initially text based (Toyoda & Harrison, 2002) but then moving on to audio based (Cziko & Park, 2003) and then video based (Wang, 2004). The promise of interacting with native-speaking interlocutors was indeed an attractive one, again with undertones of increased motivation and intercultural understanding along with opportunities for negotiation of input (Kitade, 2006), but the problem of task design along with unfounded assumptions regarding the nature of online communication resulted in outcomes that were not always as positive as might have been hoped for.

Studies looking at mobile device usage was, however, relatively limited, as student ownership remained limited to a certain extent and relied on the teacher providing class sets. This meant that all of the students were using the same technology, and there was essentially no variation in the tools that were being used by the subjects of studies utilising them. As described before, however, the downside was that the tools were closely monitored (likely to avoid anything being broken or going missing), and usage outside of the classroom was typically still on the school grounds (e.g., Chen & Li, 2010). Smaller phones began to appear with simple web browsers that enabled users to access information over a 2G network, but the volume of data that could be received was limited, as was the complexity of the sites that could be loaded by the relatively simple browsers of the time. This did allow some creative studies to be carried out, using both the SMS (and/or email) functions and the browsers. The technical affordance of being able to use text and to be able to access the Internet proved to be the start of a movement towards using more text-based communication on mobile devices, and the original primary function of telephoning gradually became a lesser-used function.

Post-2000 research saw a range of new concepts emerge – including various learning management systems (LMSs), educational gaming, MOOCs, and, of course, sophisticated mobile devices. This research was shaped by new technologies of the time, but there was also a shift

in the focus of the research, prompted in part by the critical evaluations of CALL research by Hubbard (2005) and Felix (2005). Thus, the focus came to include the specific features of CALL design and of learner engagement (Stockwell, 2012a), such as the role of learner training (Romeo & Hubbard, 2010), the development of learner autonomy (Hafner & Miller, 2011), sustainability of task engagement inside and outside of classroom settings (Appel & Mullen, 2002), and the views of students towards using their own personal devices for language learning purposes (Stockwell, 2008). This resulted in a movement away from comparing CALL to non-CALL in terms of effectiveness for second-language teaching and learning towards looking at the specific features of the environment, pedagogies, training, and support that could play a role in facilitating more effective and efficient learning.

The evolution of CALL from its beginnings as largely a replication of paper-based materials until it reached a “critical mass” whereby the finer details of design and the practice of using technologies in varied contexts were investigated has been repeated in the evolution of MALL in some respects. Early MALL research that appeared in the mid-2000s was, in essence, a replication of the resources available for computers adapted to mobile devices, often in a way that was simplified to work with lower processing power and smaller screens, resulting in “stripped down versions of their more complex desktop predecessors” (Squire, 2009, p. 71). While this was hardly surprising in some sense, it is likely to have contributed – at least in part – to the limited engagement and the failure to meet expectations regarding learning through mobile devices, a fact that was pointed out by several researchers investigating mobile learning during this period (e.g., Petersen, Divitini & Chabert, 2008; Stockwell, 2008). Of course, there are other reasons why engagement in mobile-based activities was limited, particularly where learners were using their own devices for engaging in activities. Data costs were relatively high, and when the onus for paying this was placed on the learners, there was reluctance and even resistance to using it. Obviously, Wi-Fi for mobile devices at this point was limited, meaning that self-funded data usage was essentially the only option open to learners, and this combination of factors may well have contributed to limited engagement when other alternatives were available. Studies that were undertaken with learners using devices that were provided to them yielded somewhat higher usage rates. When PDAs were provided to learners, for example, studies have shown learners to be more willing to undertake activities on these mobile devices (see Stockwell, 2014, for a discussion), but these were typically lent to learners for limited periods of time, and there

was likely a novelty effect at work that may have contributed to improved usage. Furthermore, the conditions surrounding the use of these devices was quite controlled, and as such, it was very difficult to get any reliable indication of if or how learners engaged in mobile learning in natural contexts outside of the classroom.

It is not surprising that MALL research has also been heavily impacted by developments and availability of technologies in much the same way as CALL has been. Because smartphones are largely owned by the learners themselves and carried with them throughout the day, they have enabled researchers to gain insights into the true nature of “mobile” learning, in terms of the ways in which learners integrate learning through mobile devices into their everyday lifestyles. This is something that was not really possible when devices were borrowed or when the physical limitations of the devices made it difficult to design a wide variety of activities. There were, however, attempts to overcome these difficulties based on the functions of the devices at the time with minimal outlay required by the learners. Kennedy and Levy (2008), for example, capitalised on the SMS function of mobile phones to provide learners with learning-focussed reminders of vocabulary and expressions that had been taught in class, or upcoming programs on TV that learners could watch to support their language learning. Flash cards (Thornton & Houser, 2002) and an intelligent vocabulary tutor (Stockwell, 2007) were also examples of tools designed for the precursors to smartphones available during the 2000s, and these tools could be operated at relatively low cost. There were, of course, other devices that were the subject of research during the early 2000s, such as iPods being provided to all first-year students at a US university and MP3 players being used to distribute college entrance exam lectures (see Chinnery, 2006, for a discussion). Studies that looked at the use of MP3 players for podcasting appeared in the literature for some time, and the possibilities were considered as being hopeful (e.g., Rosell-Aguilar, 2007). As with mobile phones, however, usage proved to be somewhat less than expected, with reasons being cited that it was easier to use a computer, and that learners did not know how to transfer podcasts to their MP3 players (Abdous, Camarena & Facer, 2009). This second factor highlights another major factor associated with mobile learning: the provision of sufficient training, which will be covered in Chapter 7.

MALL research over the past several years has certainly become far more diverse in the contexts in which mobile devices are being used, the range of uses, and, of course, the skills and areas that are targeted as a part of the mobile-based tasks. Vocabulary has featured overwhelmingly as the main area to be studied by researchers since its

inception, and this was particularly the case with the earlier devices described previously. Smartphones have meant that a far wider range of language skills and areas have started to be researched as well, with studies appearing that look at listening and speaking (Liu, 2009), reading (Hsu, Hwang & Chang, 2013), grammar (Wang & Smith, 2013), vocabulary (Lu, 2008), and even writing (Liu & Tsai, 2013). There remains, however, an imbalance towards vocabulary acquisition, but this is likely caused in part by the fact that it is easier to measure in research than other skills and, as such, may not necessarily be completely attributable to the technology itself.

It is in some sense inevitable that MALL follows a similar path of predominantly emulating both paper-based and CALL activities, but research that considers the technological functions of mobile technologies has started to take a more central role in recent years. Modern mobile devices, particularly smartphones, have an enormous range of functionalities – including powerful processors, high-definition touchscreens, high-resolution cameras, video and audio recording capabilities, GPS capabilities, and various wireless interactive interfaces such as Bluetooth and NFC, as well as physical data transfer through linking with cables, USB flash memory sticks, or SD cards. The fact that learners possess smartphones that already have these functions built in without needing to purchase other devices has contributed to a shift towards mobile learning, and as such, practitioners consider the specific functions or affordances of mobile devices and how they may be applied to MALL pedagogies.

From the Field: Comparing Technologies

In both my capacity as a teacher of graduate students in applied linguistics and as an editor of journals using technology for language teaching and learning, I often see the starting point of research as the comparison of new technologies with non-use of technology or, more frequently since MALL has become more widespread, comparisons of mobile technologies with other technologies. This is not something that is beyond understanding. Newer technologies do seem to invoke questions about their effectiveness, and at a first glance, comparing with what has previously been shown to be effective in some sense seems very reasonable. However, the problem is that accurate comparisons just aren't possible in the vast majority of cases. By default, using different tools will alter the learning environment, and artificially attempting to control for all variables will usually end up with comparisons that have little meaning. The advice I usually give to my graduate students is

that they should ask themselves *why* they feel a need to show one technology is more effective than another. More important research would be to examine *how* a technology – including mobile technologies – do indeed alter the environment, and to use that as a starting point for seeking out pedagogies that may lead to more successful language learning.

2.5 From Affordance-Based to Pedagogy-Based Practice

As alluded to before, a look at the research into both CALL and MALL have revealed similar patterns of evolution, starting with the emergence of a new technology, often immediately followed by a rush or publications about this new technology. This pattern has been seen a number of times as new technologies emerge – some of which have persisted (LMSs, gaming, MOOCs, etc.), and others which have all but disappeared from the research, either because they failed to reach wide adoption (e.g., laserdiscs) or because they were superseded by newer technologies that made the older ones no longer viable to use, despite reaching relatively high levels of attention (Hypercard, Hot Potatoes, etc.).

The lifespan of a technology can be mapped onto Gartner's hype cycles (e.g., Gartner, 2016) – which can give an interesting perspective into how technologies can be introduced, used (or forgotten), and linked to pedagogy. While Gartner's hype cycles have received some criticism for their lack of explanation of the phenomena they represent (Cuban & Jandrić, 2015), they do provide some insights into how technologies may be viewed by users and how they might be used (or not used). There are four main phases in Gartner's hype cycles, starting with the technology trigger, when the technology appears for the first time, up to the "peak of inflated expectations," where users of the technology may expect that more might be possible from the technology than is actually logistically feasible. This is behind Bax's (2003) argument that one of the fallacies of using technology is that a new technology is often considered as being more effective than what came before it, simply because it is newer. This can prompt users of the technology to believe that it will be able to solve the problems that were evident in previous methods of technologies and cause a rush to use it with expectations that may not be practical (see Stockwell & Reinders, 2019, for a discussion). It is hardly surprising that such inflated expectations lead one to disillusionment, which is the next step in the hype cycle. It is at this point that the disappointment regarding a technology may result in a greatly reduced use of the

technology, perhaps even to the extent that it is abandoned. Should usage of the technology continue, then comes what Gartner refers as the “slope of enlightenment,” where the way in which a given technology might be used is investigated in more detail, until eventually reaching the “plateau of productivity,” where the balance of how to use a technology together with existing tools can be settled on, and specific ways of integrating the tool to achieve maximum productivity become clear to the users.

Although Gartner’s hype cycles were not intended for education, it is easy to see how they can be applied. We can see, for example, how the early phases (from technology trigger to the trough of disillusionment) typically represent affordance-based practice, where users are still trying to identify the technical affordances of the technology and counterbalance them with their own individual goals and expectations for the technology’s use in their individual context. It is very much a trial-and-error phase, where practitioners consider various ways in which the technology might be used. This phase also coincides with what Levy and Stockwell (2006) refer to as “emergent CALL,” where new technologies are still very much in the experimental stage, and the potential of the technology is sought.

The later phases of the hype cycles (the slope of enlightenment and the plateau of productivity) reflect a narrowing of focus to pedagogy-based practice. There is already some understanding of the technology and the way in which it might be used, and, as such, researchers can direct their attention to finding the best practice for the technology, where they hope to achieve as much as possible from the technology, using a balanced pedagogy that builds upon what has been found out about the technology from the previous experimental phase. This phase corresponds to Levy and Stockwell’s (2006) “established CALL,” where practitioners already have a basic understanding of the technical affordances of a technology and now look at how more generalisable conclusions about using the technology effectively are devised.

This means that technologies in language education appear to follow a pattern that starts from the hype of what the technology is capable of, with the exploration of the technical affordances of the device and what the potential uses of the technology might be. Expectations appear to be high at this point, often to the point of being unrealistic. The release of the iPad, for example, was thought by many to be a revolution in education, where textbooks would become obsolete and learning would take on new forms of interactivity that were simply not possible through more traditional means. While there has obviously been some impact on education from devices such as

iPads, the concept of providing all learners with them at the beginning of a course of study has not really eventuated. After seeing that iPads themselves did not radically change the learning environment in any broad sense, some abandoned them altogether as a learning tool. At the same time, others have considered what features of iPads could lead to more informed pedagogy. Without understating the importance of the earlier phase of exploration with new devices, it is in these next stages where the empirical testing of these devices has led to pedagogies that lead to more stable results. Research into mobile devices, like those that have come before, needs to go beyond the potential and extend into the actual in order for meaningful progress to be made in the practice of using them.

From the Field: Hype Cycles with Technology in Language Learning Research

Technology in language teaching and learning has invariably been bound to trends. Some of these trends last for as long as several years, and others seem to be very short-lived. I recall attending many conferences over the years where there seemed to be a certain technology or concept that attracted a particularly large number of submissions, regardless of the theme of the conference itself. I can still very clearly remember one conference where there were numerous presentations about the LMS Moodle. The conference theme was fairly broad (Human Issues in CALL), so it was not so unexpected to see presentations that were relatively loosely related to the theme, but in that particular year, there was as many as one-third of the presentations or more just exploring the affordances of Moodle. Moodle has, of course, remained broadly used, but presentations that look at what Moodle can do have all but disappeared. In this sense, while Moodle has moved from the affordances stage through to the pedagogy stage on the hype cycles chart, research has become quite sparse. Actual use of tools will not always be accurately reflected in the research which appears about them, where research seems to rotate around new developments. It is often not the case of these tools disappearing but, rather, the way in which research is framed to focus more on trends than might be the case in practice.

2.6 Complexity of Research, Theory and Practice in MALL

The intersection between research, practice, and theory has been considered a central feature of most academic fields, and CALL and MALL are by no means any exception to this. There have been several

articles and books published over the past several decades that deal with one, two, or even all three of these areas, and as such, it deserves sufficient attention in any discussion pertaining to the use of technology in second-language teaching and learning. One point that cannot be denied in mobile learning, almost to the point of being obvious, is that the technology itself cannot be ignored. As much as a technology may become “normalised” – that is, so much a normal part of their repertoire of tools that learners do not need to think to pick it up in much the same way as with a pen or notepad (Bax, 2003) – it is essential to remember, as Levy (2002) reminds us, the technology is “never transparent or inconsequential.” In other words, the technology must have some impact on the context in which it is used, even if the users (learners, teachers, researchers, administrators, and so forth) are not consciously aware of it.

Technology will necessarily have an impact on research, both as a means of conducting the research and also as the focus of the research. Indeed, technology has made it easier to conduct research by giving us insights into the learning process that would not normally be necessary if not for the technology featuring in a study. Similarly, the relationship between technology and practice is also an interdependent one. Technology most certainly has an impact on practice, in that the way that most things at the planning, preparing, execution, and evaluation stages are impacted by the technology in some shape or form. This includes, of course, the media or platform through which tasks are administered and even the ways in which feedback is given to learners based on their performance in both online and offline learning activities and events. As with research, this relationship can also be a bidirectional one, and technologies themselves, in terms of the selection of a technology, the activities that are designed for use through a technology, and the timing of using which technologies for what purposes (in class, after class, desktop computers, mobile phones, Moodle, or Blackboard, etc.) are all important decisions that need to be made based on the way in which practitioners decide to undertake practice. The cross-relationship between research and practice has been discussed in so many fields over the past several decades, and it is clear that there is a co-dependent relationship, where research affects practice and vice versa. Research will more often than not be based on determining best practice, and identifying how to improve upon practice through research is often one of the primary outcomes for carrying out research in the first place. Similarly, practice will be changed based on the results of research, sometimes relating in even further research, causing an ongoing cycle of research and practice until the most effective ways of undertaking practice can be found. Although this is

an example of practice being altered by the outcomes of research, it is also feasible that practice is modified in order to make it easier to carry out research. In other words, varying practice could be a means to an end – that is, a way of carrying out research that may well have an impact once again on the way that practice is undertaken. In this way, it is clear that each of the three elements of research, practice, and technology have an interdependent relationship, where changing any one of these elements will necessarily have an impact on the other two.

The element that has not been mentioned yet, however, is theory. What is the role of theory in the use of technology in second-language teaching and learning? While this will be discussed in more depth in Chapter 6, at this point, we can say that theory has an impact on the way in which we view a learning situation, the choices of teaching approach, the choices of materials, the choices of technology or technologies, the way in which the technologies and other teaching materials are used, the way in which research is collected, and the way in which the results of such research are analysed and interpreted. Therefore, it is safe to say that theory underlies everything that happens in the teaching and learning context, and the selection of a given theory or theories can shift some elements to the foreground and others to the background. Practitioners have a need, then, to consider the theory or theories that they have selected and to consider how they might impact the views that may be held of the learning context. Views that see the complexity of the entire environment have gained more attention over the past several years (e.g., Larsen-Freeman & Cameron, 2008), and this has enabled the consideration of the fact that all of the different elements may contribute to outcomes, even when they may not always be immediately obvious.

What is pertinent in considering the role of research, theory, and practice in the field of MALL is that it would be a blatant overgeneralisation to try to put the technology into just a single category of tool. The preceding discussion makes it very clear that the technology can be used in a number of different ways, many of which are largely unrelated to one another, and hence, an overarching theory, teaching approach, or research agenda would simply not be possible. If we view the technology as a reminder tool, as we might with push notifications such as the learning reminders (e.g., Kennedy & Levy, 2008) or lists of vocabulary that learners experienced difficulty with when undertaking online vocabulary activities (Stockwell, 2013), then the ways in which the technology is used – and, of course, the teaching approaches and research methods – will necessarily be very different from when using the technology as a communication tool for text or voice

communication, such as using it in reading and composing messages in social networking (e.g., Tran, 2016). What becomes a challenge with using mobile devices for practice and research, however, is the fact that because learners are frequently engaged in activities as a part of their everyday lives outside of the classroom, using their own mobile devices, it is extremely difficult to get an accurate indication of how they use the technology to engage in language learning. The issues of research and theory are raised in more depth in Chapters 4 and 5, respectively.

2.7 Summary

As the preceding discussion shows us, teaching and researching in MALL has in many ways followed the path that CALL went down some two decades earlier. MALL has been greatly influenced by developments in technology, many of these being some of the most remarkable seen over the past century in terms of processor speed and stability, interactivity, and the ability to not only access but also create multimedia resources. The fact that mobile learning now heavily relies on learners using their own devices means that there is greater variation not only in the tools that they own but also in the skills they have regarding the individual functions and features of their own devices. While there are going to necessarily be differences in the capabilities and functions of the devices owned by learners, more importantly, the different skills and experiences that they learners possess can also lead to a wider range of improvised affordances as they use trial and error along with their own creativity to make the most of the designed technical affordances. The sociomaterial impact on learners, the technology, and what learners do with technology have the potential to shape learning, teaching, and the tools that they are using, as can be seen in Chapter 3.

MALL is an extremely complex field that has certainly already moved out of its infancy – it is more like a teenager primed to leap into the wider world of language teaching and learning while looking at its older sibling who is already making an impact on that world. What is important to bear in mind is that mobile learning is more than simply selecting or designing apps that learners can engage in alone or just, outside of the classroom, and there are infinitely broader possibilities regarding the ways in which learners can be encouraged to use their own devices in a wider variety of ways, which can enable them to take advantage of the broad range of technical affordances of the device beyond apps. Given the sophistication of the devices that learners carry around with them on a daily basis, the task of both

research and practice should be to capitalise upon these features so that learners have more opportunities to engage with the target language as a part of their daily lives.

2.8 Discussion Questions

1. Some people believe that mobile learning will replace learning through other devices. What do you think of this view?
2. Mobile-assisted language learning is still considered a relatively under-researched field, although there is already a large body of research which has been undertaken. Why do you think this idea persists?
3. A common belief is that learning through mobile devices refers only to apps. Is this view understandable or acceptable? Why or why not?

3 *Shifting Paradigms in Language Learning and Teaching*

3.1 Introduction

There is no doubt that technology has had a profound impact on the way learning and teaching are conceptualised. At the same time, understanding this impact has proven to be difficult, as encapsulated in the famous statement by Roy Amara in what is now known as Amara's law, that, "We tend to overestimate the effect of a technology in the short run and underestimate the effect in the long run" (VanDerLinden, 2014, p. 78). This also has been evident in education, and language education is no exception. What it means today to be a learner or a teacher is somewhat different from what it was as recently as a decade or two ago, and technology has been one of the greatest facilitators of this change. Coupled with a move towards learner-centredness from the more traditional teacher-centred model in the last thirty years (Cuban, 1993), teachers are relinquishing much of their control over what and how learners learn. The reasons behind this move have been varied, but one of the main tenets is that learners are diverse and that a one-size-fits-all approach provided by many teachers cannot be sufficiently flexible to meet the needs of the various students in a single learning environment (Wenden, 2002). Allowing for individuality has long been cited as one of the advantages of using technology as well (e.g., Warschauer & Healey, 1998), so it is not surprising to see that technology has in some ways become synonymous with learner-centred approaches.

Deciding on specific methodologies that enable the learner to become the centre of their own learning has proven to be difficult, and the ways in which learner-centredness is implemented depend upon the pedagogical perspectives of teachers (Beck, Czerniak & Lumpe, 2000; Isikoglu, Basturk & Karaca, 2009; Minor, Onwuegbuzie, Witcher & James, 2002). This has been compounded with the fact that teachers are often needing to make decisions about using technology in their

language learning environments without sufficient training or experience (Hubbard & Levy, 2006). Teachers are aware of the potential of technology in education but struggle to make the first steps in integrating it into their own teaching and learning. This load is not only one that is felt by teachers. While there are indeed enabling factors that technology can afford in educational contexts, institutional requirements to use technology infer an added responsibility as well (García-Peñalvo & Alier Forment, 2014), which is perceived by learners as even being a burden (Marín Juarros, Salinas Ibáñez & de Benito Crosetti, 2014). One of the underlying factors behind this burden is the assumption that learners are expected to be active users of technology, with learning technologies being provided to them based on the assumption that learners are competent and willing users of technology as part of their learning experience. Teachers may find themselves to be unwilling users of technology, often as a result of executive decisions that pressure them into using technologies that may not necessarily be the best fit for the learning environment, and employ pedagogies that they do not fully understand or particularly ascribe to (Hu & McGrath, 2011). The following discussion describes the shift in roles for both learners and teachers and explores the position of mobile technologies as both a facilitator and supporter of this shift.

3.2 Shifting Roles for Learners

For learners, education has changed dramatically in terms of the ways in which they engage in activities for learning, the outcomes of these activities, the means of assessment, and even the goals of their language learning. Change is inevitable in language teaching and learning – a fact that was made all the more obvious with the COVID-19 outbreak – and many of the broad range of language teaching approaches that have been seen over the past century. Technological developments have had a particularly marked impact on what happens in education settings, some of which are more direct and others less so. The development of jet engines, for example, meant that international travel became faster and eventually more affordable, which gave rise to opportunities to visit and spend time in countries and regions where different languages are spoken. Guided also by changes in general views of psychology in education, this contributed in a shift in teaching and learning approaches away from methods of teaching language with a lower focus on speaking such as grammar translation, and communication was given much more attention (see Ellis, 1994; Morrow, 1981; Savignon, 1983). Technologies such as cassette players and VHS recorders made it easier to bring audiovisual materials into the

classroom, and this is likely to have had an impact on actual classroom practice during the methods era of the 1950s through to the 1980s, when teaching approaches that relied on more authentic input became more widespread.

Technological advances have contributed to not only the practice of what happens in the classroom but also the goals of learning itself. Just as jet airplanes have made travel to different language communities more accessible, this type of travel has also impacted the focus of language study – that is, to use language to communicate with others from these communities. More recently, learners have also been affected by how technology is being used in education, and at the same time, have moved towards electronic tools as preferred means of using the target language. Technological developments have impacted the role of learners in terms of both the processes and products of learning a language (Levy, 2002), but this has gained even greater relevance over the past several years. Changes in the *processes* result directly to how learning takes place; technology is becoming an integral part of the entire learning process in the shape of learner management systems (e.g., Blake, 2016), MOOCs (e.g., Martín-Monje & Bárcena, 2014), virtual online communities (Sokett & Toffoli, 2012), and so forth. In contrast, *products* refer to what learners are now required to create as a part of their course of language study. Handwritten essays may be replaced by online writing methods such as blogs (e.g., Fellner & Apple, 2006; Pinkman, 2005), face-to-face collaboration may be supplemented with social networking tools such as Twitter (Leis, 2014), and paper-based quizzes may be replaced by online quizzes that are automatically graded (Fageeh, 2015).

Since information and communications technologies (ICT) have become an integral part of the interactions that many of us take part in on daily basis, face-to-face communication is no longer the only option available to language learners, and in many cases, it is not even the preferred option. People commonly engage in electronic communication with others more and more frequently, where they can interact in different languages in written and spoken forms without needing to travel at all. Language learners are becoming language users in diverse multilingual contexts – exploiting their full range of linguistic resources while using language creatively to communicate through *translanguaging* (García & Li, 2014), but also through personal development (Leung & Scarino, 2016). Expectations of acquisition of languages, then, have seen a shift from striving towards native-speaking norms to include communicating with others who may also be learners of the languages that they share (e.g., Cook, 2016). This type of communication is particularly evident in predominantly

English online communities like Facebook, Twitter, or Instagram, where people may come together to share their views about mutual interests. It is not surprising to see that the objectives for learning a language by many learners have moved towards participation in these communities, and there are learners who (initially at least) may have no great desire to travel to regions where the languages that they choose to learn are spoken but, rather, to interact online with communities of people through the language. This is particularly evident with learners of English, largely due to the strong presence of English as a lingua franca in many regards on the Internet, but there are also strong communities of Chinese, French, and Spanish, to name a few. As a result, despite the movement towards oral communication from traditional written approaches such as grammar translation, there are learners now who are honing their skills in written communication, using tools such as SNS and text messaging tools. Technology has also become central in communication, disseminating and gathering information, and even entertainment, uses which may coincide with learners' ultimate aims of using a second language.

These different products place a burden on learners to be familiar with the tools that are used to create them. While many learners are familiar with personal uses of technology, there is evidence that suggests that not all learners are willing and/or able to use them for learning purposes, despite the claims made by the so-called digital natives metaphor (Bodsworth & Goodyear, 2017), they need sufficient training to use technologies for learning (Hubbard, 2004). Creating the optimum learning environment for these learners is dependent upon this understanding that learners are not homogeneously equipped to deal with the changes in education that technology has brought with them and, at the same time, realising that the goals that the learners bring with them may differ greatly to the goals that were held by their teachers when they were themselves learners.

3.3 Shifting Roles for Teachers

As described in the introduction of this chapter, changes in technology have meant that there are different demands on teachers in many aspects, some of which have the potential to impact the very concept of what teaching is. Virtually all aspects of education have been affected in some way, including the ways that students enrol; the ways that classes are conducted, assessed, and evaluated; the ways that students search for, collect, and collate information; the ways that teachers and students interact; the ways that grades are calculated and notified to students; and the ways that teachers receive and deal with

feedback from students – just to name a few. Given this enormous shift over the past several years, it is not surprising that some teachers have had difficulty adjusting to these changes, in the midst of a need to keep up with the developments that are taking place on an ongoing basis and to be aware of the various implications that these changes entail. The impact of COVID-19 in this regard has also been phenomenal, with the majority of teachers being thrust into situations where their only alternative is to teach through technology even though this was unthinkable to most just weeks before the outbreak. The changes brought about by COVID-19 are indeed enormous, and this point could be emphasised at various points throughout this book, but I will leave this discussion to Chapter 9, where some of the considerations are discussed in more depth. Education has been in flux for some decades, and although these have been compounded by the pandemic, the issues discussed in what follows have been in the background for a very long time.

Such is the scale of the shift that technology has brought to education over the past thirty years, there are those who question the very nature of what it means to learn – such as Dikkers (2014), who suggests that there is a need to think about whether the current model of teaching, learning, and testing is relevant when learners can quickly look up information using mobile devices that are readily available to them. One common response by teachers to this change is to express fear about the relevance of their jobs in the future, and there are many teachers who worry that they can be replaced by technology. They feel threatened not only by the technology but also by teachers who are more proficient users of the technology than themselves and, in some cases, even by the learners, who they see as being more competent with technology than they are themselves. The identity that teachers possess of themselves in the learning process plays a central role in their success (see Dörnyei & Kubanyiova, 2014), and this links directly to the confidence to appropriately perform the tasks required in each of these capacities.

The motivations for teachers to use technology in the classroom are varied; some are prompted by external factors such as the institution, and some are internal, such as a desire to use a technology in their teaching and learning contexts, although this may also be prompted more by a sense of needing to rather than wanting to. The impact of teacher motivation for using technology should not be underestimated, as it will have a direct influence on what and how technologies are used and the longevity of their use (see Stockwell, 2013). For example, if a teacher is using a technology that is imposed upon them by their institution without a clear understanding of how and why

they should be using it, then there is a greater chance that they will be less likely to integrate it into their teaching in any meaningful manner. This will have a direct influence on learner attitudes to the technology, as learners are often expected to try to use it without sufficient guidance or explanation.

Teacher training with technology has been a topic that has featured quite heavily in the literature since the mid-2000s, and there has been increased awareness of the need to ensure that teachers are sufficiently trained to be able to use technology effectively. While the issue of learner training comes up in Chapter 7, it is important to bear in mind that learner training is inseparable from teacher training (see Hubbard & Levy, 2006). As with many other aspects pertaining to the use of technology in teaching and learning, it is important to bear in mind that the complexities of teacher training are compounded depending on whether it takes place in pre-service or in-service contexts (Table 3.1).

Teachers in each of these categories are likely to encounter different types of problems with integrating technology into their own contexts.

Table 3.1 Overview of teacher experience with training in technology

Category	Typical Characteristics
Pre-service teachers	Recent university graduates Have familiarity with private uses of technologies Inexperienced in teaching May have used technology in own education May have received/will receive some training in technology
Early in-service teachers	Recent university graduates Have familiarity with private uses of technologies Some experience in teaching May have used technology in own education May have received some pre-service training in technology
Experienced in-service teachers	Graduated from university some time ago May have familiarity with private uses of technologies Experienced in teaching Unlikely to have used technology in own education Unlikely to have received pre-service training in technology

Pre-service teachers are frequently younger and are generally familiar with using technologies as an integrated part of their own daily lives. They have also likely experienced the use of technology from a student's perspective and, as such, have some idea of what it feels like to use a technology as a student and understand some of the problems that learners might encounter. At the same time, they are also likely to be inexperienced as teachers, and this means that even if they understand how to use technology (for private purposes at least), they may not be aware of how technology use might fit into the larger teaching context. They may have received some training in the use of technology as a part of the pre-service education, but, as still relatively inexperienced teachers, they may experience difficulties in seeing how technology can be used in achieving certain teaching and learning goals.

Early in-service teachers who are recent graduates are likely to face some overlapping benefits and difficulties but may have different problems compared with pre-service teachers. They are likely to have experienced using technology as a learner in the same way as pre-service teachers and be familiar with using technologies for private purposes. They may have received some degree of pre-service training with using technologies, which may give them some idea of where to start when adopting technology in their teaching and learning contexts. They would be expected to have more teaching experience than pre-service teachers but may still be getting used to their context and working out relationships with administration and other, more senior teachers, which may somehow restrict their attempts to use technologies in their teaching contexts.

Experienced in-service teachers who graduated more than ten to twenty years ago would be expected to face a different range of problems. While they may be familiar with technologies for personal uses, they are less likely to have experience with using technology as a student as a part of their own studies. This does not necessarily mean that they are less competent users of technology, but that they have not been in the position of student, at least as far as their own formal studies are concerned. In contrast, however, they would be expected to be experienced teachers who know how to deliver content effectively as well as to design a range of tasks and activities, but they may feel some apprehension with regards to delivering these through technology.

Teacher training can occur at a range of different points throughout a teacher's career, but it is apparent that it is difficult for teachers to acquire the skills that they need to use technology in their teaching and learning context without some kind of support, such as formal

training or assistance from a mentor (Stockwell, 2010). Experience with technology has been a notoriously bad predictor of technology adoption, as has been suggested by Tour (2015), who points out that teachers who have experience with technologies are likely to apply the same technologies to their own teaching and learning contexts as well, and Kessler (2007), who proposes that teachers with moderate experience with technology are more likely to integrate it into their language courses than those with higher or lower amounts of experience. Therefore, it is apparent that it is not technical skills in themselves that play a role in facilitating broad or integrated technology usage but, rather, other factors that are more related to the support that teachers can receive as they make decisions about what and how technology is used in their learning environments.

The content of courses on technology for both pre-service and in-service teachers has been an area of increasing interest over the past several years, and there have been several books (e.g., Hubbard & Levy, 2006; Son, 2018; Son & Windeatt, 2017) and journal articles (e.g., El Shaban & Egbert, 2018; Son et al., 2017; Stockwell & Reinders, 2019) that allude to the importance of this topic. One consistent outcome that has been that teachers typically have difficulties in integrating technology into their teaching and learning without some degree of support (Robb, 2006; Stockwell, 2010), and being able to work with others who have experience in using technology in the early stages of experimenting makes it possible to remove much of the unnecessary trial and error and “reinventing of the wheel,” where teachers rediscover problems that have already arisen in previous studies due to failing to explore what has already been published in the literature. One thing that also becomes apparent is the fact that while teachers need to have skills in using certain technologies that they use in their teaching, it is not feasible for pre-service and in-service courses to provide information about all available technologies that teachers may wish to use. Instead, courses need to lead teachers to understanding the possibilities with technologies so that they can make decisions about what to use from the ever-increasing range of technologies that are available to them.

While the three categories presented here are intended as more indicative than definitive, they serve to illustrate the potential relationship between teaching experience and training with technology to use in teaching. Teachers who are currently in education or have recently completed education are more likely to have had some kind of hands-on experience using technology as a part of their own teaching and learning or possibly even some direct instruction as a part of their teacher training course. These experiences may form the foundation

for what they do in their own classrooms in the future, but there is the limitation that any negative experiences may translate into reduced enthusiasm for using technology in their own language teaching contexts. Regardless of experience with technology or with teaching, there are teachers who embrace technology in their teaching environments and are eager to try new tools in order to enhance their teaching and learning contexts, but there are also those who feel far more reluctant to do so, as discussed in the following section.

From the Field: Resistance to Technology

I have seen many experienced teachers with long teaching careers show reluctance to adopt new technologies into their teaching contexts. When talking with these teachers, the reasons are largely divided into three main overlapping categories. The first reason is that they don't believe that technology can add anything more to their current teaching environment than they already have. This belief generally stems from either a lack of knowledge of what technological tools and resources are available or a resistance to try to use something that they are unfamiliar with in their teaching. The second reason is that they have a fear that their learners are more capable of using technologies than they are, and as a result, they feel that they will not be able to explain how to use them or respond to questions should they need to provide technical support for the learners. The third reason is related to a lack of motivation to change current teaching practice because there is no particular external motivation to do so. That is to say, they are aware of the fact that developing and using new teaching materials will take considerable time and effort, but as long as the institution does not place pressure on them to change, they are happy to devote their available time to other duties required of them as a part of their job. Each of these are understandable perspectives, but suitable counterarguments could also be made in each case. Learning how to use new tools without training is an extremely difficult undertaking, but there are academic communities that can be sought out to assist with this effort. Similar, supporting learners with new technologies that teachers are unfamiliar with can also be daunting, but learners will typically not know how to effectively use technology for learning purposes unless they have received sufficient guidance from teachers. Finally, teachers are indeed busy, and deciding whether or not to incorporate technology is a matter of priority. Although teachers may see the value of using technology in their teaching and learning environments, they might feel that it is a low priority compared with other things that they need to do within their busy work schedule.

3.4 Resistance to Technology Use in Education

The factors influencing teachers' reluctance to using technology in their teaching environments are varied but have been linked most commonly to a lack of conviction that technology can contribute to their teaching in any meaningful method and the feeling that there is insufficient support for them to be trained in using technologies (Ertmer, Gopalakrishnan & Ross, 2001; Fang, 1996; Liu, 2011; Pelgrum, 2001; Stockwell, 2009). While resistance certainly seems to persist in specific contexts based on teachers' confidence and even the culture of the environment (Ertner & Ottenbreit-Leftwich, 2010; Wang, 2020), there is also evidence to suggest that there has been a shift in attitudes towards keeping up with changes in access to technology inside and outside of the classroom, changes in student attitudes towards technology, and changes in curriculum where technology is taking on a more central role (Ertmer et al., 2012). Maintaining a healthy distrust of technology in education, however, is likely to lead to more realistic and successful learning outcomes than will believing that technology will necessarily benefit learning (Selwyn, 2014). To this end, provided that resistance does not lead to unwarranted rejection of technology when it *could* be a viable option, taking time to consider the pros and cons of technology in teachers' specific environments would be desirable.

Resistance to mobile devices is often due to overlapping but quite different reasons from other technologies. Institutionally, management's attitudes towards the use of mobile technologies – particularly mobile phones – will have an impact on technology use. Many pre-tertiary schools (i.e., elementary, junior, and/or senior high schools) have a negative attitude towards the use of mobile phones in schools and ban their use in school, often based on fears that learners will use them inappropriately during school time. In these contexts, even if teachers desire to use the technology, it is extremely difficult to do so because they need both institutional and parental permission, which is not always easy to obtain. Studies have reported that parents have often prevented their children from studying on mobile devices because they believe that they are playing rather than studying, and this is extremely difficult to monitor (see Huang, Chen & Chou, 2016). This view is often reflected in teachers' view of learners using their mobile devices in class, as they are concerned that mobile devices will be an extra distraction in the classroom, resulting in teachers restricting or even banning their usage.

At a class level, many teachers fear that if learners have access to mobile devices like mobile phones, they will not engage in required

activities but, rather, use their devices for private purposes such as social networking or Internet surfing (Grimshaw, Cardoso & Collins, 2017). It is true that the device itself can be a distraction for “cyber-slacking” (Flanigan & Kiewra, 2018), but the classroom was not void of distractions before the event of mobile devices. It was commonplace for students to have their eyes fixed outside the window, on each other, passing messages when the teacher wasn’t looking, or simply daydreaming. Although mobile devices do allow learners to “look out the window” through the screen of their device or pass messages electronically, it is questionable whether or not it is entirely the fault of the device that learners engage in this type of behaviour. One would assume that failing to concentrate is a product of boredom and is facilitated by the technology rather than behaviour that is entirely prompted by the technology. It would be difficult to make strong conclusions one way or the other about the bilateral nature of this relationship, but at the very least, it may prove beneficial to take measures to change teaching practice rather than blanket banning a potentially useful tool for what teachers fear might happen.

While this might seem to be a rather severe perspective, as has been pointed out before, the nature of teaching and learning is changing, and teaching learners with methods that do not suit the ways in which they are accustomed to or expecting might lead to this type of boredom. In other words, if learners do not see sufficient relationship between what is happening in the classroom and the attainment of their goals (which may or may not be clear to even the learners themselves), then they are less likely to direct their attention to the task at hand. This could be a product of learners who feel that they can search for the information that they require by themselves without needing to engage class activities, but this confidence often means that they confuse the difference between “googling” for straightforward information rather than conducting research at a deeper level (Weibe, 2016). At the same time, many learners feel that they are quite capable of multitasking and can look at their phones or other mobile devices while still taking in the content of the class. There is research to suggest that when learners spent time looking at their devices while attending lectures, they were able to retain far less of the content than classmates that did not (Ophir, Nass & Wagner, 2009; see Chapter 6). Being made aware of the difference in perspective compared with the reality of using mobile devices is something that learners themselves may benefit from and may have some impact on effecting a change in learner behaviour.

Regardless of the existence of or the reasons behind learners’ undesirable use of mobile devices in learning settings, what is certain

is that the role of the teacher is changing and, indeed, must accommodate different tools and different expectations with these tools. In a world where information is readily available to learners at the touch of a keyboard or screen, the role of the teacher as predominantly a provider of information needs to undergo a shift into more of a curator of information (Potter, 2012). Curators will act in much the same way in educational terms as they might as curators of an art gallery or museum. As most people are aware, art galleries and museums have a far greater collection of items than the ones that are on display, and it is up to the curator to make decisions about what items to show and how to display them in such a way that they are easy for visitors to understand. In education – particularly with regards to the Internet, where the sheer volume of resources can be overwhelming to learners – the teacher's job includes putting together relevant resources in a way that they are comprehensible to learners and, as a result, far easier for learners to make sense of them, particularly if they are annotated with supporting information. Curation can take on yet another meaning with mobile devices. Downloading enormous chunks of information to a mobile device can be overwhelming to sift through, given the technological affordances (see Chapter 6 for a discussion of screen size, speed, storage capacity, etc.), meaning that information that is made to be available on mobile devices should be curated in a way that it is appropriate to the devices that will be used to access it (Udell, 2015). Curation is far from an easy task, as it entails the teacher understanding the range of resources that are available and the devices owned by learners, as well as sufficient preparation time. Nonetheless, if learners have access to a curated list of resources suitable for their mobile devices, it is likely they will be able to have a more enriching experience than they would if trying to seek out and sift through resources on their own.

The changes that we've seen over the past several decades have had an enormous impact on the ways in which we teach and learn not only language but also other aspects of education. Even if future changes in technologies are relatively minor in the foreseeable future, it is likely that the changes that we have seen to date will continue to transform the way we teach, impart, store and retrieve information; interact; and even form relationships between the various actors involved in the teaching and learning process.

An issue that deserves attention is the development of specific digital literacies that both teachers and learners need to develop in order to teach effectively using technology (e.g., Lawrence, 2014). The shift in literacies that are required by teachers in an era where technology is becoming central to much of what happens in and out of the

classroom is one that has appeared in the literature with increased frequency over the past several years. The term “digital literacy” has come to include a range of individual skills and related literacies such as *information literacy* (the ability to seek and organize information) (Weibe, 2016), *multimodal literacy* (the ability to deal with information that comes through multiple modes) (Lotherington & Jenson, 2011), and *media literacy* (the ability to evaluate the credibility of sources and to create new messages based on the information taken from these sources) (Buckingham, 2007). The problem of how teachers can be trained in these literacies, however, remains an extremely important point.

3.5 Mobile Learning in Formal and Informal Learning Contexts

The context in which mobile learning takes place will have an impact on various aspects of how learning takes place. While most research into mobile learning relates to formal learning contexts, the widespread availability of mobile apps for informally learning a language makes it an area greatly in need of further study. In reality, it is likely that learning in informal contexts is more commonplace in reality than in formal learning, although the nature of mobile learning in this manner makes it extremely difficult to research in a systematic manner, as will be discussed in more depth in Chapter 4.

Though it has been argued that the formal and informal are along a continuum rather than distinct categories (Hubbard, 2020), there are still several points that separate learning in informal contexts from learning in formal ones. Not the least of these is that the purposes for undertaking the study in the first place, which will typically be more intrinsic in informal learning contexts than in formal ones. That is to say, when mobile devices are used to learn for personal, intrinsic reasons, learners are likely to study in a less systematic way than might be done in formal contexts (Marsick & Watkins, 2001). This will have an impact on the selection of the tools and the ways that they are used. For example, learners in informal settings may select apps based on the order that they appear in online app repositories such as the App Store or Google Play and make evaluations about these apps from early impressions of usability. If they find a tool to be difficult to use in the first few instances of trying it out, it is possible that they will quickly abandon usage and try another app to see if it appears to fulfil their needs. On the one hand, this is not necessarily a bad way to select learning tools in one sense, in that learners can try a range of tools to

see what is suited to them. But on the other hand, there is also the danger that potentially useful tools may be abandoned before learning how to use them properly, and opportunities for language learning may be lost.

Training in how to use apps and other mobile-based tools can go a long way towards ensuring that learners can get the most out of their learning endeavours, which can help them to maintain higher levels of motivation. Related to this is the fact that informal learning contexts often lack clear goals, making it difficult to maintain motivation in the long term, which can have a negative impact on sustainability of task engagement. Formalised learning environments can make it easier for learners to have clearer short-term goals that they can work towards, rather than larger, more nebulous goals that might be associated with informal learning. For example, when asking learners about their overarching language learning goals, they often respond with “communicating with friends,” “watching movies without subtitles,” or even “chatting online.” These goals are certainly reasonable goals to have, but most learners are unable to pinpoint what is that they specifically need to do in order to achieve them, which can result in different types of behaviours with using mobile technologies for learning than might be the case in formal ones. It is for this reason that there has been increased interest in the balance between formal and informal learning, where both can contribute to ultimate success in language learning if applied appropriately (Godwin-Jones, 2018; Hubbard, 2020; Kukulska-Hulme, 2015; Lai, 2019). While there may be similar underlying long-term goals in each type of learning, teachers can play a role in directing learners towards clearer short-term goals (e.g., studying for a vocabulary test, undertaking specific listening activities, or completing grammar exercises) that will ultimately contribute to achieving long-term personal goals. These may be done in part through mobile devices or through a combination of online and offline methods, but even just having more tangible goals may lead learners to engage in activities in a different way, even if they supplement their learning with selecting apps of their own volition.

Another issue that may impact learners’ ability to engage in self-directed language learning is their proficiency level, particularly with regards to using mobile devices as a primary tool. There are apps and other mobile-based activities that are better suited towards lower levels, and the online stores seem to be filled with apps for beginning-level learners but offer very few targeted towards learners of a higher proficiency, a fact which has been bemoaned by many intermediate- or advanced-level learners who have difficulty finding

appropriate apps to assist them with their language learning. Lower-level learners may find the apps and online learning resources that are available to them will satisfy their immediate learning needs but then run into difficulties when they reach higher levels. Also, the instructional design that is embedded in learning resources designed for lower-level learners will often be of less importance than for learners of a higher proficiency. That is to say, the role of the teacher in training learners how to use the resources more effectively is likely to have a greater impact as learners reach higher levels of proficiency, as they would be expected to find that simple presentation of vocabulary lists or grammar manipulation exercises will be less useful when they are trying to develop communicative competence in the target language. Of course, the role that mobile devices can play in achieving this is also an extremely important one and will be dealt with in more depth in Chapters 6 and 8.

Finally, motivation to learn and previous language learning experience can be related to the issue of *autonomy* in language learning, an area which has received a great deal of attention over the past several years (e.g., Benson, 2013; Benson & Voller, 2013). As learners engage in language learning activities, possessing a balance of motivation to continually sustain their use of activities along with the skills needed to make the most out of their use will have a direct impact on their effectiveness in leading to language learning gains. Those learners who have had experience in using apps or other online learning resources to learn a language may find that they can apply these skills to learning a new language. However, this will depend on the availability of apps for different languages, and, of course, there is great disparity in this regard, with an abundance of resources for some languages, such as English and Spanish, and considerably less for languages such as Croatian and Thai. In this case, even if learners have experience in using online resources or apps, when such resources do not exist for the target language, those skills are no longer applicable. In this case, even motivated learners may find it difficult to maintain their motivation or to study autonomously if they have difficulty in finding resources that meet their language learning needs.

What criteria do learners typically apply to mobile resource selection such as language learning apps? There has been a small amount of research carried out to date that looks at how learners make decisions about which apps to use for learning purposes when they are left to make these decisions on their own. Some researchers have noted that learners will generally choose price as the first criteria, opting to use free options before considering paid options (Kim, 2013; Stockwell, 2008). The free option may be a free trial version of a fully paid

version, but the free version usually only includes a sample of content, limited functionality, or both, and it is often based on this where learners decide whether or not they will purchase the full version, but the position of the mobile device in the larger learning context will have an impact on whether learners decide to go ahead with purchasing it or other alternatives. It would not be an exaggeration to say that there are many learners of a language who have multiple free versions of apps installed on their mobile devices, the majority of which are rarely used.

Another problem is that in many cases, learners will typically choose apps that target the more “obvious” aspects of a language such as vocabulary or, in the case of script languages like Chinese, orthography, with little emphasis on development of other skills such as listening or pronunciation. In many cases, this is because learners are not aware of what they really need in order to develop their skills in the target language, and consequently, vocabulary just tends to be the skill that is the easiest to notice improvement and the easiest to study, where rote learning would be thought to result in at least some degree of language development. This is not a type of behaviour that is peculiar to mobile learning, as many learners will simply create vocabulary lists and study them repeatedly in the hope that this will help develop their entire language proficiency. In one sense, of course, learning vocabulary does have the potential to contribute to the development of other skills (Miralpeix & Muñoz, 2018), but learners are more likely to retain vocabulary items when there are opportunities to see them used in context (e.g., Feng & Webb, 2019), which is rare in mobile-based self-study vocabulary apps.

What criteria, then, should learners apply to selection of apps or other resources for their private language learning? This is a difficult skill to acquire without some support, and this is likely one of the reasons that many learners do not find it easy to sustain unaided learning with mobile devices. One of the greatest limitations for learners is understanding what aspects of the language that they need to focus on to learn the language successfully. As described earlier, when left to their own devices, learners will typically focus on vocabulary, which is likely to be also the area in which they possess the greatest experience. And this is where there appears to be the greatest concentration of apps and resources available, particularly those that would be suited to mobile devices.

As stated before, learning in formal learning environments takes on a rather different form when compared with informal ones. Perhaps the most obvious of these is that in formal learning contexts, learning goals are typically set by teachers, and the burden for finding and

evaluating resources is placed more heavily on the teacher rather than on the learner. This, of course, depends on the teacher's ability to choose from the wide range of available resources. The primary advantage that the teacher would be thought to have over learners is an educated view of what skills the learners need to acquire in order to be successful in their language learning and, as such, may be more likely to suggest a wider range of resources that learners can use in a systematic way in order to have sufficient opportunity to learn different aspects of the language rather than simply focussing on vocabulary.

In addition to this, learning through formal contexts where the use of mobile devices is a central part of the learning tools will usually mean that there is a more solid support framework in place. This support can take several forms, but the most likely areas in which this support would be apparent would be in the selection of the appropriate learning tools suited to the learners' level and needs and in the training to provide the learners with the skills to use these tools in an effective manner. The types of training will be dealt with Chapter 7, but there is already evidence to suggest that providing with learners with ongoing training that not only shows the functions of the technology but also how to use these functions for language learning purposes can lead to a higher degree of task engagement and advances in language skills (Stockwell & Hubbard, 2014). This type of training in formal contexts may have an impact on learners' ability to engage in tasks and activities in informal contexts as well, but this will obviously depend on the type of training the learners are provided with. To that end, formal training is potentially a step leading to the ability to also use mobile resources in informal contexts. Not only that, but perhaps it would be more appropriate to say that a part of training in formal contexts should include an element of training learners to be able to make appropriate decisions about what tools to use and how to use them without support from teachers in more informal learning situations.

A look at the ways in which mobile technologies have been combined with non-mobile ones can give us some insights into the ways in which mobile technologies can complement the range of tools that are available to learners. For example, MOOCs have been optimised for accessibility through not only desktop and laptop computers but also through mobile devices. One of the tenets of offering this type of access is to enable learners to progress easily and seamlessly regardless of where they are (Sharples, 2014). Sharples, Delgado Kloos, Dimitriadis, Garlatti and Specht (2015) argue that the nature of mobile devices makes them conducive to providing the type of seamless learning that

Sharpley describes as, “connecting learning activities distributed over time, space and social settings” (p. 2). While our understanding of learning through MOOCs is still deepening as the potential of how the mobile aspect of learning through technology can relate to learning through other technologies (see Jitpaisarnwattana, Reinders & Darasawang, 2019), Sharpley and his colleagues propose that the mobile part of the learning should be a link between other platforms, which can allow learners to maintain their motivation to engage in learning on an ongoing basis. In this way, they consider that MOOCs may not be something that is conducted entirely through mobile devices, but that mobile devices make it easier to keep learners engaged in the learning and allow them to have access to tools that can keep them connected to the content and to others involved in the learning process such as discussion forums or supporters (Friðriksdóttir, 2018). In this way, it is believed that the use of mobile devices may contribute to lowering attrition rates in MOOCs through keeping a closer relationship between the learner, the content, and the learning community.

Using mobile devices as a part of flipped classrooms has also attracted attention. There is still a general lack of a standard when it comes to describing what flipped classrooms are (Abeysekera & Dawson, 2015), but in essence, flipped classrooms enable learners to engage in activities outside of class that allow for in-class discussion to deepen understanding of points learners have encountered in their own learning. The role of mobile devices in flipped classrooms is varied and can include promoting problem-based learning, individual and group collaborative project-based learning, peer assessment, and peer competition (Hwang, Lai & Wang, 2015). These activities are anticipated to help learners to stay engaged even when they are not directly supervised by the teacher and, thus, to extend the amount of time that learners spend on task and perhaps even encourage learners to engage more in self-directed preview learning (Wang, 2016). While researchers have suggested that flipped learning may be a means of helping learners to achieve autonomy (Han, 2015), it should be noted that flipped classrooms, despite being based on a solid concept of encouraging learners to take advantage of time more efficiently and effectively, also require a certain degree of self-directedness to be successful in the first place, and if learners lack this, it is unlikely that they will be able to benefit from this potential (Zainuddin & Perera, 2019). Thus, the use of flipped classrooms may be seen as being a chicken-and-egg situation, where it is uncertain how autonomous learners are at the outset when engaging in this approach and how

their autonomy may be developed through the use of flipped classrooms. What can be said, however, is that mobile devices would be thought to play an important ongoing role, in that they can keep learners' minds on tasks at any time and location, changing the idea from learning as being something that just takes place in the classroom.

Educational gaming is another area where mobile devices are thought to be able to play a role in language learning, and suggestions have been made that games can help motivate learners to engage in activities when other methods lose their appeal (e.g., Liu, Navarrete & Wivagg, 2014). For example, Hwang, Shih, Ma, Shadiev, and Chen (2016) found that having learners complete games on mobile devices helped learners to develop their verbal skills more than when the activities were completed through traditional methods, as they engaged in the activities with greater regularity, giving them more opportunity to reflect on their speech. It is this aspect which is considered as one of the areas of most potential for educational gaming through mobile devices such as mobile phones. If the game has sufficient educational content, encouraging learners to engage in it regularly, due to the "addictive" gaming element, learners are more likely to benefit more from it. A major difficulty that developers face, however, is how to maintain the balance between the educational component and the gaming component. If the focus on the game is more on the educational side, then it might not capture the learners to engage in them sufficiently, and if the game is more focussed on the entertainment side without enough educational content, then even with extensive usage, it is unlikely to bring with it sufficient learning benefit. There have been several examples of mobile-based learning games that have been developed by researchers over the last few years. The context-aware Handheld English Language Learning Organization (HELLO) tool from Liu and Chu (2010) included a gaming element which led to enhanced motivation compared with when the games were not used, and Palomo-Duarte, Berns, Cejas, Doderio, Caballero, and Ruiz-Rube (2016) suggested that the gaming element of their interactive app designed for Spanish learners of German called *GuessIt!* caused learners to engage in tasks more regularly. While both of these examples show the potential for gaming to sustain learner interaction in activities, research into educational gaming is still plagued by the fact that many of the studies tend to be carried out over relatively short periods of time, and the long-term effect on engagement is still largely unknown.

3.6 Summary

While it is impossible to cover here all of the changes that have taken place in teaching and learning as a result of developments in technology over past several decades, the discussion aims to illustrate the complexity of the various elements that are intertwined to form the educational context that many of us find ourselves in today. Learners are being expected to take on more and more responsibility for their own learning, and while some embrace the freedom that this might entail, others feel frustration and apprehension regarding how to go about their learning without sufficient guidance and support. Similarly, teachers are finding themselves in a position where pressure is being placed on them to provide this support without the technical skills and training necessary to deal with problems that may occur with these technologies. Teachers who have received education using technology themselves will be more likely to have less resistance to incorporating technology into their own teaching, but experience with learning through technology as a learner is not an accurate indicator of innovative or appropriate use of technology as a teacher.

Technologies – and mobile devices in particular – have become such an integral part of our everyday lives that we surround ourselves with them at most times. Taking advantage of this constant access to these technologies can help bridge the gap between formal and informal learning contexts, which can ultimately reinforce one another through consolidating both long-term and short-term learning goals. The range of options available to teachers and learners is constantly increasing, which necessitates adjustment to the changing environment regarding the roles of both formal and informal learning in the larger learning context. Many of the changes in education in the foreseeable future are likely predictable to a certain extent as we familiarise ourselves with the availability and functionality of the technologies that we use on a day-to-day basis. What paradigm shifts will be brought by emerging technologies such as virtual and augmented reality, wearable technologies, and artificial intelligence (see Chapter 9), however, remain to be seen. Education is a field that will continue to make the most of technological developments, and the face of teaching and learning will reflect these with constantly evolving roles for educators and students alike.

3.7 Discussion Questions

1. The belief that increased access to learning resources through mobile devices will result in learners being able to study effectively

is widespread. What do you think of this view? Why do you think so?

2. Imagine that you have a colleague who is reluctant to use technology in the classroom, despite there being abundant resources available. Do you think that you could convince your colleague to adopt technology? If so, how? If not, why not?
3. Can you think of a concrete example of where formal and informal learning could be effectively integrated with one another? What do you see as the role of mobile devices in achieving this?

4 *Research Considerations in MALL*

4.1 Introduction

One of the primary purposes of research is to better understand why something happens. This can be to get insights into the ways in which people think, why people behave in a certain way, and the factors that influence this thinking and behaviour. Regardless of the shapes or forms it may take, the ultimate objective of research into teaching and learning is to understand the process better in order to improve upon it. Despite this seemingly straightforward goal, research itself is an extremely complex process that needs to take into consideration the entire range of factors that may affect the outcomes. Egbert (2005, p. 5) describes research as being “studies that take an analytic approach by looking at one or more ... variables [of context, task, tool, language, and people] in any number of ways or studies that look at the system of which these variables are a part, at their interactions and complexities and their effects on each other.” The relationship between instruction and learning has long been considered a complex one (Ellis, 1990), and given the nonlinear nature of learning a language, research is likely to “only provide us with an understanding of individual pieces of the language learning jigsaw, not the whole puzzle” (Nunan, 1992, p. 52). Indeed, understanding how each of the complex parts of learning a language fit together is a tremendously difficult undertaking given the extraordinary range of factors that play a role in the process. One of the key problems of any kind of research that relates to second-language acquisition is defining what SLA actually is and how to measure it (see Norris & Ortega, 2005). That is to say, researchers need to decide what aspect or aspects of the target language to focus on and how these will be measured in an objective and systematic manner. At the same time, the focus of language learning research may not be directly related to the acquisition of language itself at all, and researchers may be identifying the underlying

factors that lead to better teaching and learning practices such as motivation, autonomy, strategy use, cognitive processes, social relationships, and so on.

Needless to say, research involving technology has also rarely been limited only to how it contributes to development of a second language. In a comparative sense, early research did focus heavily on L2 development but also included a relatively vague definition of “motivation,” often with the ultimate goal of seeking to “prove” that learning through technology was more effective than non-use of technology. While comparative studies do still exist – albeit to a lesser degree – studies that look at the acquisition of certain elements of the target language have remained a part of the research that is seen in CALL/MALL research. Naturally, the same problems of measurement of language development exist in studies incorporating technology. Under strictly controlled conditions such as within a classroom situation where the parameters and variables are easier to fix, valid methods of measurement of SLA may be possible to a certain degree if there is sufficient replication (Porte & McManus, 2019; Weideman, 2017), but over the past several years, the use of technologies, particularly mobile technologies, entails a movement into naturalistic settings where the countless extraneous variables become extremely difficult to control for. Of course, this does not mean that there is no place for experimental research in MALL, but it highlights the need to realise that learning is something that does not happen in a vacuum, rather taking place with real students with real everyday experiences and problems that need to be taken into account as a part of the larger context (see Chapter 5 for a discussion of the intricacies of language learning).

This is one of the primary reasons why research into MALL has proven to be somewhat complex when compared with much of the earlier research that was carried out in CALL, although these difficulties are rather different from those that were encountered when ownership of personal computers was still relatively rare. Perhaps one of the most obvious differences is that in the early days of CALL, learners were comparatively unfamiliar with using these technologies, and the teachers were the relative experts. The range of resources was limited, and therefore, learners basically only had access to what teachers provided them with. The past several years has seen an exponential increase in the variety of materials that learners can access on their own devices with ease – such as websites, apps, and other resources – and decisions about what to use as supplementary materials for learning are often made by the learners themselves with little or no consultation with teachers. Another difficulty is associated

with the mobile technologies themselves. By the nature of the device, there is often an expectation that a significant proportion of learning through mobile devices will take place out of class. While this does not preclude using devices in classroom settings (and there have been several very interesting studies that have looked at the use of mobile devices in class), it is exponentially more difficult to determine how learners are using their mobile devices in uncontrolled and unsupervised settings. Teachers can get some indication of the ways in which the learners are using the mobile devices in the classroom, but obviously, classroom uses do not necessarily translate into uses outside of the classroom. Classroom usage is likely to be closer to teacher expectations, particularly if learners know that they are being observed in class. Moreover, when learners encounter difficulties in using technologies in the classroom, then they can quickly put up their hand and ask the teacher or any support staff for assistance. Outside of class, on the other hand, providing support when learners encounter difficulties is not an easy undertaking. It can be difficult to predict the times at which the learners will be using devices, and regardless of how much developers may have prepared for potential problems, there invariably will be problems that were not anticipated. Unless the scale of technology use is large enough, such as when it is used at an institutional level as a major component of many of the courses which are being run (such as in a MOOC or in distance learning), support will be provided only at limited times of the day. If a class teacher is using technology as a part of a course that is not supported by the institution centrally, much of the responsibility for taking care of difficulties that may arise falls almost exclusively on the shoulders of the teachers themselves. This has immediate implications for research, as the degree of available support will have an impact on how learners engage with the technology, particularly when difficulties are encountered. Support might be readily available in the classroom, but if such support is limited and learners do not know how to deal with problems, this will very likely influence both usage and attitudes regarding the technology.

Moreover, as learners are typically using their own devices, it is logistically difficult to track usage. Early studies into MALL typically involved using loaned mobile devices such as PDAs (Trinder, 2005) or iPads (Yang & Xie, 2013), and while this was certainly investigating mobile devices, the learning that took place was rather limited in its mobility. In many of these studies, usage of the devices was closely monitored, and functions were often locked to ensure that learners remained on task. While, of course, this type of approach to research is also necessary, its applicability to natural everyday use of mobile

devices outside the classroom is somewhat limited. In MALL, teachers and researchers are primarily interested in what learners do with their devices outside of the classroom, including how they make use of the small spaces in time that they have available to them (e.g., Kukulska-Hulme & Traxler, 2005). Needless to say, when learners are using their own devices, it is more difficult to get an indication of what they're actually doing with them outside of class and, indeed, in many cases, what they are doing with their devices inside of class too. As described in Chapter 2, this has wider implications for classroom management, but unless some type of tracking is available to researchers, they need to rely on reported data to get an indication of how the learners use their devices when they cannot be observed directly. The limitations associated with the types of data collection methods will be discussed in more detail later in the chapter.

4.2 The Focus of Research in MALL

The focus of research into learning through mobile devices has changed dramatically over the years, but it has always been broad in its focus, both looking at attitudes towards devices and examining preferences in the ways in which they can be used. Research into mobile learning has gone through a series of phases, starting with the earliest research in the late 1990s that sought to see whether or not it was possible to use mobile devices as a part of language teaching and learning. As is the case with a number of new technologies as they appear (see Chapter 2), much of the early research in mobile learning focussed on the affordances of the technology in terms of what can be achieved through mobile devices, often as a comparison with other technologies such as the computer, with activities carried out using non-technological means, or both. Comparative research has long been a focus of much of the research into technology usage, where computers were compared with textbooks or other so-called traditional teaching and learning techniques. As Burston (2003) conjectures, one of the primary reasons for this type of research was to prove that the amount of time, effort, and cost that went into purchasing expensive technologies were worthwhile, which he labelled the "burden of proof" (p. 219). These types of comparisons have persisted to a certain degree, despite the obvious difficulties in replicating paper-based materials with technology and, more importantly, the problem that simply replicating paper-based materials usually fails to take advantage of the affordances of the technology. These kinds of contrasts plagued research into CALL for many years, as teachers attempted to justify the use of technology but were gradually replaced

with studies that explored appropriate pedagogies for using technology in teaching and learning environments.

It is both surprising and anticipated that these comparisons have resurfaced with the advent of mobile-assisted language learning – it is surprising that it is still believed that comparisons are possible, and at the same time, it is anticipated that teachers, researchers, and administrators feel the need to confirm how new technologies fare compared to existing benchmarks. In saying this, some comparative studies do give insights into learner behaviour, depending on how they are carried out. An example is Huang and Lin's (2011) study of mobile reading, in which they examined learner attitudes towards reading on computer, paper, and mobile phones. They found that learners demonstrated a preference for paper over both computers and mobile phones, and that phones were particularly inconvenient for reading as a result of the small screen and the need to scroll through the pages. It should be noted, however, that the study was held before smartphones reached the mainstream, meaning that not only were the screens smaller than standard phones of today, but it was also necessary to press and hold down keys in order to scroll through pages. Other attempts to compare mobile devices with alternatives have also appeared in the literature, such as Lin's (2014) comparative study of Taiwanese learners of English as a foreign language using desktop computers and tablets for extensive reading. Lin's study showed that learners engaged more actively in extensive reading activities on the tablets compared to desktop computers, a result which at first glance seems to contradict the results of earlier studies. The major defining difference with earlier studies is that the learners in Lin's study used tablets rather than mobile phones or PDAs, and the larger screen of the tablets may have been easier for learners to engage in the reading activities compared to the smaller screens of mobile phones and PDAs. Coupled with the results of other studies, this type of research is significant in that it shows that mobile learning is not homogenous, and the choice of device will affect how it is used.

The combined results of the two studies described here provide some insights into students' perceptions of using mobile devices for a specific function – in this case, reading – but they also shed some light on the hidden complexities behind comparative studies. It is evident that learners prefer larger screens when reading in the target language. In one sense, this is already an anticipated outcome, but if the purpose of engaging in mobile learning is to take advantage of the small spaces in time that appear during the day, many of which are unplanned, it would seem that expecting learners to have access to tablets with the same frequency as their access to mobile phones would be overly

optimistic. Therefore, one might argue that the primary argument is not whether learners prefer tablets over mobile phones for engaging in reading but, rather, how to encourage learners to make use of the devices that they carry with them to both learning and non-learning locations to enable maximum opportunities before learning. In other words, knowing whether learners will choose portability over non-portability provides foundational information that suggests that learners are, in principle, prepared to use their mobile devices for learning, but the more difficult question to answer is not what but how.

Aside from comparisons, another area of MALL where a large amount of research continues to be conducted pertains to attitudes towards mobile learning. This type of research has been central since mobile learning first began to appear in the early 2000s, but twenty years later, attitudes of learners and teachers to mobile learning remains a popular theme. It might be anticipated that fundamental motivation behind conducting this type of study is determining whether or not learners and teachers are psychologically, emotionally, and pedagogically equipped to engage in using mobile devices as a mainstream part of their language learning and teaching, respectively. Of course, this is not to say that these types of attitudinal studies do not have a place in research. Understanding teachers' and learners' willingness (or lack thereof) and the reasons behind these attitudes can help researchers to seek ways of dealing with the problems that may eventually lead to more active engagement.

It is not surprising that learner attitudes towards mobile learning have been of interest to researchers. The almost universal ownership of mobile devices by learners makes them an attractive tool for teaching and learning, as they can be used in environments even where there is minimal institutional infrastructural support for technology. Intuitively, many teachers hold the view that learners are typically enthusiastic about using their mobile devices for language learning, and attitudinal surveys have shown that learners frequently hold a positive disposition towards mobile learning, although often in the absence of reports of actual usage (e.g., Dashti & Aldashti, 2015; Wong, Wang, Ng & Kwan, 2015). Studies that include data examining learner engagement in learning using mobile devices, however, have shown mixed responses with limited out-of-class usage being reported in formal learning situations (Kim, Rueckert, Kim & Seo, 2013) unless sufficient training was provided about how to use the apps and resources provided for them (Stockwell, 2019) or when there was mobile-based social support from teachers and other students (Tran, 2018). In self-directed use of mobile devices as a support for

formal education, learners see their devices as a way of personalising their learning and, to a lesser degree, accessing authentic resources and connecting with others (Lai & Zheng, 2018). What is apparent is that attitudes to learning through mobile devices can also be extremely complex and are dependent on a range of factors that go beyond familiarity and accessibility of mobile devices. Learners' own perceptions of the usability of the device and their confidence in their own abilities with the device, along with the provision of support and training and the social and cultural context in which the device is used will all have an impact on the way in which it is viewed.

It could be assumed that one of the major reasons why researchers have focussed on learner attitudes towards mobile learning is that they are interested in identifying the factors affecting task engagement or, perhaps more importantly, the reason for the lack of engagement which has plagued many of the studies into mobile learning. While these types of attitude studies can provide some insights into the background causes, identifying why learners have tended to be less enthusiastic about engaging in tasks and activities on mobile devices than anticipated by teachers requires a more global investigation of the context in which they are used. One example of this is provided by Stockwell and Liu (2015), who identified quite different patterns of engagement with online vocabulary and listening activities that learners could choose to carry out on their mobile devices or computers. Participants in the study were based in Japan and in Taiwan, and the course design that was adopted in both countries was controlled to be as similar to one another as possible. They found that the learners in Japan were far more likely to engage in activities on their mobile phones when compared with the learners in Taiwan, and the Japanese learners engaged in the activities relatively consistently across the semester, whereas the Taiwanese learners completed the activities predominantly at a rush at the end of the semester.

There appear to be two main reasons for these differences. The first of these relates to the *tempo-spatial* context raised by Lai and Zheng (2018). In Japan, the vast majority of students commuted to university using public transportation, mostly trains, but in Taiwan – at least in the more rural city, where the study was conducted – almost all of the students commuted to university using motorbikes. This difference meant that the learners in Taiwan did not have access to a potential learning opportunity that was available to the Japanese learners. The results of surveys, server records of login times, and the information provided by learners when they log in to the system, confirmed that most learners in Japan used this commuting time as one of the major opportunities for engaging in the online activities. The Taiwanese

learners, on the other hand, were far more likely to engage in the activities in their homes using their computers. The decision to use their computers rather than their mobile phones to engage in the activities is likely a result of the fact that they just didn't have an appropriate place where they could use their mobile phones to engage in the activities.

The second factor that is thought to have played a large role in learner attitudes towards the mobile-based activities is the *teacher*. The way in which the teacher frames learning activities to technology-based activities will invariably have an impact on learners' attitudes and readiness to engage in them (Lamb, 2017; Stockwell, 2013). Learners in Japan were recommended to complete the activities as they would assist them in weekly quizzes, while the learners in Taiwan were given a grade for the number of activities they completed before the end of the semester. This difference in the ways in which the teacher framed the activities brought about rather different views on the part of the learners, where the emphasis for the students in Japan was the relationship between the activities themselves and the weekly vocabulary quizzes, while for the students in Taiwan the focus was the completion of activities to receive a grade. This likely resulted in the usage patterns exhibited in the study, where the Japanese learners engaged in the activities relatively consistently, and access logs showed that there was a tendency for many of the learners to use their mobile phones for the activities shortly before the quizzes. Because the ultimate goal for the Taiwanese learners was the completion of activities before the semester, the learners likely rushed to complete them immediately before the deadline. This tendency also could have had a direct influence on a selection of the platform; because the learners engaged in the activities for extended periods of time in blocks, it was more efficient for them to use their computers than their mobile phones.

The example provided here helps to illustrate the potential weaknesses of simply asking learners about their attitudes towards mobile learning. Actual engagement appears to be far more likely a result of contextual factors than simply a positive predisposition towards using their mobile devices. The differences in the engagement between Japanese learners and the Taiwanese learners described in this study may have been less a result of attitudes towards mobile devices as learning tools than a combination of a lack of suitable times and places to complete the activities on mobile devices and the ways in which the activities were framed by the teacher. Studies looking at learner attitudes have their place, but it is essential to bear in mind that positive learner attitudes are not necessarily a predictor of engagement in mobile learning, and that attitudes are prone to change depending

on contextual factors. Understanding this greater context can mean that the meaning behind learner attitudes can be explored more comprehensively in order to facilitate more consistent engagement.

In addition to learner attitudes, teacher attitudes towards mobile learning have also featured quite centrally in research over the past several years. Studies looking at teacher attitudes to technology both in language education and in education in general featured in the literature as access to computers and mobile devices became more widespread, and these have continued to appear relatively consistently (e.g., Canals & Al-Rawashdeh, 2019; Levy, 1997; Liu, Lin & Zhang, 2017; Pelgrum, 2001; Weibe & Kabata, 2010). The main focus of such studies has been to identify the barriers to the integration of technology, and the majority paint a similar picture of the problems faced by teachers: a lack of training, a lack of support, and a lack of appropriate resources. The issue of teacher training has also been a recurring theme in research since the mid-2000s, as the need for appropriate training in technology has come to be more widely recognised (Hubbard & Levy, 2006; Kessler, 2007; Kessler & Hubbard, 2017). This will be dealt with in more depth in Chapter 8.

Although attitudinal studies regarding learning through mobile devices have in some ways mirrored those of broader uses of technology in language education in terms of the problems faced by teachers, there have been some obvious differences as well. Not the least of these differences is that studies looking at mobile learning also deal with the potential for distractions of mobile devices in the classroom (Grimshaw, Cardoso & Collins, 2017). Indeed, it could be argued that one of the reasons for the mixed views towards mobile learning has been the fear that the device will be a distraction that could have a negative impact on what happens both inside and outside of class. A large proportion of teachers agree that mobile technologies can lead to better learning but, at the same time, are concerned that learners will be distracted from the class content if they use technology in class (Lisenbee, 2016). The distractions of using technology in class (see Chapter 3) are to a certain degree valid ones, but holding overly negative views of technology usage in the classroom without considering the ways of dealing with them can lead to failing to take advantage of a potentially powerful in-class learning tool.

Another aspect that relates to teacher attitudes towards technology is how they evaluate their own abilities with regards to their learners and vice versa. In a study of teacher and learner attitudes in Sweden, Lindberg, Olofsson and Fransson (2016) confirmed that many teachers believe that, because learners can use technology at home, that they are also capable of using it at school. What was particularly

interesting, however, was that learners felt a gap between the abilities of different teachers and that this may have an impact on how technology is used for education purposes. Moreover, learners commonly did indeed use their smartphones for non-educational purposes but felt that smartphones were not capitalised upon as much as they could be for learning purposes. This result suggests that there may be gaps between the perceptions of both the teachers and the learners, leading to dissatisfaction on both sides. For this reason, research that looks at attitudes of teachers towards learners – and learners towards teachers – is an essential part of laying the foundation for solid pedagogy. However, there is a need to go beyond the attitudes towards mobile learning to explore the appropriate pedagogies utilising mobile devices to establish best practice.

There have been several meta-analyses and syntheses of MALL research carried out over the past few years which take stock of what has been achieved thus far. The focus of these studies has been very broad. For example, Hwang and Wu (2014) investigated the impact of mobile devices on learner motivation and achievement in reported research between 2008 and 2012 and suggested that there has been a small impact on both. Duman, Orhon and Gedik (2015) identified research trends in MALL research and named vocabulary as overwhelmingly the most researched language area from 2000 through to 2012, with nearly four times the amount of research than that carried out on listening. Burston's (2012) comprehensive sixty-nine-page annotated bibliography of implementation studies MALL in practice remains a benchmark in determining the nature of research conducted with mobile devices between 1994 and 2012. Certain language skills and areas have been the topic of meta-analysis as well. Lin and Lin (2019), for instance, investigated the effectiveness of MALL on English vocabulary learning through a meta-analysis on thirty-three studies from 2005 through to 2018, concluding that mobile learning could have a positive impact. However, some studies have been quite critical of MALL research, such as Burston (2015), arguing that just nineteen out of 291 research studies were designed solidly enough to determine any potential learning outcomes from language learning with mobile devices. The broad range of research carried out through synthesis of previous research allows us to see the depth and breadth of the research being conducted with mobile devices in language teaching and learning, along with the potential shortcomings.

Other research has focussed more specifically on discreet elements of MALL, such as language skills and areas. For example, Çakmak and Erçetin (2018) investigated the effect of gloss type on text recall and incidental learning of vocabulary on mobile phones, and Chen,

Carger, and Smith (2017) examined the impact of mobile devices on learners' narrative writing. There has also been research which has attempted to explore how the affordances of mobile devices may be used in language teaching and learning, such as the GPS function for location-based educational games (Holden & Sykes, 2011) or augmented reality (Richardson, 2016). Finally, research that seeks to draw relationships between mobile learning and brain function – such as the work of Hong et al. (2017), who identified areas of the brain that activate while using mobile-based games through MRI scans.

The examples provided here are far from exhaustive and represent just a small part of the complex studies that have been carried out into MALL over the past twenty years or so. The scale has varied from small studies exploring a single language skill through to large-scale syntheses of research carried out over tens or even hundreds of studies. As with most research into second-language teaching and learning, MALL research depends very heavily on sound research design, and knowing how research has been conducted can assist potential researchers to understand the options that are available to them in conducting research and how to avoid some of the potential pitfalls.

From the Field: Choosing a Focus in MALL Research

Being in a position where I supervise MA and PhD students, I see many students who are interested in MALL but struggle to find a focus. Invariably, the starting point for most tends to be learner attitudes towards mobile learning, where they want to see how learners feel about using mobile devices for language learning. Attitudes towards learning are certainly important, but they are highly dependent on underlying factors such as skills and experience with technology and the quantity and quality of training that they have received. As a result, I try to encourage my students to go beyond the superficial side of attitudes and to look at these factors that contribute to the formation of attitudes towards mobile learning. At the same time, when students discuss "MALL," they are rarely able to articulate what is encompassed in the term beyond apps, and they frequently view learning through mobile devices as exclusively an informal, out-of-class undertaking. Choosing a research topic is very dependent upon having a clear understanding of what MALL is, how it is conducted, and knowing who the learners are. If the ultimate aim of MALL research is to enhance learning with the support of mobile devices, then research needs to be carried out that explores the various barriers faced by teachers and learners – many of which are now widely documented – to determine how they can be overcome to lead to better practice.

4.3 Data Collection Methods in MALL Research

This section provides an overview of some of the data collection methods that have been used in MALL research. It is largely descriptive at this point, and the limitations, while touched on here, will be dealt with in more detail in the following section. The discussion will look at surveys, interviews, learning journals, and various forms of tracking (server logs, app records, and so on). The features of the various studies – including the contexts in which they were conducted, the focus of what the studies sought to find out, and the implications for our understanding of MALL through these collective methods – will be discussed.

As has been alluded to earlier, MALL research has been limited to a large degree by the means through which studies are conducted. When mobile devices were less commonly owned by learners themselves, devices were often provided in class sets and were frequently under constant supervision of teachers to prevent damage and/or loss (see Stockwell, 2016, for a discussion). This in itself meant that the research was largely limited to supervised usage of devices where teachers could deal with difficulties as they arose, but it also meant that usage was not what could be termed as natural in some sense. It should be pointed out that this type of research is not without meaning, but it does mean that it is not necessarily representative of how learners will engage with devices in naturalistic settings. This is an issue of great importance, because it has an impact not only on task design, but also on how and when support should be provided (e.g., Lai, 2016). As a result, this section will discuss the potential limitations of research into the use of mobile devices for language learning, along with some suggestions for how to deal with these difficulties. The section will discuss issues with installing tracking software on mobile devices that are owned by the learners themselves and describe possible ways in which tracking can be carried out, such as logs maintained on the server (Stockwell, 2008). It will also describe the problems with relying on reported data, look at research into the use of surveys (Hsu, 2013) and learning journals (Lin, 2014), and discuss ways of overcoming the issue of seeing patterns of usage with mobile devices outside of class. Learner usage data then will need to take the form of data that can be recorded and stored externally or reported data, each of which face potential limitations.

While there are several different ways of describing research methods, the categories outlined by Nunan (1992) make a very accessible starting point for ease of understanding the different types of research methods that might be applicable to MALL. The descriptions

that follow, then, are based loosely on these categories but have been adapted and reordered to suit the ways in which they might be used in MALL. There are, of course, other ways of categorising these data collection methods, but this does allow an easy overview to assist in identifying the different options that are available to researchers, along with examples of how they have been used in MALL research.

4.3.1 *Observations*

While the main focus of MALL research has been on the use of technology outside of class, there have been several examples of studies that investigate how mobile technologies are used in class as well. Through such studies, researchers have primarily sought to determine (1) how learners use the available technologies and (2) how they interact with one other and/or the teacher. A difficulty associated with classroom observation with mobile devices, however, is that while it is possible to see what learners do outside of the device, it is extremely difficult to determine what is actually being done with the device itself. This has been a problem with learning outside of class to a degree as well, meaning that introspective methods (see the following subsection) will often need to be used in conjunction with other observation methods to get a clear indication of what learners are doing on their devices. Two of the most commonly used methods of carrying out observation in the classroom are *observation checklists* and *interaction analysis*. Both methods provide insights into what learners do in class, but, of course, each has its own strengths and limitations. Observation checklists mean that classroom behaviour can be quantified, meaning that it is possible to determine if a behaviour is random or systematic, and the circumstances surrounding it can also be examined more reliably. Interaction analysis depends on their being interaction among learners or between the learners and the teacher but will require there to be some means of recording this interaction. This interaction can be audio or video recorded, or if the interaction takes place electronically (such as chat, SNS, or email through a mobile or other device), a digital record will most typically remain on completion of the interactions.

Observation checklists typically involve researchers setting up in advance a list of behaviours that they wish to check for – such as learners referring to their devices, asking the teacher for help, talking with others, and so forth – and tallying a record of each time that these occur. The researcher may also choose to include further information for when these behaviours take place – such as the individual circumstances, the duration, or other information that they feel is relevant to

how, when, and why the learners or teacher engaged in this behaviour. Classroom observation regarding MALL has been unusually rare, but there are a small number of examples that have appeared, such as Stockwell's (2019) observation of learners' in-class discussions of their out-of-class activity with their mobile phones for learning vocabulary and Kassem's observation of how in-service teachers used vocabulary apps in class as a part of their teaching. In both cases, checklists made it possible for the researcher to keep track of predetermined behaviours – in these examples, of both teachers and learners – to determine why and how they engage with mobile technologies in class.

Classroom interaction analysis usually occurs through taking a record of the interactions that take place in class and applying a coding frame to analyse them for certain features. This can take the form of criteria that have been decided from the outset in order to confirm an existing theory, or it can take the form of *data-driven research*, from which a working hypothesis is derived from a preliminary analysis of the data (Riazi, 2016). When data are oral, researchers may choose to transcribe the interactions in order to make the analysis easier to carry out, but with data from text-based communication, the logs of communication will normally be more readily available. With video-based data, analysis may become multimodal, where the use of language is correlated with other observable behaviours. There have been examples of analysis of classroom interaction involving mobile learning. For example, Van Praag and Sanchez (2015) observed in-service teachers to explore how they used mobile technologies as a part of their language classes using video recordings, supported by research notes to point out what they termed as “critical events” that were considered as being directly relevant to their research. Regarding student behaviour in the classroom, Wong, Chin, Tan, and Liu (2010) examined learners' use of an app designed for learning Chinese idioms in class, using video recordings and field notes, coupled with interactions of the online discussions that took place between the students. It should be kept in mind, however, that the problem with observations of classroom interaction is that the participants who are being observed are unlikely to engage in behaviour that is completely natural until they have had time to get used to being monitored. According to what Labov (1972) has termed the observer's paradox, although researchers wish to see natural behaviour, participants will typically not exhibit this behaviour during observation, at least in the short term. To attain data for classroom analysis, researchers need to spend sufficient time in the learning environment for learners to feel comfortable enough to revert back to their natural behaviour (see Section 4.3.3).

4.3.2 *Introspective Methods*

As described in the previous section, understanding what learners do with their mobile technologies can be difficult, with researchers being limited to what they can physically observe or record. While this can be said of in-class behaviours, tracking learner behaviour outside of class is even more difficult, and researchers are usually forced to rely on introspective methods to get further information from the learners themselves about what they do with their mobile devices for learning. Introspective methods have gained popularity over the past several decades as researchers have argued that they provide insights to learners' behaviours that cannot be observed, and often also allows them to infer learners' reasoning behind their behaviours (Taylor & Sobel, 2015). According to Gass and Mackey (2009), introspection makes it possible to "observe internal processes in much the same way as one can observe external real-world events" (p. 1). Two commonly used introspective methods in order to achieve this are *learning diaries* and *retrospection*. Each of these has its own individual merits and, at the same time, also its limitations.

Learning diaries, or learning journals or reflective journals, which have been in use in language teaching and learning for several decades, are "a first-person account of a language learning or teaching experience, documented through regular, candid entries in a personal journal" (Bailey, 1990, p. 215). They are, in essence, records that subjects keep of the activities that they carry out pertaining to their learning or teaching. Ideally, diaries are filled in on an ongoing basis, and as they carry out an activity related to their study, they keep a record of this activity. Learning diaries enable researchers to have insights into what teachers and learners are doing outside of class, and they also enable respondents to include some degree of reflection on their activity. Having respondents reflect on their activities through learning diaries has been associated with positive change in behaviour, in that learners can consider their own learning practices and how to improve upon them (Rampazzo & Aranha, 2019), but learning diaries may also serve to alter learners' behaviour because of the increased salience of their activities, making it difficult to determine "natural" activity outside of class. A specific problem with using learning diaries with mobile learning is that unless participants have access to the diary (be it paper-based or electronic) at all times that they can access mobile-based activities, it is difficult to keep records consistently. In such cases, participants will fill in the diaries later, often recording several instances collectively. Learning diaries have been used in research into

mobile-assisted language learning in both formal and informal learning contexts. Stockwell (2007), for example, used learning journals to investigate when and where learners engaged in mobile-based vocabulary activities as a support for their classes, while Chen (2013) asked learners to keep a record of how they used tablets outside of class. For informal learning, Benson, Chappell and Yates (2018) asked a student who had arrived recently in Australia to keep track of her learning experiences (including those on mobile devices) using an electronic diary.

Retrospection, also known as *stimulated recall*, also has a long history in SLA research, with examples seen in the literature from as early as the 1950s (e.g., Bloom, 1954). It consists of providing subjects with some prompt or reminder of an event and asking them to recall the mental processes that took place at that time. Participants are able to think about certain behaviours that they engaged in during a task or other activity and to verbalise these. According to Bloom (1954), if there are sufficient stimuli to remind the participants about their behaviour, they are able to vividly and accurately recall the events that took place. It is this accuracy, however, which is the downside of stimulated recall (Nisbett & Wilson, 1977, cited in Nunan, 1992), and the gap between the event and the recall task can make the data unreliable. One of the most common tools used to stimulate participants' recollection of events in general education has been video (Gass & Mackey, 2009), but the use of video to observe mobile learning has limited value, particularly if the goal is to keep a record of what individual learners are doing on their devices. Depending on the goals of the research, taking video recordings is possible during class time, but this is not an option to researchers outside of class, unless there is software installed on learners' mobile devices which enables researchers to record learner activity on their devices (see the following point on ethnographic methods). There have been studies using stimulated recall in mobile learning, and these have been carried on both teachers and learners. Van Praag and Sanchez (2015), for example, identified teachers' behaviour and attitudes towards the use of mobile devices in their language learning classes. Classes were video recorded, and teachers were asked to comment on their activity in class and how they responded to the ways in which learners used their mobile devices in the classroom. Records of online video interaction between participants has also been used as a means of retrospection. Lee, Hampel, and Kukulska-Hulme (2019) asked learners of English to reflect on their use of gesture in video-conference-based discussions and to verbalise the reasons behind the different gestures that they used during their communication.

4.3.3 *Ethnographic Methods*

Ethnographic research is typically more longitudinal than other types of observational research (Polio & Friedman, 2017) and generally involves the researcher getting into the everyday lifestyle of the subjects to examine their practices. It is often associated with examining the social aspects of language learning and usage and can take multiple forms such as observation, surveys, questionnaires, and introspective methods. With regards to mobile learning, however, studies using longitudinal ethnographic methods have been relatively sparse. One such study is by Palalas and Hoven (2013), who investigated learners, teachers, and administrators over a seven-month period to understand how mobile devices were used for language learning in the broader environment. They utilised multiple methods of collecting data as a part of their study, including interviews, focus groups, surveys, researcher observations and reflections, email data, and a project site called Wiggi. Another example of such a longitudinal study was by Tay (2016), who explored the ways in iPads were used by learners over a three-year period. The data collection methods used semi-structured interviews, online surveys, classroom observation, and end-of-semester examination scores, the results of which concluded that learners who actively used their iPads as a part of their learning more actively engaged and collaborated, and outperformed their peers who did not.

Given the difficulties in being able to observe natural behaviour with mobile devices for learning outside of class, researchers may also use the virtual footprint left behind by learners from the resources that they access. Learning management systems such as Moodle allow for tracking methods such as access records, which can give researchers access to information about how learners behave in a non-intrusive manner. Stockwell (2010) for example, was able to track the usage of learners over three four-month periods to examine when learners accessed vocabulary and listening activities from either their mobile devices or PCs, finding a discrepancy between post-questionnaire data and actual usage logs. The problem with such a method is, however, that only records of usage within the site are recorded, and if learners download resources to their devices and then use them offline, there is no way for researchers to keep track of this. One form of ethnographic research, the case study, allows researchers to look into the lives of a small number of learners in naturalistic settings, but it has been criticised for its lack of generalisability, which may lead to theory building. Despite this limitation, case studies do have some promising aspects that can make them invaluable in gaining insights into the

multifaceted learning that learners may engage in (see Chapter 9). Ethnographic research into mobile learning remains one of the areas where research is still extremely necessary in order to see how learners use their devices naturally, and then to build pedagogies around these behaviours.

4.3.4 Elicitation Techniques

Nunan (1992) includes four types of data collection methods in his description of elicitation techniques; production tasks, surveys, questionnaires, and interviews. With the exception of production tasks, these elicitation techniques are extremely prevalent in MALL research. While tasks have featured heavily in the literature in recent years with some variation in definition (e.g., Ellis, 2003, 2012; Nunan, 1989), in the context here, Nunan (1992) is referring to tasks to encourage learners to produce naturalistic language to identify specific language forms, either to determine their degree of competence in the form or to examine the ways in which particular forms are used in natural language.

Through technology, it follows that computer-mediated communication (CMC) would be the most logical choice through which to facilitate communication to elicit production task data, and there have been some examples of this over time with mobile devices. One example is from Kiernan and Aizawa (2004), who used the SMS function of their students' phones to complete information-gap activities task face to face, on PC email, and on mobile phone SMS to examine the differences in the language used the learners in each of the three modes. Perhaps unsurprisingly, they found that learners produced the most language in the face-to-face interactions and the least using mobile phones. A more recent example is a study by Li (2018), who requested advanced learners of Chinese to engage with Chinese native speakers through WeChat, an online messenger program used widely in China, and examined the type of language that is produced during the interactions by learners of Chinese language. Examples of both formal and casual language styles were evident in the data, along with wide of use memes and stickers. While this is in some ways closer to ethnographic enquiry (see Section 4.3.4), depending on the tasks that the learners are required to engage in, it is feasible that focus can encourage production of certain language forms in order for the extent to which they have been acquired to be examined.

Surveys, the second elicitation technique, are used to determine the state of things as they are and require/expect no intervention on the part of the researcher in altering the respondents' environment.

Surveys are used quite extensively in the social sciences – such as psychology, sociology, and economics – and the collection of data is typically carried out through a combination of questionnaires and interviews, both of which are described in more detail in what follows. Surveys have also featured in education research, including investigations of how technologies are perceived or used within given contexts. Pelgrum (2001), for example, sought to identify obstacles to technology integration faced by teachers in schools. He carried out a comprehensive study of schools in twenty-six countries, finding that while there were both commonalities and differences across international borders, lack of access to technologies, lack of teacher skills, and difficulties in integrating technologies into instruction were the most significant problems faced by teachers. More specifically, there have also been studies investigating mobile learning in general in higher education. For example, Al-Emran, Elsherif, and Shaalan (2016) examined student and instructor attitudes to mobile learning in Oman and UAE and found that teachers generally held positive views towards using mobile learning and that more than 80 percent of learners were using their mobile devices as a part of their university study. Surveys have also been used widely to examine technology in language education. For instance, a study by Kessler (2007) investigated how graduates of a TESOL program view CALL and learn the skills that they need to integrate technologies in teaching, finding that most gain their knowledge of CALL through informal sources or from personal experience, and that this experience has a greater positive influence on their attitudes towards CALL than formal preparation in their teacher training. With regards to MALL, Hsu (2013) looked at EFL learners in different countries to identify their perceptions towards mobile learning, revealing that general sentiments towards mobile learning were positive, despite some cultural differences. Bradley, Lindström and Hashemi (2017) investigated the ways in which newly arrived Arabic speakers used their mobile devices to help them to integrate into society around Sweden, and Steel and Levy (2013) examined how learners in Australia used their mobile devices in and out of the classroom as a part of their language studies. The key point of each of these studies is that the researcher is investigating the state of things as they already are and has not intervened in the environment as to cause any change attitude or required respondents to use technology in any way that may have an impact on the ways that they view and use it.

Questionnaires can be used to identify three different types of data: *factual* questions aim to find out background information about respondents; *behavioural* questions ask about what respondents have

done or they are doing; and *attitudinal* questions seek to find out what respondents are thinking and investigate their opinions, values, beliefs, and interests (Coombe & Davidson, 2015; Dörnyei, 2003). All three of these types of data collection methods are apparent in MALL research as well. Factual questions may seek information about learners' background or experience with MALL, behavioural questions may try to find out how learners have used or are using MALL and/or MALL activities either generally or as a part of some treatment required by the researcher, and attitudinal questions may investigate how learners feel about using MALL. There would be very few studies that would not include questionnaires as at least one of the data collection methods in MALL studies, and they are often used to find out further information about what has been collected through other methods. For example, Stockwell (2008) found that learners of English expressed a desire to use their mobile phones as a supplement to the listening activities that were used in class, but server records revealed that the vast majority of learners elected to use their computers rather than their mobile phones in order to do the activities. The questionnaires that were administered at the end of the semester could shed some light into the learners' reasons for doing this, revealing that apart from the physical limitations of the device, there were several environmental factors (distractions while studying while in transit, a need to use their computer for other subjects so they continued using the computer for the vocabulary activities as well, etc.) that were possible to determine just from the server log data.

The question formats that may be used in questionnaires will be either closed (i.e., fixed responses such as true or false, selecting a number from a Likert scale, or choosing from a set of options) or open (i.e., where the respondents can answer freely in their own words). Both formats have advantages and disadvantages. Closed questions are easier to answer and to quantify but are typically formulated from anticipated responses by the researcher (Fife-Schaw, 2006), which means that the information that can be yielded from them is limited and insensitive to responses that were not expected. Open questions provide the opportunity for more information, consisting of responses that may be as short as a single word, but longer responses may include sentences or even paragraphs. While questionnaires are an easy way to elicit information from participants, they are not without limitations. Respondents are likely to respond differently depending on whether the questionnaires are anonymous or not, and surveys that allow the respondent to be identified may lead them to provide more positive responses, particularly when there are potential (actual or perceived) repercussions for negative responses.

Another problem with questionnaires is that unless the respondents are particularly engaged in the issues being raised in the questionnaires, it is very easy for questions to be left blank or filled in rather briefly, and response rates to surveys are generally quite low, particularly if they are offered online or in some environment where respondents may opt out of responding. The validity of questionnaires is also a point that researchers need to be aware of, as the intended meaning behind a question may not be clear to respondents, particularly if the survey is administered in a second language, where nuances may not be as clear to the respondents as hoped, or if the survey has been translated into different languages, where slightly different connotations may be attached to questions than intended. Hui and Triandis (1989) talk of what they refer to as “acquiescence bias” where respondents from some cultures may tend to respond more positively than their actual attitudes, possibly with the intention of saving the face of the researcher. While, of course, care needs to be taken in the design of questionnaires, there is also a need to ensure that they are analysed appropriately. This includes ensuring that small samples are not overgeneralised and that the outcomes from surveys are not interpreted too positively or negatively. Finally, rather than taking random snippets of responses from open questions to illustrate a point, it is necessary to ensure that such responses are indeed representative of the sample, which can be confirmed with content analysis to quantify frequencies of responses to see how much comments reflect the perspectives of the group. A response may seem to researchers to occur more frequently than it really does if it is particularly salient to the to them, and only through a systematic analysis of open responses is it possible to ensure that samples are not being used in a way that misrepresents the sentiments of the sample.

In order to dig even further into the learners’ attitudes or perceptions, interviews can make it possible to clarify responses in surveys, which are often rather brief in the responses that can be obtained, even if the surveys themselves are relatively extensive. Through asking the learners directly in either one-on-one or small-group interviews, it is possible to find out more about learners’ perspectives, which may be used to change practice or lead to further research questions.

4.3.5 Experimental Methods

While this may be said of other methods to a certain extent, one of the primary goals of experimental research is to build or confirm a theory,

where data are collected in order to confirm some predefined hypothesis regarding the language learning process.

As described previously, experimental methods are difficult to implement in MALL, given the fact that much of what happens through mobile devices takes place in the small chunks in between various other daily tasks – such as getting on the right train, ordering a caffè latte, or finding a seat in a cafeteria – that are likely to impact on the learner’s frame of mind when the start to engage in the activities. Extraneous factors such listening to announcements to ensure disembarkation from the train at the right station, the arrival of said caffè latte, and bumping into friends while in the cafeteria are all likely to have an impact on the learning process, making generalisations that could be translated into effects of the technology difficult. Experiments are dependent upon the controlling of external factors as much as possible, but this becomes logistically difficult to do when learners are integrating mobile learning into various aspects of their lives.

Given that experimental research is often a form of comparative method study, another problem raised by Ellis (2012) is also relevant, where he describes the problems of comparing “methods” at both global and local levels. As alluded to earlier (see Chapter 1), CALL and MALL were often seen as being homogeneous, as is reflected in expressions such as “CALL/MALL method” that still seems to appear in the literature from time to time. This type of nomenclature has parallels with the “global” comparative method, where one “method” is compared with another, such as Audio-lingualism against (De) Suggestopedia. Apart from the fact that, to my knowledge, there is no such thing as a particular “CALL/MALL method,” it still has problems in that it makes claims about teachers being required to teach in a particular prescribed way, following the principles and techniques that are laid out for them. Quite obviously, there is no prescribed way of integrating technologies – mobile or otherwise – into a learning environment, and comparing “CALL/MALL methods” with “non-CALL/MALL methods” requires a definition of a standardised teaching approach that does not exist. At a local level, the variation that can arise becomes even more apparent, and this is of more relevance to what is seen in recent research into the use of mobile devices.

One point that needs to be iterated here is that any research that involves human subjects needs to take into consideration the ethical concerns that it entails (Israel & Hay, 2006). When research involves learners using their own devices in their own time, then these concerns become even more pertinent, and care needs to be taken to ensure that

personal information pertaining to learners and their habits is kept securely and not used in a way that could possibly disadvantage them. As Traxler and Vosloo (2014) point out, however, although informed consent is the “ethical gold standard” (p. 24) when it comes to research regarding people, the interaction and exchange of information through mobile devices comes more fluid and abstract, meaning that the boundaries are not as clear cut as they once were. It then becomes the responsibility of the researcher to ensure that all measures are put in place to protect their subjects, even though this may be as much of a challenge as collecting the data itself in many regards.

4.4 Summary

As the discussion in this chapter shows, research into mobile learning is extremely complex and broad in focus. The path which MALL research has followed has resembled that of CALL in many regards, particularly with respect to comparisons with other methods. Comparisons in themselves need not be to show that mobile learning is superior to methods that have come beforehand, and indeed this type of research could even be considered as being counterproductive. What is of more relevance with regards to research into MALL is to identify what leads learners to more active and meaningful engagement, and how this knowledge can be applied to the development of appropriate pedagogies.

The range of research methods that are available to researchers is also broad, encompassing that which is directly observable and those behaviours that occur away from the eyes of researchers that require more innovative ways of investigating. Indeed, as has been mentioned at several points throughout this book, given that one of the primary aims of mobile learning has been to take advantage of small snippets of time and to weave learning into the everyday lives of learners, understanding what happens in the broader learning context should remain one of the prime foci of MALL research. Using an appropriate mix of methods can provide researchers with the insights that are needed in order to achieve this.

4.5 Discussion Questions

1. What do you think the most important consideration is when deciding on a method for your own research? Do you think that this can impact on how your research is conducted?

2. How can you avoid problems with your research that could impact on your timeline (i.e., could delay that completion of your research)? What steps could or should be taken at the outset of your research?
3. What specific issues do you think need further research with regards to language learning through mobile devices? Find one article published within the last two years and see if there are any outstanding questions remaining at the end of the study.

5 *Theory in MALL*

5.1 Introduction

As described in Chapter 2, theory underlies virtually all aspects of teaching and learning, including the choice of content, design of materials, teaching approaches, interactions between teachers and learners, assessment, and even evaluation. Language teaching – with or without technology – is a very practical undertaking, and theory is often sought as a means of validating both research and practice (Levy & Stockwell, 2006). Although language teaching fits into a wide range of disciplines, it is widely associated with applied linguistics. Applied linguistics is not limited to language teaching and learning (Simpson, 2011), but as Davies and Elder (2004, p. 1) remind us, one of the central questions of the field is, “How can we teach languages better?” Li (2018) rather pointedly argues, however, that applied linguistics has suffered branding as largely atheoretical, largely due to its practice-oriented nature. This branding does the field little justice, given that it has not only borrowed concepts, methods, and theories from other disciplines, but it has also been developing and refining its own theories, which have formed the foundation of research and practice for decades. It would be anticipated that most language teachers would have at least some basic training in theory relevant to their teaching needs, and knowledge of language acquisition theories is a fundamental element of the ACTFL Program Standards for the Preparation of Language Teachers (ACTFL, 2013).

CALL has been similarly labelled as lacking in theoretical grounding, which may be expected given that studies have revealed it is not uncommon for research in the CALL literature to fail to provide any explicit information about theory (Hubbard, 2009). However, a lack of reference to a formalised theory does not necessarily mean that there was no theory present, but that its contribution may have been too tacit to warrant mention by the designers,

regardless of the role it may have played in the design of the research or teaching environment. Perhaps not surprisingly, the view of theory in MALL has resembled that of CALL, with theory taking on an important but sometimes less visible position in published research. According to Duman, Orhon and Gedik (2015), theory was mentioned in around two-thirds of the studies in their sample of MALL research from 2000 to 2012, and around half of the studies included theories related to learning. The remaining studies that included theories were concerned with the attitudes towards technology or with the design of the interface, indicating multiple purposes for the inclusion of theory in research on technology in language teaching (see Levy & Stockwell, 2006, also described later in this chapter).

These issues are explored in the following sections, including the concept of theory, formal and informal theories, and theories associated with language teaching and learning with technology. Theories which may be applied to MALL are also described here, including theories for formal learning, theories of technology, theories associated with the complexity of the learning environment, theories of motivation, and social models of language learning. The range of theories that are presented here are certainly not meant to be exhaustive but are presented as examples that may be relevant when considering the use of mobile devices in language teaching and learning.

5.2 Informal and Formal Theories

The practical nature of language teaching and learning can often mask the vast range of underlying complexities involved. When entering their educational environment, teachers bring with them views of what language is, views of what elements need to be taught, views of how languages are learned, and views of the role of the teacher in facilitating learning, all of which culminate into what happens in (and out) of the classroom. As stated before, it is highly unlikely that teachers would be entering their classrooms without any kind of theory about what language is and how it is learned. Rather, it is possible that they simply may not be using a *formal theory*, basing their research and practice on *informal theories* (Breakwell & Rose, 2006), also referred to as *implicit theories* or *mindsets* (Ryan & Mercer, 2012). These informal theories may be an intuition or a “gut feeling” on which teachers base their views and practices, and most educators bring some kind of informal theory into their learning contexts. The theories educators use might be independent of formal theories for a variety of reasons. It may be that they lack the training to know what relevant theories exist, but it is more likely that they may be consciously

interpreting multiple theories from different perspectives, adding new elements that have not been included in existing formal theories based on their own experiences and intuitions.

Starting with an intuition or hypothesis and further developing this into a theory is a natural part of progression of theory development. Van Ryn and Heaney (1992, p. 217) suggest that “most social and behavioural science theories are rooted in the same fertile ground that engenders most of human creative endeavours: common everyday experience.” It is the instincts that are held by many teachers that form the starting point for many of the accepted theories that we see in language teaching, and to this end, they are an essential part of research and practice. Since informal theories are not bound to more formalised or established theories, they can provide the foundation for deeper insights into the learning process than might be gained through over-reliance on existing models (Sixsmith & Simco, 1997). The largest problem with informal theories, however, is that they are often applicable to a single narrowly defined context with outcomes that have yet to be confirmed by sufficient data in varied contexts to eventually lead to building a formal theory.

Formal theories are developed over time and provide a rational explanation for a given phenomenon based on incremental empirical evidence (Hanson, 1995). As more evidence is collected through carrying out replicative studies, it becomes possible to build theories that can be applied to wider contexts. Theory building is an important part of deepening our understanding of phenomena, because it “provides a framework for analysis, facilitates the efficient development of the field, and is needed for the applicability to practical real-world problems” (Wacker, 1998, p. 361). Theories are, however, typically based on assumptions from which new ideas are derived and tested, and the biggest limitation with theory building is that it is tied to its fundamental assumptions, regardless of how complex the theory may become (Hanson, 1995). In other words, it is very easy to fall into the trap of making claims based on theories and accepting them as fact, but it is essential to remember that the very nature of theories means that they are interpretations of phenomena, and these interpretations may vary greatly depending on the underlying assumptions from which they derive. Obviously, a single study is unlikely to lead to any significant development in theory; rather, studies need to systematically build on previous work to test the robustness of hypotheses until they reach a point that they become accepted as a valid theory. The applicability of theory to research and practice is described in more depth in what follows.

5.3 Fundamentals of Theory

In general, views towards theory in education have polarised towards more positive or negative perspectives. On the one hand, there are those who argue that practice simply cannot exist without theory, even if these theories are not explicitly stated or even fully formulated, while on the other hand, there is the view that focussing too much on a specific theory or theories can make it possible to have a biased or skewed perspective of the environment, the data, and subsequent conclusions. Advocates of the importance of the role of theory in research and practice would likely agree with Breakwell and Rose, who argue that “knowledge is said to be power, and knowledge is stored elegantly and systematically in the form of theory. Consequently, good theory is both powerful and of practical relevance” (2006, p. 2). This claim includes some fundamental points that underlie the functions that theory may take on. Theory provides us with the central concepts and assumptions that are necessary to carry out research, without which it would be very difficult to make sense of data (Neuman, 2003). Thus, without some form of theory, not only would understanding data be a challenge, but researchers would have little idea of what to investigate and for what reasons. From this perspective, the argument of theory as being of integral importance in terms of the knowledge that is held within it and its relevance to carrying out practice and/or research becomes apparent.

Because formal theories have been confirmed with empirical data collected from varied contexts, most people would agree that formal theories tend to have greater face validity than informal ones. As theories become more widely disseminated, they can be either further confirmed or disconfirmed through systematic research studies. This enables a theory to evolve over time and to ultimately be shaped into something which can account for and predict potential outcomes. Wacker (1998) describes formal theories as having four basic elements, these being (1) conceptual definitions, (2) domain limitations, (3) relationship building, and (4) predictions. Conceptual definitions lay the foundations for the theory in terms of the meaning and applicability of the terminology that is used in describing the theory. If we imagine we are formulating a theory of social interaction, for example, we will need to define what is meant by both “social” and “interaction” in the context they are used within the theoretical framework. Domain limitations restrict the scope of the theory to enable it to be applied to a given situation, such as who the participants within an interaction might be and the nature of the interactions. Relationship

building examines the ways in which the theory can build upon existing theories and allows for links to be drawn between them to collectively provide insights into the phenomena from a broader perspective. Finally, according to Wacker's (1998) definition, theories should allow for predictions to be made regarding the concepts that have been defined as a part of that theory. In other words, based on our hypothetical theory of social interaction, we should be able to make predictions regarding the outcomes that would be achieved through social interaction, and these predictions would, for the most part, hold true within the given domain. It is this predictive element that some argue is one of the most important outcomes of theory. The contribution of theory to understand why things happen can be used to improve future practice through enabling necessary intervention to bring about desired results (Breakwell & Rose, 2006), and this is one of the attractions of having a solid theory on which to base research and practice.

In contrast, however, theories have been criticised for failing to explain the complexities of the fields to which they are applied. They can oversimplify the interrelationships between the elements and how these elements interplay with one another to result in the given outcomes, and it is difficult for a single theory to sufficiently account for complex phenomena (e.g., Burns & deVillé, 2007). This perspective does not condemn theory per se, but, rather, it illustrates a need to understand that one theory is unlikely to be able to incorporate all competing factors that may be at work on many different scales of time and space (e.g., Mitchell, 2009; Sawyer, 2005). A concept which has gained ground that seeks to address this issue is *complex systems theory*, also known as *dynamic systems theory*, which explores the apparently random relationship between the interplay of factors, where an outcome does not necessarily correlate to the sum of the components. In a complex system, a seemingly insignificant change in one component can have unseen and unexpected impacts on other components. Examples of archetypal complex systems are global climate, economies, and immune systems (Mitchell & Newman, 2002), which are an intricate interplay of components where the outcomes are extremely difficult to predict despite understanding the workings of the individual components. While complex systems viewpoints have the potential to bring about a dramatic change in the way in which systems are conceptualised and unify cognitive theories and pedagogical research (Wilensky & Jacobson, 2014), they are still relatively new to educational circles and very much away from the mainstream.

From the Field: Using Theory in Research

Teaching in postgraduate programs, I find that my students are often at a loss as to how to apply theory to their research when writing their dissertations. There largely appear to be two patterns that I see students falling into when seeking theory. The first is that students already have their research plan in mind but have not looked at relevant theories until their study is conceptualised. This often results in students looking for theories that suit what it is that they already have in mind for their research, and they often select theories based on the compatibility with their predicted results. The second pattern is that students have read theories and find one that they feel is particularly appealing, and this forms the foundation of their research design. While there may be arguments on both sides, it is difficult to say that either one of these methods is particularly good or bad. Looking for theories that match the data that students are working with can allow them to work from the data and then look for theories that might explain their results, but at the same time, it may lead them to dismiss theories that could be very relevant. Similarly, working from established theories can provide a solid foundation for the work that they are doing, but at the same time can focus their view on one aspect of the data that might cause them not to see other outcomes. The key point is that students need to be aware of the fact that theory can indeed shape how data are collected and analysed, and bearing this in mind from the outset can allow them to have a balanced view of their own research contexts.

5.4 Theory in CALL

It is well known that theories associated with language teaching and learning have traditionally followed trends in psychology, originating mostly from behaviourist theories and shifting to the cognitive and social theories that still form the basis for much of our understanding of the language learning process today (see Mitchell, Myles & Marsden, 2013). Theories undergo trends and fashions, and there has been ongoing evolution of these theories over the past century reflected in the differences in teaching approaches that have come and gone, particularly evident in the methods era (see Richards & Rodgers, 2014). Despite the decades of change, a question that has been at the root of language teaching and learning has been, “What must a theory of language learning seek to explain?” A straightforward answer to this question has proven to be surprisingly elusive, and views on the strengths and weaknesses of the various theories that have been posulated over the years have continued to be quite strongly divided.

One of the main problems is that different theories explain different parts of the language learning process, and these theories are essentially looking at an enormously sophisticated phenomenon from the different perspectives that represent an area of particular interest to the researcher. For decades, there have been theories that have addressed the need for input, such as Krashen's *comprehensible input theory* (Krashen, 1981), and others that stressed the need for interaction that required both input and output, such as Long's *interaction hypothesis* (Long, 1983) and Swain's *comprehensible output theory* (Swain, 1985, 1995). Other theories have placed more emphasis on the context in which learning takes place such as *sociocultural theory* (Lantolf & Thorne, 2006), or on consciousness and awareness in learning as in the *noticing hypothesis* (Schmidt, 1999, 2009), and still others on the construction of knowledge inside the minds of the learners as in *constructivism* (Kaufman, 2004). Theories have also attempted to explain the motivation of learners to engage in language learning such as *self-determination theory* (Deci & Ryan, 2009) and the *ideal self* (Dörnyei, 2001; Dörnyei & Ushioda, 2011). Attempts to bring together various competing models are seen periodically, such as Robinson's (1995) discussion of attention, memory, and noticing, which brings together related theories to show how they complement or contrast one another, but these discussions still struggle to bring together the wide range of theories that exist to describe the teaching and learning process in all its dimensions.

Theories have been broad in their collective focus and have endeavoured to account for the various intricacies of the language learning process, but learning outcomes are "multifinal," meaning that essentially the same stimuli lead to different conclusions (Hanson, 1995). As has been seen in other disciplines, there have been efforts to account for the inability of theory in applied linguistics to account for the nonlinear nature of learning with a complex systems theory perspective (Larsen-Freeman & Cameron, 2008), and this has gained a good deal of support over the past decade. There would be very few who would refute any of the aforementioned theories in terms of how they can account for at least some part of what learning a language means, but at the same time, none of them would be able to paint a complete picture of the process. What becomes clear from this discussion is that theory most certainly does feature heavily in applied linguistics, which has consisted of adopting, adapting, formulating, and reformulating theories in order to understand and enhance the language learning process. To this end, to return to our question from the beginning of this section, we could argue that a theory of language learning needs to explain *part* of the entire picture, to shed light on our

understanding of at least one aspect of the intersecting aspects involved, but this part also needs to fit together with other theories in the larger ecology of language learning.

Considering the criticism that applied linguistics faced regarding a lack in theoretical rigor from other fields (Li, 2018), it is somehow ironic that CALL has received similar treatment from many applied linguists (Coleman, 2005). The field is sometimes seen as being comprised of a group of tinkerers with technology who can provide lists of software, websites, and apps for language learning based on unsubstantiated and often anecdotal evidence. This is likely the result of the strong technocentric focus of CALL over the years, with the majority of studies in CALL using technology – typically new technologies – as the point of departure for their research (Stockwell, 2012a). Theory has, of course, featured in CALL research, and indeed there has been great variety in the range of theories that have been applied and the rationales for their use. While theory played a lesser role in early discussions of the use of technology in language teaching and learning, research since the 2000s has not only included theories from applied linguistics but has also expanded to include psychology, learning theories, sociology, human–computer interaction, instructional design, and information systems (see Hubbard, 2009, for a discussion). Over the past several years, there have been a few attempts to take stock of theory in CALL (e.g., Hubbard, 2009; Hubbard & Levy, 2016; Levy & Stockwell, 2006; Stockwell, 2014), and these have highlighted the complexity of theories involved in the inherently interdisciplinary nature of CALL.

Theory has had a wide-ranging impact on the way in which CALL is viewed with respect to other fields, in particular SLA, and how to consolidate SLA theory and CALL has been discussed since the early days of CALL (see Doughty, 1987). CALL is seen as having obvious relationships with SLA, but it maintains enough elements of its own to deserve separate consideration in some capacity, particularly with regards to the role of the technology (Stockwell, 2012b). Since CALL is at the intersection of two major fields – language learning and technology – each of which representing an enormous body of research, theory needs to consider both of these larger fields to account for the impact that each one has on the other. Hubbard and Levy (2016, citing Hubbard, 2009) define CALL theory as representing the “set of perspectives, theoretical models, frameworks and specific theories that offer generalisations to account for phenomena related to the use of digital technology in the pursuit of language learning objectives; ground and sustain relevant research agendas; [and] inform effective CALL design and teaching practice” (p. 25). The perspective

presented here is very much in agreement with Levy and Stockwell (2006), who propose that theory may be used for *design* of CALL, for *teaching* with CALL, and/or for *researching* with CALL. Design may refer to the development of a new tool – or *artefact* – such as a program, a website, or an app, but it may also refer to the creation of a learning environment that uses these artefacts. Theories selected for design may relate to second-language learning but also to instructional design, which is often embedded within the design of these artefacts. Teaching will typically rely mostly on theories associated with language learning, but other theories associated with psychology, sociology, or cognition may well be relevant to the methods and approaches that teachers adopt. Finally, research will often involve theory testing, which – as stated before – is a necessary step for validating theory. It is feasible, of course, that the theories selected for design, teaching, and research are compatible, but it is also possible for there to be conflicting views that need to be consolidated in carrying out teaching or research that involves technology.

Based on the interdisciplinary nature of CALL, Levy and Stockwell (2006) suggest that CALL practitioners often draw on multiple theories – *theoretical pluralism* – which may be necessary to gain the greatest insights into the complex interplay of concurrent processes (see Mitchell, Myles & Marsden, 2013). As Hubbard and Levy (2016) point out, to date, there have yet to be any native theories that have been born from the CALL field itself, but rather, CALL practitioners have borrowed, instantiated, adapted, combined, synthesised, constructed, or refined other theories and applied them skilfully to both research and practice. Building on Kramch's (2002) ecological perspective of language acquisition and language socialisation, Blin (2016) proposes that technology in language teaching and learning forms a *CALL ecosystem*, which she defines as consisting of “interacting components including language learners, teachers and other users of the target language, technological devices, applications and platforms, and multimodal material/semiotic artefacts and resources, all of which participate in a language learning/use activity, as well as the social processes and semiotic practices that characterise the way that human actors interact with one another and with other components of the system” (p. 39). Not surprisingly, and Blin herself makes this connection too, but the relationship with a complexity theory perspective is immediately obvious. Technology comprises one component of the entire whole, and its potential to have an impact on other components within the whole system – sometimes unpredictably – is reminiscent of the nature of a complex system and helps to explain the existence of multiple complementary and competing theories in CALL.

A MALL ecosystem can become even more multifaceted when considering that learning with mobile devices entails a range of additional environmental factors, particularly if this learning takes place in varying contexts that fit Traxler's (2007) gaps in time and space, as described in the following section.

5.5 Theories Relevant to MALL

It is to be expected that there is quite some overlap in the theories applied to MALL when compared to those used in CALL. Durman, Orhon and Gedik (2015) in their overview of MALL research from 2000 to 2012 found similarities with CALL in the theories related to language learning – namely constructivism, social constructivism, and sociocultural theory, collaborative learning, and task-based learning. Other theories were more closely associated with the design of tasks using mobile technologies, such as ubiquitous learning, situated learning theory, dual-coding theory, cognitive theory of multimedia learning, cognitive load, multimedia design principles, and learning memory cycles. Theories related to attitudes towards mobile technologies were also evident in the studies in their sample, namely the technology acceptance model (TAM) and the unified theory of acceptance and use of technology (UTAUT). It is interesting to note here that the categories of theory for design, teaching, and practice proposed by Levy and Stockwell (2006) were also clearly evident in MALL studies, providing support for there being similar processes at work when implementing mobile devices in language teaching and research as has been seen more broadly in CALL.

While there are several necessary commonalities between CALL and MALL, to determine which theories are likely to be more applicable to MALL, it is important to note the differences between learning through more “traditional” fixed or portable (as opposed to mobile) technologies such as desktop and laptop computers and mobile devices such as mobile phones and tablets. Of these, the two most pertinent aspects would be *mobility*, meaning that learners can use them as ongoing tools at all times quickly and easily at any time that they choose, and the *interactivity* of the device with the surroundings, other devices, or people. Both of these factors combine to mean that learning opportunities can be found in the contexts in which the devices are used. The ecosystem of language learning with technology is a primary area of interest to practitioners, and understanding how theory has been applied to MALL can help to give insights into better practice in teaching and research. Moreover, the motivation that learners have for using mobile devices and how motivation can be

affected by using technology – and vice versa – can have a direct impact on how, when, where, and why learners will use these devices for learning both inside and outside of the classroom (Stockwell, 2013). Autonomy is also a key element of discussion surrounding the use of MALL, which is closely related to both motivation and training (which is discussed in Chapter 6) and the social relationships that take place between teachers and learners (see Chapter 2; Lewis, 2014). The discussion that follows focusses on four elements which are thought to be of particular relevance to MALL, looking from differing but nonetheless interrelated perspectives. These are *theories of technology*, *theories of complexity*, *theories of motivation*, and *social models of language learning*, as described in turn in the following subsections.

5.5.1 *Theories of Technology*

As technology is quite often the most visible part of the learning environment (Stockwell, 2009), it is logical that there has been a lot of focus on this technological aspect. There are four areas where theory regarding technology in MALL is thought to be of particular relevance: theories surrounding the design of tasks and activities, theories of attitudes towards technology, theories of digital literacy, and theories of the impact of technology on users. These are illustrated in Table 5.1, but it should be noted that not all of these theories have featured in MALL research to date, and some remain at the theoretical level without empirical evidence in either CALL or MALL, while others have appeared only in studies relating to CALL only.

As the table shows, MALL research that applies theories associated with task design do appear in the literature, and this provides evidence of a solid understanding of the importance theory as a foundation for the design for mobile devices. Although these can overlap, design here can refer to design of tasks themselves and also to the design of artefacts or even interfaces. Chapter 8 explains design in more detail and how this can be related to practice in MALL.

Research that explores learners' willingness to use mobile technologies for language learning has also been the focus of research a number of studies being carried out over the past several years (e.g., Chen, 2013; Dashti & Aldashti, 2015; Kondo, Ishikawa, Smith, Sakamoto, Shimomura & Wada, 2012; Yang, 2012; see also Chapter 4). Notably, there is a lack of empirical research that looks at the application of digital literacies specifically to MALL, despite the fact that there has been quite a lot of discussion of the importance of digital literacies in education and language learning in general (Dudeney, Hockly & Pegrum, 2013; Hauck & Kurek, 2017). Lastly, theories

Table 5.1 Theories related to technology relevant to MALL research

Category	Theory/Model	Examples
Design	Multimedia theory	Lee, Lo, and Chin (2019) ²
	Item response theory	Browne and Culligan (2008) ² ; Chen and Chung (2008)
	Dual coding theory	Lin and Yu (2017)
	Situated action	Suchman (1987, 2007) ¹
	Spaced learning	Zhang, Song and Burston (2011), Nakata (2017) ²
	Cognitive load theory	Hong, Hwang, Tai and Lin (2019)
	Activity theory	Lin, Lin, Liu, Kou, Kulikova & Lin 2019
	Situated learning	Huang, Yang, Chiang and Su (2016)
	iPAC framework	Burden and Kearney (2018) ¹ , Koenraad (2019)
	TPACK	Mishra and Koehler (2006) ¹ ; Hsu (2016)
	ADDIE model	Branch (2009), Heift and Caws (2014)
Attitudes towards technology	TAM	Al-Emran, Mezhuyev and Kamaludin (2018)
	UTAUT*	Gan and Zhong (2016)
Digital literacy	Sociocultural literacies	Mills, 2016 ¹
	Critical literacies	Hafner (2014) ² , Pegrum (2014)
	Multimodal literacies*	Lotherington and Jenson (2011) ¹
	Media literacies	Buckingham (2007) ¹
Impact of technology	Distributed cognition(s)*	Stockwell (2014) ¹
	Dual systems theory*	Davazdahemami, Hammer and Soror (2016) ¹

Note: Examples that refer to the theory without examples from CALL or MALL are marked with a “1,” and studies that relate to CALL but not MALL are marked with a “2.”

that examine the impact that mobile technologies can have on learners’ cognition and attention has yet to be seen in either MALL or CALL research, but these have the potential to shed light on the ways that technologies can alter the ways that learners think and behave as a result of mobile technologies. A selection of these theories is explained in the following paragraphs. Considering the wide range

of theories listed here, only a sample of these theories has been explained in depth, marked with an asterisk in the table.

Unified Theory of Acceptance and Use of Technology (UTAUT): Originally developed by Venkatesh, Morris, Davis, and Davis (2003), this was designed to go beyond just the simple acceptance of technology such as that proposed by Davis (1989) in the Technology Acceptance Model (TAM). The TAM has faced criticism for its lack of ability to predict learner behaviour with technology (Legris, Ingham & Collette 2003; Turner, Kitchenham, Brereton, Charters & Budgen, 2010), so the UTAUT aimed to explore further the mental processes behind viewpoints of technology in order to predict usage as well. It is based on four core variables – namely performance expectancy, effort expectancy, social influence, and facilitating conditions – as well as four moderating variables – gender, age, experience, and voluntariness of use. It is based on eight models: (1) Theory of Reasoned Action, (2) Technology Acceptance Model, (3) Motivation Model, (4) Theory of Planned Behavior, (5) Model of PC Utilization, (6) Innovation Diffusion Theory, (7) Social Cognitive Theory, and (8) Combined Technology Acceptance Model and Theory of Planned Behavior. It represents an extremely thorough examination of the major models that may contribute to learner views and usage of technology, but does lack cultural reference, which may be an important contributing factor in examining how technology is used and viewed (Im, Hong & Kang, 2011). Though not without limitations, it has great potential as a means of explaining learner willingness and actual usage of technology.

Multimodal literacies: The concept of multimodal literacies is based solidly on work by Kress (2003, 2010) and Cope and Kalantzis (2000) and argues that humans are constantly engaging in communication that goes beyond just the written word and includes pictorial and audio modes as well. Lotherington and Jenson (2011) maintain that the affordances of technology naturally facilitates multimodality in communication, and humans require literacy skills in order to make sense of the multimodal cues that they are constantly exposed to during this communication. Although multimodal messages are a normal part of human communication, when these messages are facilitated through technology, they can become filtered as a result of the technological limitations which can result in a different communication experience. Needless to say, this is highly applicable to MALL in that the size of the screen and methods of inputting can have an effect on the way in which communication is carried out, but mobile technologies also mean that learners can have constant access to multiple modes of information that may be applicable to their

learning. This is particularly important for communication that is required through certain modes, and having access to required information in that mode and supported by other modes can result in facilitating successful communication or providing useful learning opportunities. The prerequisite for this, however, is having sufficient literacy to decode, comprehend, and encode multimodal information appropriately.

Distributed cognition(s): The concept of “distributed cognition” was proposed by Hutchins (1991) as a part of organisational management, and its educational applications were explored as *distributed cognitions* by Salomon (1993). In short, the underlying tenet of distributed cognition is that we support our internal cognition (i.e., in the human mind) through external tools around us (i.e., using something to help facilitate our thinking). An example of this would be a computer keyboard, where we have the letters written on each of the keys to remind us where each of the keys are. When we are typing, we are largely able to type sentences without looking at the keyboard, but we may find that we refer to the letters written on the keyboard to help us to remember which key is which if we suddenly wish to type a single letter out of context. We currently use mobile devices to store information that we no longer commit to our own memories, and in this sense, we are distributing our cognitions between our biological brain with the mobile device to reduce our own mental load. This has also been known as the *Google effect*, where people can easily use mobile devices to carry out Internet searches for information rather than trying to commit it to memory. The reliance on this readily available information has been shown to reduce how much is committed to memory (Sparrow, Liu & Wegner, 2011), suggesting that having constant access to information can diminish our willingness to remember facts. The impact on education and assessment has been raised as a concern. Dikkers (2014), who points out, “Can we proactively continue to define teaching and learning as relevant? For instance, consider the need for massive amounts of memorized data . . . if a student can find information in less than thirty seconds, is it worth testing them on it?” (p. 111). To this end, the impact of distributed cognition on MALL may be considered as potentially a large one, although the need to memorise data will differ depending on the content, and other uses such as searching for expressions through accessing corpus data can be a way in which mobile devices can change the ways in which learners memorise or seek out new information.

Dual systems theory: A theory that can help to explain the distractive effect of technology that can detract from time spent on task both inside and outside of class (see Chapter 7) is *dual systems theory*.

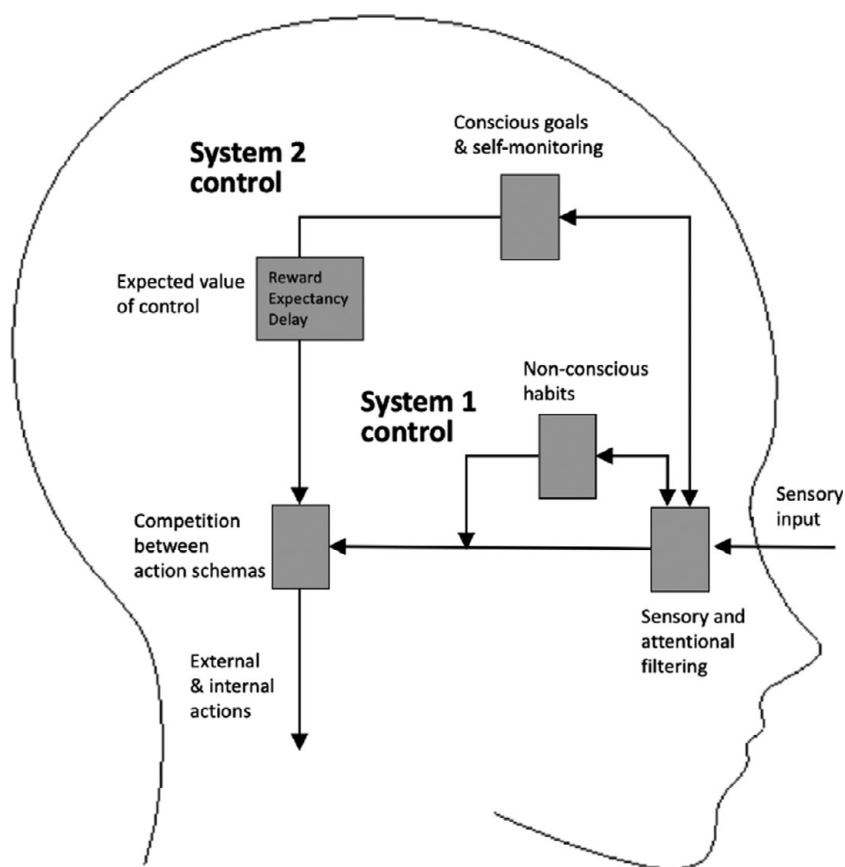


Figure 5.1 A dual systems theory model (Lyngs et al., 2019, p. 5)

While this has yet to appear broadly in research on language teaching and learning, it has attracted a good deal of support in research on human behaviour since it was proposed by Evans (2003, 2008) as a model of dual processing of cognition and reasoning. As Lyngs et al. (2019, p. 1) describe, the central tenet of dual systems theory is “a major distinction between swift, parallel and non-conscious ‘System 1’ processes, and slower, capacity-limited and conscious ‘System 2’ processes” (see Figure 5.1). A person engaging in a task or activity such as listening to a lecture or reading an academic paper diverts his or her conscious attention towards this task using System 2. The temptation to check a smartphone notification is part of System 1, which is the

unconscious level of processing by which a person might perform actions such as scratching a mosquito bite, and it has been used to account for mobile phone and Internet addiction (Davazdahemami, Hammer & Soror, 2016).

In order to *not* undertake an unconscious or habitual action, System 2 must be working at a conscious level to prevent scratching the mosquito bite or yielding to the temptation to check the smartphone notification through *self-regulation*. This self-regulation is, according to Lyngs et al. (2019), affected by the person's capacity limitations or their emotional state or level of fatigue. Capacity limitations dominate if the task that the person needs to undertake is not sufficiently represented in their working memory at the time of the action, whereas a poor emotional state or high level of fatigue results in lower levels of self-regulation. This theory has immediate and obvious implications for MALL. Providing tasks and activities that overload the capacity limitations of learners means that there is a greater likelihood that learners will be unable to fully activate their working memory, thus requiring them to exercise a greater degree of self-regulation over their temptations to check their mobile devices. Making learners aware of the fact that poor mood and tiredness can lead to a higher susceptibility to distractions can also help them to plan their time and avoid interrupted task engagement times as a result of a failure to regulate their unconscious system.

5.5.2 Theories of Complexity

Theories that have been proposed to explain the nonlinear nature of many phenomena of the natural sciences have been termed as complexity theories. These find their roots in chaos theory, which is a branch of mathematics which seeks to identify the dynamical systems that underly seemingly random irregularities and disorder. It is also related to the *butterfly effect*, coined by Lorenz (1972), who posed the question, "Does the flap of a butterfly's wings in Brazil set off a tornado in Texas?"

Complex systems theory: The past decade has seen a surge in discussion of the complex nature of language learning, largely prompted by the highly influential work by Larsen-Freeman (1997, 2002), who argued that current theories of language learning fail to explain and predict the learning process consistently. Larsen-Freeman and Cameron (2008) contended that a limitation with theory in language teaching and learning is that it removes the dynamic aspects of the systems they seek to describe, treating them as being fixed points that are not subject to change. The application of complex systems has

spread rapidly into a range of areas within language teaching and learning since then, with the concept being applied to areas as varied as motivation (Dörnyei, MacIntyre & Henry, 2015), teacher cognition (Feryok, 2010), learner agency (Mercer, 2011), developmental cognitive neuroscience of language learning (Hohenberger & Peltzer-Karpf, 2009), and even language itself (Ellis & Larsen-Freeman, 2009). To a certain extent, looking at language learning through a complexity theory lens helps to ease the tension that exists between theories and to explain unexpected outcomes, but it has also faced criticism regarding its applicability to language and language learning and for lacking in detail to form a foundation on which to analyse second-language acquisition data (e.g., Gregg, 2010). Hiver and Al-Hoorie (2020) have, however, explored the possibilities of carrying out research based on complexity theory principles and have described the problems associated with generalisability and causality, outlining numerous quantitative and qualitative methods in order to make sense of the interplay of factors in a language learning ecology.

It has only been relatively recently that areas such as motivation in language teaching and learning have been viewed from a complex dynamics system theory perspective (see Dörnyei, MacIntyre & Henry, 2015), but this perspective on motivation has enabled researchers to look beyond snapshots of a learner's motivation at a given point in time, and to account for the fluid nature of motivation over time (Schumann, 2015). Larsen-Freeman (2017) discusses three lessons drawn from discussion on complexity theory that continue to challenge research into second-language teaching and learning; (1) the applicability and appropriateness of dichotomies, which may serve to oversimplify phenomena; (2) whether and where boundaries can be drawn between a system and its environment; and (3) the difficulties in generalisation caused by rejection of simple linear causality. Given the complexity and dynamism of language learning, it is not surprising that criticism regarding linear and structured theories that try to account for the language learning process has taken on increasing prominence. The complex systems theory approach to analysing language teaching and learning environments provides us with insights into this immensely intricate process but also makes us keenly aware of the need to conduct systematic research to refine our understanding of the relationships between theories of second-language teaching and learning.

5.5.3 Theories of Motivation

Motivation and autonomy are perhaps two of the foremost areas which have relevance to mobile learning, although many claims

regarding the motivational aspects of technology have been excessively optimistic views (see Chapter 2). The past few years have seen various newer theories that attempt to explain the factors behind the learner being involved in learning for a longer or more consistent period of time, and research has focussed on both the internal and external factors that are thought to be relevant to learning and have considered how to promote motivation that will result in more sustained language learning with the hope of leading to more successful learning outcomes.

Motivation has seen a renewed interest in the past two decades or so (e.g., Dörnyei, 2001; Dörnyei & Ushioda, 2011), and theories of motivation seek to identify the reasons behind learners engaging in studying other languages, the features of the language learning environment that may have an impact on the way they view the language and learning materials, and the way that learners engage in particular activities. As Grabe (2009, p. 176) points out, the “beliefs, values, and expectations [learners hold] are influenced by a range of external social and contextual factors,” which can link directly to effective learning behaviour (Guthrie & Wigfield, 2000). From this discussion, the importance of motivation is immediately obvious, as it can have a direct influence on what learners do inside and outside of the classroom in both the short and long term.

Expectancy-value theory: Originally proposed by Eccles, Adler, Futterman, Goff, Kaczala, Meece and Midgley (1983), the underlying precept of this theory is that motivation for engaging in a task is based on a combination of the beliefs that a learner holds about their ability to acquire a new skill, their expectations in the future with regards to their prospect of developing further in the skill, and the value that they place on the tasks and activities that are assigned to them in meeting these expectations. There are parallels with expectancy-value theory and self-efficacy theory or agency developed by Bandura (1997), who argues that motivation to engage in learning activities will depend upon the “beliefs in one’s capabilities to organize and execute the courses of action required to produce given attainments” (p. 3). The relevance of this theory for second-language learning – including of course learning through mobile devices – is clear. The relationships that learners draw between their current (and future) ability to acquire the target language and the tasks that are assigned to them is likely to have an impact on the way in which they engage in the activities. There have been several cases of learning through mobile devices where learners were encouraged to engage in language learning activities outside of class, but actual usage did not reflect usage indicated in pre-surveys (e.g., Kim, Rueckert, Kim & Seo, 2013). One possible

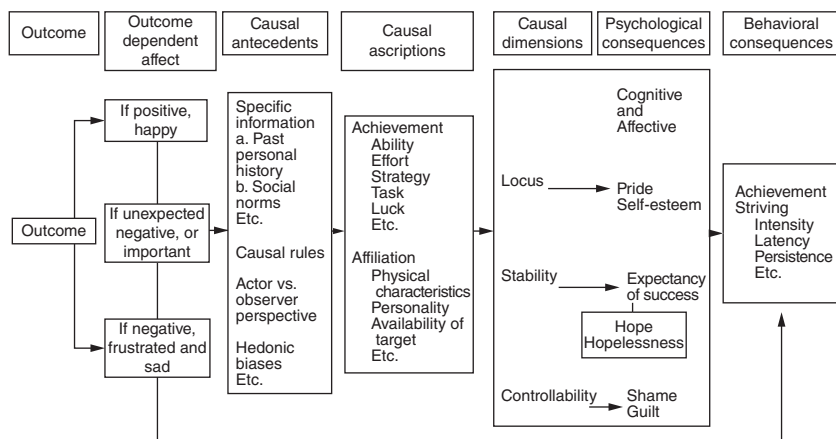


Figure 5.2 Attribution-based theory of intrapersonal motivation (Weiner, 2010, p. 34)

explanation for this is that the learners failed to see the relevance of the particular activities that were assigned to them and their view of their ability to acquire the target language. Understanding the potential relationship between learner beliefs and expectations can play an important role in determining learner engagement in the activities.

Attribution theory: Another theory that has featured quite prominently in studies regarding motivation, attribution theory, looks at the factors that the learner may consider as contributing to their successes or failures in the learning process. It has undergone several revisions over the years, including various degrees of detail, but the interpretation of this theory by Weiner (2010) as shown in Figure 5.2 shows the complexities that might be involved in developing and maintaining learner motivation. To some degree, it does include overlap of the expectancy-value theory within the interplay of factors, but these are made more explicitly clear in attribution theory.

Self-determination theory: As described before, the idea of extrinsic and intrinsic motivation has been one of interest to psychologists for several years, and the idea of how internally driven motivation can impact the learning process forms the foundation of self-determination theory (Deci & Ryan, 1985, 2009; Deci & Moller, 2007). Gardner and his colleagues (Gardner, 1985; Gardner & Lambert, 1972; Gardner & MacIntyre, 1991) describe intrinsic motivation as referring to the desire to engage in some activity for the satisfaction that the activity brings, whereas extrinsic motivation is engaging in some activity in order to achieve another goal that is not directly related to

the activity itself. As Williams, Mercer and Ryan (2015) point out, treating external and intrinsic motives as a dichotomy is overly simplistic and can mask the interplay that can take place between them. Self-determination theory is made up of three basic human needs – autonomy, competence, and relatedness – and it focusses on the motivations behind a learner's actions to understand the degree to which they are self-determined.

Directed motivational currents: This is a more recent theory of long-term motivation and is proposed by Dörnyei, Muir, and Henry (2016). This theory includes a combination of factors from the learner to the teacher, in that the teacher can play a role in leading learners into the motivation that they need to engage actively in their learning. One area where there has been a noticeable lack of attention in motivational studies has been with regards to longer-term motivation, which is what Dörnyei, Henry, and Muir's motivational currents attempt to address. The basic premise of the theory is that learners will see some kind of goal that they can work towards, and during that time, that they become engrossed in the learning process so that they are able to carry out their learning with a relatively high degree of autonomy. The theory differs from flow theory (Csikszentmihalyi, 1990), which tends to be shorter in duration where the learner may "lose themselves" in the moment. Motivational currents, as the name suggests, sees learners pulled along by their motivation, such as preparing for a test or a trip to the target language country, in such a way that they do not find the study to be particularly burdensome as long as they are in the current. The directed part of this theory is that the teacher can play a role in setting the learners into the current that keeps them engaged in their learning for a longer period of time.

5.5.4 Social Models of Language Learning

The dangers in viewing language acquisition in isolation from the social environment was proposed by Kramsch (2002), who argues that all of the relationships that learners engage in can have an impact on not only the language that they acquire but also the way in which they acquire it. Social models of language learning are far from new to the literature and are open to varying interpretations that place different emphasis on the cognitive and social elements that may comprise social interaction. Interpersonal undertones may even exist in theories of language learning that are not typically associated with social models, but the inherent social nature of these approaches is undeniable. A stronger cognitive influence is evident in interactionist accounts of second-language acquisition such as the interaction

hypothesis proposed by Long (1981, 1996), which argues for the need for interaction as a means of acquiring language through provision of conditions for negotiation of meaning (e.g., Pica, 1991) and comprehensible output (Swain, 1995). This has often been pointed out that the interaction account provides the theoretical foundation for many studies using CMC (e.g., Hubbard & Levy, 2016), where learners may interact with other learners, native speakers, or the teacher as a means of acquiring lexical and syntactic aspects of the target language. The relationship between technology and language learning, and in particular the social aspect of exposure to contextualised language through the Internet, has been recognised for many years now (Kramsch, 2002), and it is not surprising that these social models have come more to the forefront over the past several years. Two models that are of particular relevance, sociocultural theory and language socialisation, are described next.

Sociocultural theory: Approaches that look more at the social aspects of learning are typically based on Vygotskian perspectives that advocate that learning arises from social interaction, where novices can bridge the gaps in the knowledge they need to perform required tasks through interaction with relative experts (Vygotsky, 1978). Sociocultural theory, popularised by Lantolf and others (e.g., Lantolf, 1994; Lantolf & Pavlenko, 1995), is one such theory that is based on Vygotskian principles and views language learning as a social interactive process. Although there has been debate about what sociocultural theory actually refers to (Lantolf & Thorne, 2006), it has been used quite widely in research into learning through technology (see Levy & Stockwell, 2006, for further discussion).

As Hubbard and Levy (2016) point out, social interaction can be augmented through the use of virtual technologies that enable learners to carry out collaborative activities remotely using interactive virtual environments such as *Second Life* (e.g., Wang, Deutschmann & Steinvall, 2013) or other types of online role-playing games (e.g., Peterson, 2012). Another perspective often associated with social aspects of learning is collaborative learning (Nah, White & Sussex, 2008), where learners may benefit from the social interactions that may arise from learning with others.

Language socialisation: The goal of language learning for an increasing number of learners is to engage in communication with others almost exclusively through electronic means, primarily some form of social media. The distinction between these types of digital communication and face-to-face communication has been the focus of research very much since digital communication became more

mainstream (e.g., Collot & Belmore, 1996), and various differences have been identified. In saying this, the language of the various forms of electronic communication through the tools that are used on a day-to-day basis is in a state of flux, and the standards of what constitutes appropriate conventions for communication are subject to evolution over time. For example, early email messages closely resembled written letters, with formal greetings (e.g., “Dear Mary”) and closures (e.g., “Sincerely”). While these conventions are still used for certain forms of email communication, it is also appropriate to send replies to messages that are as simple as a single word of acknowledgement that is void of any formal greetings whatsoever, or people may respond by intentionally threading replies to specific questions throughout their email messages.

Many of the conventions that used to be a part of synchronous textual communication have seen a steady decline, such as smileys that are made up of characters like :-) (which, interestingly, I had to force back into text after my word processing software automatically changed the text into ☺). These symbols have largely been replaced by the use of emoticons or stickers to express language in lieu of prose, although there are also cultural and social dimensions associated with the selection of different non-linguistic symbols such as emoticons, emoji, and stickers, meaning that there are different sociocultural rules associated with the selection of these symbols and the meanings that are attached to them, all of which need to be acquired by language learners who communicate through digital means such as messaging apps or social media. Language socialisation refers to how speakers of a language modify their language to meet the conventions of a social context (Garrett & Baquedano-López, 2002; Schieffelin & Ochs, 1986). Children engage in some kind of language socialisation from the time that they first become cognizant of communicating with others, and this phenomenon is also evident in second-language learners (e.g., Duff, 2012). In this way, users who interact through the multifarious social media and communication tools will invariably follow the general practices of that particular tool, imitating styles and emoticon and sticker use they see. For language learners, the examples that they see in social media or more personal messaging can provide valuable data to guide them in how to make linguistic and non-linguistic choices in their interactions.

5.6 Summary

As pointed out in a number of places in this chapter, theory plays a very central role in what happens in both research and practice.

Theory can shape both research and practice and can have an impact on how we view a situation and how data are collected, analysed, and interpreted. From this point of view, we need to have a clear idea of what theory or theories we are applying, as well as our rationale for adoption of such theories. The adoption of technology does not preclude this need, and the relevance of theory to MALL will depend very much on what elements of language are to be acquired and the methods that are used to teach these language elements. Very much like CALL, MALL is likely to see theoretical pluralism, where theories will need to take into account the design process, the pedagogical principles for learning either inside or outside of class, and the conducting research that explores the impact that MALL can have on the teaching and learning environment. The link with social and interactional aspects that mobile devices afford, as well as the propensity for distractions that also seems to go hand-in-hand with using mobile devices in learning contexts, both inside and outside of the device, will also need to feature quite prominently in discussions of theory. Understanding how mobile devices form a part of the daily lives of users will be key in determining which theories are applicable and, through systematic exploration of learner behaviour in mobile learning, can play a role in further predicting patterns of engagement in language learning tasks and activities.

Theory – both in the theory-building stage and after it has gained acceptance – has an impact on research, not only the focus of research, but also the ways in which data are collected and analysed, the interpretation of the results, and the conclusions drawn from them. Theory forms the foundation for essentially all that teachers require of learners inside and outside of the classroom and the ways in which this is viewed and analysed by researchers. It has been argued that applied linguistics is becoming a field where there is a widening gap between researchers and practitioners (Kramsch, 2015), and care needs to be taken to ensure that the practice-oriented focus of language teaching and learning is not forgotten at the expense of theory that is not applicable to better practice. As described earlier in this chapter, correct use of theory can help to predict patterns of behaviour that can be shaped through the “necessary intervention” prescribed by Breakwell and Rose (2006). The hypothesis testing which goes behind theory development adds to the credibility of research and practice, which is founded on appropriate theories for the given environment. Taking the time to fully appreciate the theories which have been postulated regarding a particular phenomenon can deepen the awareness of researchers and teachers, which has the potential to bring about better practice in both.

As Richards and Rodgers (2014) point out, a teaching approach should be based on both a theory of language and a theory of learning, and both aspects interrelate to form the basis of how a language is learned. These theories may not be completely formalised, and informal theories have the potential to provide insights into a given phenomenon, but formal theory allows for more solid testing of the theory allowing a greater understanding of the phenomena (Price, 2012). In MALL, theory is gradually taking a key role in research and practice that is being undertaken, but care needs to be exercised to ensure that we don't fall into the trap of "theory shopping" (Levy & Stockwell, 2006), where theories are sought and applied post hoc to collected data in order to somehow validate anticipated results of a study, often in lieu of a theory being considered at the outset of a study.

Understanding and appreciating formal theories is key to avoid reinventing the wheel, but it is helpful to not become overly dependent on current theories without questioning their applicability to a wide range of contexts. The role that technology plays, and in particular mobile devices, can have a profound impact on research and practice and, ultimately, also on theory. Theories of MALL need to consider the way that mobile devices alter the ways that we as humans function in their entirety, including how we seek out and process information, engage in social relationships, set and meet short-term and long-term goals, and understand and represent the world around us in textual, audio, and visual formats. Based on this foundation, it becomes possible to see how learning has been transformed, and through solid theory building, we can start to predict how best to take advantage of mobile devices to benefit learning from a broader and more stable perspective.

5.7 Discussion Questions

1. How do you think that theory affects language teaching? If you have experience as a language teacher, have you ever felt that your practice has been impacted by one or more theories? Why or why not?
2. If you are to use a theoretical foundation for your research, at what point do you decide what theory or theories will be relevant? Why do you think so? How do you choose an appropriate theory?
3. Is it possible to use conflicting theories as the framework for a teaching or research approach? Why do you think so? Try to give examples.

6 *Physical, Psychosocial and Pedagogical Issues*

6.1 Introduction

Despite the various benefits that mobile devices can bring into teaching and learning contexts, there have continued to be issues that have surrounded their integration into education as serious tools. The physical characteristics of mobile devices such as screen size and input methods have long been issues when considering their applicability for learning, but they have often been considered as a necessary trade-off in order to maintain their portability. In addition to this, however, there are also psychosocial issues, considering the position of mobile devices in the minds of the learners and teachers. The perceptions of mobile devices are often a product of the social context in which they are used, and this will likely vary depending on the region, the socio-economic status, the age group of the users, and so forth. Finally, the issue of pedagogy when learning through mobile devices is discussed, and the range of considerations that need to be kept in mind to enable successful integration of MALL in terms of achieving learning goals and sustaining task engagement. While these factors are discussed separately, it is pertinent to bear in mind that there are certainly overlaps between them that have a mutual impact on one another.

The physical, pedagogical, and psychosocial considerations regarding using mobile technology for language teaching and learning each has the potential to have a large, but rather different, impact on the language learning and teaching process. As has been raised several times in the previous chapters, there are several concerns with using mobile devices that must always be kept in mind. The affordances of mobile devices for learning – such as their portability, the technological affordances, and the fact that they usually carried to both learning and non-learning locations – have been discussed in some depth in Chapter 2. However, the limitations associated with using mobile devices in educational contexts cannot be ignored, as will be discussed in more depth later in the chapter.

It is essential to realise that from the learners' perspective, their mobile devices – in particular mobile phones – may not even be considered as being “proper” education devices in the first instance. Balancing between these learning and non-learning uses may contribute to facilitating further engagement in language learning tasks and activities on learners' mobile devices (Steel, 2012). As has been alluded to in Chapter 3, there are differences in the roles that mobile devices will assume when looking at formal and informal learning situations, partially in the ways in which mobile devices may fit into the larger learning environment. Looking holistically, mobile devices can fulfil specific functions within the learning context, but it can be assumed that they will typically be complemented by other technological and non-technological means.

Informal learning is likely to be made up of learners seeking out for themselves different learning resources and deciding on the timing of when, where, and how to use these resources. In formal learning contexts, it would be expected that teachers would provide learners with direction regarding what to use and when. There may be some freedom in the timing, but for the most part in formal contexts, learners will most likely be given guidance regarding when they are expected to complete mobile tasks and activities, typically governed by the constraints of the course.

6.2 Physical Issues

The physical aspect of mobile devices is the most visible concern regarding mobile learning, and it is therefore not surprising that the small screen size and input methods are often cited as limitations of MALL in the literature (e.g., Kim, Rueckert, Kim & Seo, 2013; Rosell-Aguilar & Kan, 2015). However, as Koole (2009) points out, there are other factors that are thought to have an impact on the way in which mobile devices are used, ranging from these two primary issues to storage capacity, processor speed, battery life, compatibility, and network access, as is shown in Figure 6.1.

6.2.1 *Screen Size*

The issue of screen size is, to some degree, an inevitable one, given the fact that portability is an indispensable part of what makes a mobile device mobile, and as such, there will necessarily be limitations as to the size of the screen itself can reach. Improving resolution of the screen means that more information can be included on small screens, however, and smaller text and pictures can be used without detracting too much from readability.

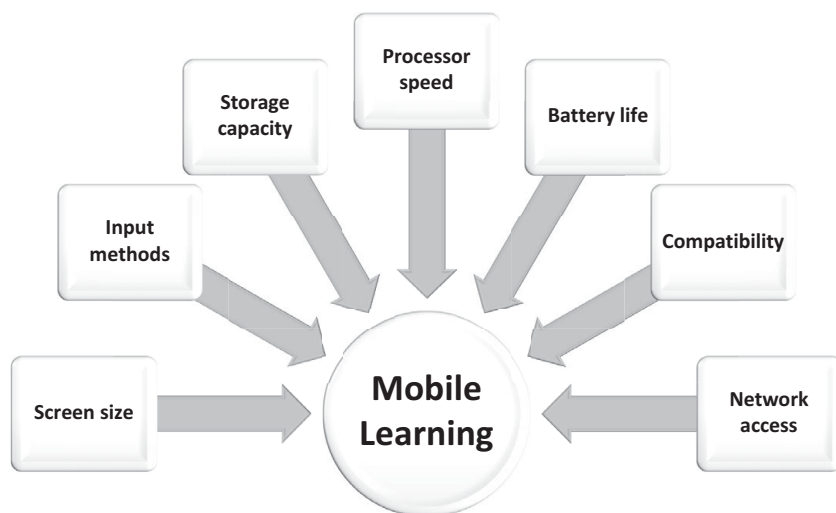


Figure 6.1 Physical issues of mobile learning

6.2.2 Input Methods

The on-screen keyboard is a feature of most modern smartphones that makes it much easier to type than the keypads on the pre-smartphone GSM phones, but the very nature of it being “on-screen” means is that it must occupy a certain percentage of the screen in order to be usable. Functions such as autocorrect can counteract less accurate typing, but as most people have experienced at some point or another, the accuracy is sometimes questionable, and sometimes correctly typed words are replaced by incorrect alternatives. The accuracy of voice recognition has improved dramatically over the past few years, making it a viable means of converting ideas into text, taking a little or none of the screen space. Finally, it may be possible to supplement the native inputting methods by using a Bluetooth or other wireless keyboard.

6.2.3 Storage Capacity

Storage capacity on modern mobile devices is another area where we have seen great improvement in recent years, largely as a result of a reduction in prices of storage media that may make up a part of the built-in storage. Many mobile devices come with lower-end or higher-end pricing options, where the lower-end models typically have far less storage compared to the higher-end models. The more

expensive models may be too expensive for many students who then opt for the lower-end models and, therefore, have smaller amounts of storage available to them.

6.2.4 *Processor Speed*

The last several years have brought significant changes in processor speed of mobile devices, at least in terms of the experience of users. While, of course, there are fundamental differences in the ways in which processors in PCs and mobile devices such as smartphones operate, most users do not notice speed differences unless using multiple apps simultaneously. It should be pointed out that that nearly half of the world's mobile phone owners still have a GSM phone (Statistica Research Department, 2018), meaning that care should be taken to avoid assumptions that all users will have access to fast processors in their devices.

6.2.5 *Battery Life*

An ongoing problem with mobile devices has been the life of the battery, which has become more of an issue with more powerful processors and larger screens that require more power to keep running. Mobile phones will most likely be used primarily for personal purposes, and some learners have voiced concerns that they may not have enough battery for these personal uses if they use them for learning tasks and activities during the day (Stockwell, 2010). Despite some recent improvements in battery life with newer battery cell, processor, and display technologies, extended use of a mobile device still seems to drain the battery at a level that would make it difficult to use it for a full day without a charge, and this is also likely to influence users' decisions regarding using it for learning, in that they may choose to use it only when they have charging capabilities on hand.

6.2.6 *Compatibility*

The period of transfer from GSM phones to smartphones also brought difficulties with compatibility, and the totally different software and hardware made it difficult to design comparable applications between them. Even with an extremely high penetration rate of smartphones being achieved in many locations around the world, compatibility problems have also been evident in mobile learning applications, with apps typically designed for iOS or Android devices. One method that has been used to attempt to deal with the problem of incompatibility

between native apps for iOS and Android is the use of *web apps*, where the app relies on web technologies such as HTML5 that can be accessed over the Internet. The downside is that this type of system will generally rely on having an Internet connection for the bulk of operations (although it is possible to store data on the device itself), but the trade-off with compatibility may well make it a viable option when learners are coming to teaching and learning environments with differing technologies.

6.2.7 Network Access

Studies have shown that learners exhibit resistance to using their mobile devices for learning purposes when there is the possibility that they will have to pay for network access above and beyond their monthly limits (e.g., Stockwell, 2008). Thus, the use of applications requiring heavy data usage would naturally be expected to see a detrimental effect on engagement if learners do not have access to Wi-Fi.

The physical affordances will vary to some degree depending on the type of device being used, but there has been a shift towards a merging of devices to mobile phones and tablets (and hybrid “phablets”) as multifunctional tools rather than using different devices for different purposes. MP3 players, PDAs, web browsers, and so on are now functions included in modern smartphones and tablets, meaning multiple devices are no longer necessary. Additionally, most modern mobile devices have evolved to also allow for functions such as GPS, QR codes, cloud computing, access to business software (word processing apps, etc.), and various communication tools, all of which offer opportunities for teaching and learning. The issues raised here are discussed from the physical dimension only, as it is possible to consider different psychosocial and pedagogical views even with regards to the same physical characteristics, which are dealt with in the following subsections.

6.3 Psychosocial Issues

As the name suggests, the discussion in this section can be largely divided into two main parts – the psychological dimension and the social dimension. While there are features that are unique to each category, there is likely to be overlap between them. It is useful to see the different ways that people use technologies in different cultures and contexts, as there are also dangers in assuming that people from different backgrounds will have the same expectations about the way

in which technology is used to accomplish various goals. In the context of this section, the psychological dimension refers to the view of mobile devices for learning, first of all in more general terms, but also more specifically with regards to the different functions of mobile devices for learning purposes. Earlier research has shown learners see the potential for learning devices, but when presented with other options, they are likely to choose a computer over a mobile phone, mainly as a result of the view of a mobile phone as a more private device rather than as a learning tool (Lai & Zheng, 2018; Stockwell, 2008, 2010). Comments from some of the participants in Lai and Zheng's study made their standpoint regarding technology clear, such as "I usually use the laptop for study and the mobile for entertainment" and "I would say, on the laptop I'm motivated to learn, but on the mobile, I'm just killing time" (p. 309). These show learners' distinction between mobile phones and laptops with regards to their usefulness for learning.

While saying this, the view of mobile learning is a fluid one, and there is ongoing research that suggests a shift of opinions about learning through mobile devices. There would be little doubt that there has been a significant decrease in the resistance towards using mobile technologies for work or educational purposes in recent years, but the view of mobile devices as a serious learning device is still somewhat limited for many. Learners have shown a greater willingness to use their mobile devices than in the past, but effective usage is dependent upon having sufficient training to know how to use them specifically for learning purposes, as described in the following section.

Psychological aspects of using mobile technologies can also include the psychological impact of the learning environment itself. The location and a context in which mobile devices are used will likely impact the way in which users feel about engaging in learning through them. For example, if a learner is using a mobile device in public transportation such as a train or bus, there are likely to be various distractions that could affect their state of mind. The time taken for a learner to redirect their attention back to the task after a distraction – such as looking up to check a station or moving out of the way for others to get on or off a train or bus – is also an important factor. While this is also partially related to the pedagogical factors (see Section 6.4), it is also a psychological one, in that it may affect some learners more than others depending on their individual dispositions. The study of attention has been the focus of a number of theories over the past several decades, with concepts such as the experience of *attentional involvement* (Abuhamdeh & Csikszentmihalyi, 2012) and *mindfulness* (Rechtschaffen, 2014) that seek to understand what is happening

when attention is focused intensely on the moment. They allude to the fact that people can get “lost” in the moment of a task they are engaged in if there is sufficient intrinsic motivation, and outside distractions are rarely noticed.

Distractions are not only limited to those in the surrounding environment but also may occur as a direct result of the device itself. Given that most, if not all, mobile devices have constant Internet access, it means that they are constantly checking servers for new information such as a new email, a message to a service such as WhatsApp or Messenger, or a post to a social networking service such as Facebook or Twitter. If new information is detected, this will typically result in a push notification being delivered to the phone. It is this push notification that has the potential to be a distraction to the learner, who may have their attention temporarily sidetracked by the notification even if they decide to attend to it later. When learners direct their attention away from what they are doing to deal with other things, it means that they need to take time to redirect their focus back to the task at hand. It has been estimated that this loss of attention can result in quite severe losses of time, possibly even as much as 31.1 percent (Galluch, Long, Bratton, Gee & Groeber, 2009). While push notifications may also have legitimate pedagogical functions (see the section on pedagogical issues in Section 6.4 and Chapter 8), the potential to interrupt learner engagement also needs to be factored into considerations regarding MALL. Firstly, if an individual task is short enough and, to some degree, complete within itself, then it is possible that learners will finish the task at hand before checking the message. Furthermore, if learners are aware of the fact that taking time out of their learning to check notifications has a potential detrimental effect on the learning outcomes, then it may be possible to discourage them from constantly checking these notifications. If they realise that it is difficult to control themselves regarding checking messages, they may temporarily disable the notifications so that they can use their time more effectively without disruption. One problem with this is, however, that multiple studies (e.g., Roberts, Yaya & Manolis, 2014) have shown that there are many people who suffer anxiety if they do not access their devices constantly; in fact, the prevalence of this is so widespread the medical term *nomophobia* (i.e., no mobile phone phobia) has been coined to describe it. When users have their mobile devices in their hands, the temptation to check messages through push notifications or to even pre-empt these notifications by checking individual social media may be too great to prevent them from being constantly distracted as they use their devices.

While not directly related to the physical characteristics of the device, there are psychological considerations that emerge as a direct

consequence of the device's physical features, and some of them may also have sociocultural implications. Users may feel less comfortable, for example, about using a larger mobile device such as a laptop computer on busy public transportation for fear interrupting people around them. They may also be aware that others can see the screen of their mobile device in public places and feel uncomfortable that others may look over their shoulders to see what it is that they are studying. This might be enough to discourage a learner from using the device, even though they have access to learning materials and the time and motivation to use them.

The psychosocial aspects can also have an impact on the classroom. It is now considered quite normal for learners to bring mobile devices into class in many contexts – most broadly at tertiary level, although they are also seen at lower levels as well, depending on the policies at a national, regional, or local level. As is described in Chapter 2, teacher attitudes towards the use of mobile devices in their classrooms are extremely varied (Berger, 2017), ranging from general acceptance through to strong resistance, even to the extent of banning technologies in the classroom altogether (e.g., Fried, 2008; Yamamoto, 2007). One of the largest concerns expressed by teachers regarding the use of mobile devices in the classroom is that learners are distracted by the device and unable to concentrate on class tasks and activities. A large proportion of learners seem to hold on to the belief that they are quite capable of successfully multitasking in class and are able to suitably pay attention to the class with little negative impact from using their technologies. Even if they are aware of the potential harmful effects of multitasking, some learners still do engage in multitasking. The results of research have also provided a rather bleak outlook on the ability to learners to process the information which is presented to them if they are engaged in other tasks, such as texting, web browsing, online shopping, and social media, and even notetaking on computer. Ophir, Nass and Wagner (2009), for example, conducted a study to identify the effects of multitasking of university students who indicated that they concurrently used multiple media. The result of the study revealed that the vast majority of students were unable to filter out the distractions effectively, clearly indicating the simultaneous use of media had a detrimental effect on learners' abilities to maintain their concentration on the task at hand. They showed that heavy media multitaskers were more likely to exhibit negative effects as a result of engaging in tasks not directly related to the primary task, suggesting that the more that learners engaged in off-task activities, the lower their ability to direct sufficient attention towards it. Tassone, Liu, Reed and Vickers (2017) found that learners who engaged less in

multitasking in class were able to attain a higher GPA (grade point average), and even though the majority of learners in the study knew that multitasking could result in decreased grades, they still regularly engaged in it.

Multitasking does not only have to be limited to off-task activities, and learners can also suffer from trying to complete multiple learning tasks or activities at the same time. Contrary to the perspective proposed by Prensky (2001) that college students of the digital generation would make the most of digital resources for their learning, Flanigan and Kiewra (2018) have argued these resources can, in fact, have a debilitating effect. This is largely a result of the fact that multitasking is actually the switching of attention between tasks rather than actually engaging in multiple tasks at the same time (Rosen, 2012), meaning that there are losses that occur as learners switch between these tasks. Newell (2017) describes three primary efficiency costs that the so-called digital generation may suffer from when they attempt to engage in multitasking. The first of these is “switch costs,” which refers to the time required to “switch” their mindset to the goals of the new task. Secondly, there are “resumption lags,” meaning that it takes time to return to the original task after engaging in the secondary task. Thirdly, they may also suffer from a “restart cost,” where there is reduced efficiency in the original task as they attempt to see where they left off and to take stock of what they need to do next. Learners select what they wish to direct their attention towards and then shift it to something else in an ongoing process of *selective attention shifting*. Their attention is diverted away from the primary task so they can do other things for some period of time before returning to the original task again. After returning back to the original task, it will take time for the learner to backtrack over what they’ve done and to recommence what they were doing.

In the classroom, the most prevalent form of multitasking is cyberslacking, and despite the obvious negative effect on the ability to concentrate on class activities, the trend of engaging in extraneous activities during class time by students appears to be gradually increasing. Respondents in Lauricella and Kay’s (2010) study of more than 500 students at a university in Canada indicated a large amount of private use of portable devices during class time. Almost a third of the students stated that they spent more than half of class time engaged in instant messaging, sending non-academic emails, playing games, or even watching movies. McCoy (2016) found that American college students admitted to spending more than one-fifth of class time using a digital device for non-class-related purposes. Of the reasons cited by students for this behaviour, nearly two-thirds of students claimed that

they wanted to stay connected, and two-thirds said they use their digital devices to fight boredom. Nearly 90 percent of students were aware of the fact that they were not paying attention if they used the digital devices in class, yet nearly 60 percent of respondents indicated that they felt that their devices were not much of a distraction. At first glance, it might appear that these two results are conflicting, but this is not necessarily the case.

Why is it that learners engage in this behaviour despite being aware of the potential dangers? A possible explanation for learners engaging in non-task-related uses of mobile technologies in class could be related to their addictive nature. There has been quite a lot written in the past few years about the potential addictive effect of mobile technologies and the impact that this addition can have on people's lives. It has been suggested that people spend an average of three hours online every single day, with a quarter of users spending between two to three hours looking at their smartphone screens every day (Alter, 2017), checking and sending text messages, or using Facebook or other social networks (Rosen, 2012). Addiction to the Internet has been a much-discussed aspect of user behaviour since the early days of the Internet in terms of *cue reactivity* and *craving* (Brand, 2017), but the rise of social media in the early 2000s seems to have further exacerbated this problem, with addictive reactions to cues such as the visual or audio notifications of incoming messages. Given the fact that now most people have access to their social networks through a combination of desktop and mobile technologies at any time, the issue of addiction to social networking services such as Facebook is further compounded (Wegmann, Stodt & Brand, 2018). This addition extends not only to the cutting down of sleeping hours, but now it is commonplace for there to be a general lack of concentration even when users need to keep their attention directed on what they are doing (see the discussion in Chapter 5 on *dual systems theory*).

Raising learner awareness towards the dangers of multitasking, however, is proving to be a difficult undertaking. Learners in study by Tassone, Liu, Reed and Vickers (2017) were asked to provide details of their multitasking in class before and after being subjected to a presentation about the detrimental effects of multitasking. The results showed that there was no significant difference between the learners who viewed the presentation and those who did not. This indicates that learners are not multitasking because they are unaware of the dangers, but, rather, they make a conscious decision about dividing their attention between the multiple tasks that they are engaged in and are willing to sacrifice some elements of attention that might be used for learning purposes for their own private purposes.

It would seem, then, that if multitasking is something that cannot be avoided, then perhaps there are ways of taking advantage of what we know about learners' tendency to use their mobile devices for multitasking to make better use of their time and attention, which is being split between the requirements of the classroom and the use of their devices. Given that mobile devices have very much become the norm in many classrooms, as Verbeek (2013) points out, the most important consideration now is the "how" that exists between the "yes" and "no" with regards to allowing learners to use their mobile devices in class. In other words, if mobile devices are to be present in the classroom, how can learners' attention be kept focussed on where it will benefit them the most? The issue of attention has featured in psychology discussions for decades, and the most basic assumptions underlying attention are that it is limited, it is selective, it is subject to voluntary control, it controls access to consciousness, it is essential for the control of action, and, of most importance for education, it is essential for learning (Schmidt, 2009). As mentioned in Chapter 2, this has wide-reaching implications for the ways in which classroom teaching could or should be conducted. It would appear that people today interact with information differently from how they did in the past, switching rapidly between information sources at a shallow level and not taking the time to read thoroughly (Carr, 2011). At the same time, the addiction-like thirst for new information that is available through newsfeeds and social media also keeps people fixated on their devices even in circumstances and environments that are inappropriate or even detrimental.

There are those however that argue that addiction to technology can be desirable, particularly from a business perspective (Alter, 2017). This addition has been cited as a rationale for the use of educational gaming, where if learners are suitably addicted to game, then it is possible for them to learn – perhaps subconsciously – as they engage in it for extended periods of time. This is a complex issue that does have the potential to keep learners engaged in activities for longer, but the complexities, such as the balance between educational and entertainment aspects, and the nature of tasks and activities that can successfully be gamified should not be forgotten.

The social dimension looks at issues such as the digital divide, an issue which can be exacerbated by the need placed on many learners to provide their own devices, and it is not uncommon to see gaps between those with more sophisticated devices and those that have devices on the lower end of the spectrum. This has implications at both the individual level and the societal level, where access to the Internet through mobile devices may lead to a certain view of mobile learning

that differs from regions where this access is easier to obtain. Further, in a society where learners engage in private social uses on their mobile devices, there are often demands placed on learners to use their social networking tools for learning as well, which can have varying attitudinal and even ethical concerns (Blyth, 2015). This can also relate to the expectations for teachers to become more accessible to students through their own provide social networking links that may infringe upon their privacy.

There are also dangers in assuming that people from different cultural backgrounds will have the same expectations about the way in which technology is used to accomplish various goals. This has been particularly evident in behaviours associated with autonomous behaviour (Schwienhorst, 2003) and self-directed use of technology (Lai, Wang, Li & Hu, 2016), but this is to be expected to a certain degree, given the influence of learners' culture on their views of autonomy in language learning (Palfreyman, 2003). The position of technology within different societies can also shape their opinions of it, ranging from commonplace to exclusive, and this is often a reflection of the costs and accessibility of hardware, software, and Internet connectivity. If teachers are in environments where they have differing cultural perspectives from their learners, there is a very real possibility that this could lead to a mismatch in views of learning through technology for both in-class and out-of-class purposes (Chik & Breidbach, 2014). For example, teachers who come from backgrounds where technology is a normal part of the classroom may find resistance from learners where mobile devices are viewed more as personal tools. Finally, developing critical digital literacies (see Chapter 5) can also play a crucial role in alleviating cultural stereotypes (Pegrum, Dudeney & Hockly, 2018) and is an important part of understanding of sociocultural differences for both teachers and learners alike.

6.4 Pedagogical Issues

There are several potential ways to view technologies for learning purposes (see Chun, Kern & Smith, 2016), but to look at mobile device as a separate entity from more traditional technologies, this section will look specifically at those points that are related to the mobility, ubiquity, interactivity, and augmentation that mobile devices make possible. Here the discussion will move from the physical aspects as to what the implications are for second-language teaching and learning.

One of the key issues associated with mobile learning that should be kept in mind is that it is important to consider the affordances of

mobile devices and to design tasks and activities that can take advantage of these affordances. Despite arguments of some researchers that mobile learning should not just be a replication of computer-based or paper-based tasks, it is possible to argue that there is indeed a time and place for such replications, provided that these uses make a part of a larger picture of learning through mobile devices. For example, having access to static information that one might find on a piece of paper or a PDF file from a mobile device means that this is available in locations where one may not carry paper documents or computers as a result. Although the required resource may be rather “analogue,” providing access to it through a mobile device in lieu of other options is, in one sense, taking advantage of the affordances of a mobile device. The ability to use a mobile device to search for information that may be required at unexpected times is another way that mobile devices can be used in ways that may have typically been associated with more traditional means, such as looking up a term in the dictionary or performing an Internet search. The fact that modern mobile devices can be used to store a large amount of information, to provide instant access to resources online suggests that they can take on a reference role, or even to communicate with others in the target language directly or indirectly assists learners in completing required tasks or activities.

Mobility and ubiquity will certainly include this ability to have constant access to information and to be able to use the mobile device as an ongoing reference tool, where learners are able to search for information as required to serve a particular learning need (cf., Dikkers, 2014), as well as the potential of using mobile devices to have assistance in learning within the contexts in which they need to function using the target language in situated learning contexts (see Chapter 5). One of the strengths that has been widely cited regarding mobile learning is that it takes learning out of the classroom and moves it into real contexts where learners can have access to information or resources when they need them. Kukulska-Hulme (2015), for example, has investigated migrant learners in Europe with their use of an app designed to help them to cope with daily life situations in the target language.

As was alluded to in Chapter 1, an increasingly common use of technology outside of the classroom is the completion of some kind of task or activity that the teacher can track progress of remotely, meaning that there is no particular need for learners to actually submit anything to the teacher, but, rather, the teacher can see if learners have engaged in the activities by checking usage logs, such as in Moodle, Newsela, or Quizlet. At first glance, this seems to be an extremely

efficient way of determining (and encouraging) learner engagement, as the teacher can see quite clearly whether the learners have completed the required tasks and activities. This often takes place in class “sets,” meaning that the completion rates are presented to the teacher in a way that can be imported into an LMS or other grading system without the need to go through each of the learner records individually. At a time when teachers are constantly faced with time pressures, this appears to be (and in some ways actually is) an extremely time-efficient manner of carrying out homework, as it reduces the immediate need to have students submit paper-based records that can easily be lost or confused.

The primary problem is, however, the ways in which these are actually completed by the learners. The records provided by many systems may be limited to just determining if something was completed, or the teacher may not place emphasis on scores but instead just encourage learners to simply complete them so as to not pressure them too much. This is an understandable approach, but it is often based on the assumption that learners are at the very least putting some effort into the completion of the activities. As Fischer (2012) found, however, tracking learner behaviour revealed a very different outcome from what was anticipated and expected by the teacher. Tracking software was installed in computer laboratories where learners of French undertook multiple-choice activities in class. The results revealed that many of the learners were completing multiple choice by simply clicking the answers in order, often times without taking the time to read the texts that were provided for them. The ultimate outcome for learners was not to learn from the activities but to simply finish them. There was little or no connection in the minds of these learners with actually learning from the activities; they believed that the tasks were done satisfactorily if they merely completed them, regardless of the manner in which this was done.

Obviously, the reasons behind teachers’ choices to use technology for teaching and learning can also be highly complex. Not the least of these reasons are the internal and external motivations behind technology, which can heavily impact how it is used and the longevity of its use (Stockwell, 2013a). While, of course, there are many teachers who see technology as a means of enhancing the teaching and learning environment, it is not uncommon for teachers to adopt mobile technologies to keep up with trends (Hsieh & Tsai, 2017). In contrast, teacher attitudes also mirror some learner attitudes, particularly with regards to not seeing MALL as serious learning (Tai & Ting, 2011). The attitudes of other players such as parents are also important in how teachers apply technology, and the decision by teachers to avoid

learning through mobile devices may be based on perceived negative views of parents (Colpaert, 2010). Thus, the decisions about whether to use mobile technologies can be influenced by a range of real and perceived factors that may ultimately provide or deprive learners of learning opportunities. Teachers need to understand the context but also take care not to be overly biased by perceptions without confirming their real impact.

6.5 Summary

The complexity of the physical, psychosocial, and pedagogical issues associated with MALL is decidedly difficult to summarise, particularly given their mutual dependency. The physical limitations of the screen size and input methods are also the facilitating features that makes mobile devices mobile, but in one sense, many of the physical features are gradually becoming less of a problem. Technological advances are contributing to a drop in the impact of aspects such as storage capacity, processor speed, battery life, and network access, but compatibility remains a problem between different operating systems. More difficult to deal with, however, are the psychosocial and pedagogical issues, which are largely related to views of mobile technologies in social contexts and how teachers frame MALL activities, often based on these social contexts.

One ongoing concern with mobile devices is the potential for distractions, which may occur both inside and outside the device. The distractions inside the device refer to notifications or other information that pops up on the screen which may interrupt learners from what they are doing with their device, either by causing them to cease what they are doing to deal with the notification or by interfering with their concentration on the task that they are currently engaged in, even if they do not stop what they are doing. A second internal distraction of mobile devices is their potential to interrupt with learning opportunities around the learner, where learners fail to take advantage of these learning opportunities, such as texting or other activities that they attempt to multitask as they listen to a lecture. The addictive nature of the constant access to mobile devices can result in reduced productivity and ultimately have a negative impact on learning. Distractions which take place outside the device are dependent upon the location in which learners decide to engage in activities, and in-transit locations will be far more likely to be fraught with distractions than fixed ones. How to prepare for environments with distractions is largely the responsibility of the teacher through providing activities and tasks that can be completed in small chunks where continuity is

preserved even if there is an interruption. Knowing how to deal with these distractions as they occur, however, is very much the responsibility of the learner, and prevention can be as simple as planning for appropriate locations in which to engage in mobile learning activities.

Making the most of the physical affordances of mobile technologies is one of the key factors in maximising learning opportunities, but equally important is ensuring that learners are aware of what they should be doing and why, which is described more in depth in the following chapter. If the primary objective of engaging in activities is framed as simply completing them, it is likely that the process will not be given adequate attention by learners, which may detract from potential learning opportunities as well. Finally, teacher views and learner views of mobile technologies may be misaligned, particularly when teachers believe that learners are more enthusiastic about using their mobile devices for learning than they really are. As will be discussed in Chapter 7, sound pedagogical practice is based on teachers' and learners' mutual understanding of what is expected of the learners, which can only be achieved through guidance and support. The interplay between the physical, psychosocial, and pedagogical factors results in a need for teachers to find the right balance in their specific context, understanding the learners and the available technologies, and designing the correct pedagogy for achieving learning goals based on this understanding.

6.6 Discussion Questions

1. If you could make one change to the physical characteristics of a mobile device that would make it more suitable for language learning, what would it be? Explain your answer.
2. Imagine that you are a teacher giving a lecture, and you have several students in class who are looking at their mobile devices and do not seem to be concentrating. What do you do? Why?
3. How would you respond to arguments that mobile phones will completely change language teaching pedagogy? If you agree, why? If not, why not.

7 *The Learner in MALL*

7.1 Introduction

The key point of any educational context is to always keep in mind who the learners are. The design of a course, lesson, teaching resource, task, or activity will necessarily depend on who will use it. As has been alluded to at several points throughout this book, learners will come into learning situations with extremely diverse backgrounds in terms of their skills, preferences, goals, and motivation. This is equally if not more important when designing language learning contexts that use technology. What experience do the learners have with using technology for language learning? What is their perception of these technologies? How do they view their preparedness for learning with technology? What skills do they possess from either learning or non-learning situations that might be of assistance to them in learning? How capable are the learners of engaging in learning activities without constant supervision or support? These are some of the fundamental questions that teachers and designers need answers to in order to optimise the teaching and learning environment for their learners, regardless of whether learning takes place predominantly face to face, at a distance, or via self-access.

The previous chapter looked at the physical, psychosocial, and pedagogical aspects of mobile learning, and the results of previous studies have suggested that the pedagogical aspects are perhaps the most influential in determining success in the use of mobile devices. Of these, it appears that learner training is key in ensuring that learners understand the reasons for engaging in mobile activities as well as what is expected of them from teachers. There is evidence that suggests providing ongoing training in technological, strategic, and pedagogical aspects can have a very powerful effect on the ways in which learners view mobile learning activities and how they engage in them (Stockwell & Hubbard, 2014). Much of the training that is currently

described in the literature indicates that the focus is primarily on the technical aspects and that this training largely takes place only at the beginning of a course or project. The chapter argues for ongoing training that guides learners to develop autonomy through evaluation of the strategies they employ and sharing their learning experiences with others to help them reflect on their learning.

7.2 Identifying the Learner in MALL

Any discussion pertaining to learning through mobile devices must have at its core a clear image of who the learners are. If MALL is to meet the needs of the learners who will be engaging in it, designers need to understand as much as possible about the backgrounds, skills, goals, and preferences of the learners (see Levy & Stockwell, 2006). It is logistically difficult, if not impossible, to design a one-size-fits all MALL environment, and the stakes for engaging in mobile-based learning activities are high, as the design of the activities will have an enormous impact on when, how, and how long learners will spend using them. For example, learners who only engage in mobile learning when it is a recommended or a required part of a course will likely have very different needs from learners that spontaneously weave self-initiated learning through mobile devices into their everyday schedules. One of the main features that might be considered as distinguishing the learners in these environments is the degree of *learner agency*. Based on social cognitive theory (Bandura, 1986, 1991), agency is defined as the “capability of individual human beings to make choices and to act on these choices in ways that make a difference in their lives” (Martin, 2004, p. 135). Bandura (2001, p. 1) suggests that this agency – human agency – is manifested in three modes: *direct personal agency*, *proxy agency*, and *collective agency*. Direct personal agency refers to an individual having direct control over his or her own life, proxy agency relies on others to act on a person’s behalf to achieve desired outcomes, and collective agency is the product of socially coordinative and interdependent effort. The parallels between the human agency that Bandura refers to and learner agency are immediately obvious. Learner agency is defined by Dörnyei and Kubanyiova as “learners’ proactive investment in the learning process” (2014, p. 35) and is included in related concepts such as learner autonomy, learner identity, action-based teaching, self-efficacy, and self-regulatory learning. Learners who possess direct personal agency can take responsibility for their own learning and seek ways of doing this without relying on others, whereas learners relying on proxy agency have a dependency on others to assist them in

achieving their learning goals. Collective agency considers the larger environment, and in educational situations, it may include the general sociocultural context that brings about change in learner behaviour and learning outcomes.

It is not difficult to see how each of these learners may behave in contexts where mobile devices are used. Learners who spontaneously look for ways to improve upon their language skills, such as seeking out language learning apps, resources, social groups, and so forth, would be considered as having direct personal agency. These learners possess a high degree of self-regulation, and in many cases, they also have clear goals for what they want to achieve as a part of their learning. As described later in the chapter, they are largely *autonomous* – that is, they have both the motivation and skills required, or as Woolfolk, Winne, and Perry put it, they have “the *skill* and the *will* to learn” (2000, p. 384, cited in Martin, 2004, p. 135). A study by Demouy, Jones, Kan, Kukulska-Hulme and Eardley (2015) describes distance learners that would likely fit into this category. They were enrolled in language courses but actively sought out extra resources on their mobile devices that would help them with their studies. Subjects indicated that they used their mobile phones for other daily functions, so it seemed to be a natural extension for them to apply them to their language learning as well.

Perhaps the most commonly researched learners are those who are enrolled in formal language courses, but they receive support from others (usually the teacher) to guide them to use their mobile devices in a way that can benefit their studies. These learners would be considered as having proxy agency, in that many are largely unable to achieve learning goals without this support. The ways in which teachers can support learners with learning through mobile devices are varied. The importance of the provision of sufficient training has gained considerable attention over the past decade, prompted largely by work by Hubbard (2004). This will be described in more depth in the following section, but it is argued that learners will rarely, if ever, have the skills they need to engage in learning activities using technology without appropriate training (Romeo & Hubbard, 2010; Stockwell & Hubbard, 2013). There have been some examples of the positive effects of training on learning outcomes, such as Kondo et al.’s (2012) study of Japanese learners of English receiving in-class training with Nintendo DS mobile devices, resulting in improved TOEIC scores on listening and reading. In their investigation of learners of French as a foreign language using Duolingo on their mobile phones, García Botero, Botero Restrepo, Chang and Questier (2019) found that learners were unlikely

to engage in Duolingo tasks if they were just invited to do so, and that learners rarely used the built-in self-regulation function of the app of their own accord, but that engagement increased if they received training and scaffolding, resulting in increases in test scores of French writing. Another way is to support learners' agency with mobile devices is to open channels of communication between the teacher and the learners. Tran (2018), for example, connected with her students outside of class using a mobile-based messaging system called Line, giving them tips for when and what to study, assisting with technical difficulties in using online resources, and prompting informal discussions about what to do and see around the city.

The final mode of agency raised by Bandura (2001) is collective agency. Learners might experience this type of agency when the environment has been shaped for learning. One example of this the MASELTOV (Mobile Assistance for Social Inclusion and Empowerment of Immigrants with Persuasive Learning Technologies and Social Network Services) project, which was designed to provide language learning and cultural support for new migrants through their mobile phones (Gaved, Jones, Kukulska-Hulme & Scanlon, 2012; Kukulska-Hulme, Gaved, Paletta, Scanlon, Jones & Brasher, 2015). The project was multifaceted, involving a geo-social radar to find nearby volunteers using the GPS function, a tool for converting text in photographs into another language called TextLens, peer-reviewed language learning activities, a mobile navigation tool, an ongoing profile of the learners needs, and social networking and game-based learning functions. Through this support system, learners were able to have access to learning that would not normally be available to them, with the infrastructure providing the collective agency required to learn how to function within a new environment.

Having a complete understanding of who the learners are in MALL is complex, and the discussion here is intended to be illustrative rather than exhaustive. Learner agency – and related concepts such as self-regulation, autonomy, and self-efficacy – can manifest itself in different ways in different learners at different points throughout their lives (see Mercer, 2011). Agency is, of course, just one part of the overall network of factors that lead to when, how, and why learners will take the initiative for their own learning, with motivation and their belief in their own capabilities also playing important roles (Gao, 2010). It is highly likely that at some stage, learners will likely need assistance to learn how to use the available tools for their language studies, although the results of this training may not be evident until sometime in the future. As Hubbard (2004) points out, most learners do not possess sufficient skills to make the most of the resources that they

have access to, and this appears to be particularly evident for those in formal language study. Training is discussed in more detail in the following section.

7.3 Rationale for Training

Learner training has attracted a great deal of attention in second-language teaching and learning through technology over the past several years, largely prompted by the seminal paper by Hubbard (2004), who pointed out the potential of learning though technology was largely limited by the fact that learners were not provided with sufficient training in using technology for learning. Learner preparedness with regards to technology has been a key issue at the root of the push for learner training, with studies such as Barrette (2001) and Winke and Goertler (2008) concluding that learners simply lacked the skills to use the technological resources that were available. Studies have revealed similar trends with regards to mobile learning, with Stockwell (2008) finding that learners did not perceive their mobile phones as learning tools despite the fact that they had access to learning materials. It is not surprising that learner training has gained a great deal of support as a means of preparing learners for effective use of technology (Lai & Morrison, 2013), in light of findings that learners often failed to complete activities as teachers expected and simply went through the motions of doing assigned activities without a clear understanding of what was required of them (Fischer, 2012). As has been mentioned at several points throughout this book, teachers frequently hold exaggerated views of their learners' ability to use technology for learning, yet many of the technological tools that are provided are simply not used in the ways that teachers or designers planned (Karlström & Lundin, 2013).

Hubbard (2013) makes a strong case for the need for learner training with technology through arguing that there are four major misconceptions that lead to beliefs that training is not necessary. The first of these is that if technology or technology-based tasks are designed properly, then learners will automatically be successful. While at face value this statement seems logical, there is substantial evidence that good design should include a training element within the technology but that this is rarely present. The second misconception is that learners will eventually "gravitate to the most effective uses" (p. 164). He refutes this from both technological and non-technological research, showing that learners frequently fail to use techniques that have been shown to optimise learning. Thirdly, many believe that the current generation of so-called digital natives already possesses strong skills with technology and

therefore does not require training. As Stockwell and Hubbard (2013) and Kurek and Hauck (2014) argue, there is little evidence – if not counter-evidence – for this, and although they may be “net savvy” for certain purposes, these rarely correspond to learning purposes. Finally, many believe that normalisation as described by Bax (2003) has been achieved to the point that technology is now a “normal” part of our daily lives, and therefore, specialised training is not necessary. Hubbard argues, however, that only a very small number of established technologies have reached a stage of what might be referred to as normalised, and as such, there is still very much a need for training with the majority of technological tools, with new ones appearing at a tremendous rate.

As the foundation of his model of training, Hubbard (2004) outlines five principles. The first of these is that teachers will better understand their learners if they experience CALL as learners themselves. Secondly, learners should be given teacher training, that is – they should be given information about the underlying principles and models of learning that teachers base their selection of activities on to help guide learners to take responsibility for their own learning. The third principle is that training should be provided cyclically and in small chunks rather than in longer sessions. As learners’ understanding of their learning changes, the same strategies and suggestions will hold a different meaning to them, so recapping points is helpful to consolidate them. Fourthly, learners will benefit from collaborative debriefings during which they can have a chance to talk to one another about their experiences and reflect on their learning behaviours. Lastly, Hubbard argues that strategies should be provided for learners that go beyond the intended uses of a technology by the developers so that they can use them to help achieve their language goals. These principles have formed the basis for a model of training that takes learners through different phases as they familiarise themselves with the technologies and strategies for using these technologies. This model will be described in what follows with regards to its applicability to MALL.

7.4 Training, Sustained Task Engagement and Autonomy

Given that the acquisition of a second language has been inextricably linked with sufficient exposure to the target language, it is not surprising that sustained task engagement in language learning has long been an issue of concern to teachers and researchers, who are constantly looking for ways to motivate learners to spend more time on language learning tasks and activities (e.g., Busse & Walter, 2013; Deci & Ryan, 1985; Dörnyei, Muir & Ibrahim, 2014; McGroarty, 2001; Ushioda, 1998). The issue of motivation is covered in more detail in

Chapter 5, but it is evident that sustained task engagement and motivation are intimately linked, as is learners having sufficient skills in order to complete tasks required of them. It could be argued that there is often confusion regarding sustained task engagement and autonomy, where learners are assumed to be autonomous if they engage in activities on an extended bases outside of class time. This confusion is in part because learner autonomy is a largely misunderstood concept, and interpretations of autonomy vary. Some take a broader view, such as Holec (1981), who defines autonomy as “the capacity to take charge of one’s learning” (p. 16), while others provide far more detail, such as Murray (2014), who operationalises it as, “learners taking on the responsibility for goal-setting, material selection, activity and strategy implementation, progress monitoring and outcomes assessment” (p. 5). Both definitions give us an idea of what autonomy *is* – that is, the learner being at the centre of his or her own learning – and they also provide some hints of what it is *not*. Autonomy is obviously not having learners simply engage in online tasks outside of class time following the directions of the teacher, even if they do so actively for extended periods of time. However, the concern is that this kind of sustained task engagement is often mistaken for autonomy, primarily because learners appear to exhibit many of the behaviours that are generally associated with autonomy, such as (short-term) goal setting, some degree of material selection, strategy implementation, and progress monitoring.

There are numerous studies that observe these behaviours and make claims that learners have developed autonomy based on these observations, even though there is typically very little in the way of follow-up studies to see whether or not learners continue to engage in learning activities after the period of investigation is over. One of the main points that separates this behaviour from autonomy is that it is still dependent on external factors, such as completing course obligations or preparing for an exam, and engagement often ceases as soon as the pressure has been removed. It would be difficult to claim that this is indeed autonomy in the sense that many teachers – sometimes rather tacitly – are hoping their learners will achieve. Furthermore, autonomy should not be treated as a constant across cultures, where perspectives may vary depending on cultural norms (Holliday, 2003), and behaviours may be interpreted differently based on learners’ cultural backgrounds. In some cultures, learners are conditioned to closely follow teacher instructions, and going beyond defined boundaries may be seen as inappropriate, meaning that the ways in which they engage in tasks may be impacted by their cultural norms.

Task engagement itself is a complex concept, as is evidenced by Philp and Duchesne (2016), who define engagement as consisting of four interrelated dimensions: cognitive engagement, behavioural engagement, emotional engagement, and social engagement. Cognitive engagement refers to the mental effort or sustained attention in completing a task. If a learner fails to engage in a task cognitively, the expected benefits of completing it would be considered to be minimal. Behavioural engagement is the amount of actual time that is spent on completing a task and is likely what is broadly held in mind when referring to task engagement. The amount of time spent on a task is important, but without cognitive engagement, this could well be time that is not used effectively. Emotional engagement has multiple interpretations, but it may be considered as including the degree of enthusiasm, interest, and enjoyment (ranging from negative to positive). They argue that it also can refer to purposefulness, autonomy, and the feeling of connectivity with peers. A higher degree of emotional engagement can link to more time on task and, hence, be associated with sustaining the time spent on the task. The last point, social engagement, is linked to emotional engagement, but focusses more on the collaboration between learners. This social dimension may be nested in a dynamic social constructivist model, where “the learner(s), the teacher, the task and the context interact with and affect each other” (Williams & Burden, 1997, p. 46). There is evidence to suggest that learners both seek out and benefit from being a part of a community of learners (Murphy, 2014), and this can directly impact upon their attitude and engagement in tasks as a result of expectations they place on themselves compared with their peers (Yashima, 2014). Meaningful sustained task engagement, then, is a product of these dimensions. Emotional and social aspects can have a facilitating effect to keep learners spending time on task, but it is not just the amount of time but rather the quality of the time spent to ensure there is adequate cognitive stimulation as well.

The confusion with autonomy, however, stems from expectations that once a learner can engage in given tasks on a sustained basis – even without prompting from the teacher – they have become autonomous. Inferring autonomy from isolated behaviour with one type of learning is somewhat of a jump that is difficult to confirm. At best, this could be referred to as *task-specific* autonomy and it may not necessarily be a reflection of *global* autonomy. On the one hand, learners with task-specific autonomy are capable of functioning within the scope that task, including performing some of the functions that Murray associates with autonomy such as goal setting and material selection. On the other hand, global autonomy is closely associated

with agency. While, of course, task-specific sustained engagement is still desirable, the aim of many teachers would be to develop their learners' global autonomy, where they can seek out new learning opportunities that are not dependent upon the teacher or specific materials or tools. One problem is that although task engagement can be observed (e.g., Stockwell, 2010, 2019), measurement of global autonomy is elusive. Determining whether learners are autonomous requires longitudinal investigation in naturalist contexts, and while surveys (e.g., Vanijdee, 2003) and short-term observation (e.g., Mideros & Carter, 2014) can provide insights into task-specific autonomy, the relationship with agency is difficult to determine through these methods. This is not to say that knowledge of learners' ability to engage in tasks in a sustained manner is not valuable information for teachers, and sustained engagement that takes place without direct teacher intervention would be a forerunner to autonomy. Through sustained engagement, learners start to develop behaviours that can eventually lead them to realise that learning a language is something that takes time. This does not necessarily mean that sustained task engagement will necessarily lead to autonomy, but it is likely one of the conditions necessary to achieve it.

It is very easy to have unrealistic expectations about learners' abilities to carry out assigned tasks. Particularly at higher levels of education, teachers will often expect that learners have already developed their own learning styles and repertoire of strategies that they are competent in applying to the completion of the tasks they are assigned, and that failure to do so is largely associated with a lack of the second condition for autonomy – motivation. There are, of course, cases where motivation is the underlying cause for a lack of engagement in tasks, but it may also be the result of not possessing the required skills to carry out the activities in a way that the learners may see as beneficial. There must be sufficient time put into providing explanations about precisely how to undertake activities, bearing in mind that there will likely be a large variety of attitudes and skills possessed by learners. Once the learners have these skills, they need time on task to hone them, and this is where the interdependence between training, sustainability, and autonomy becomes apparent (see Smith, 2003, for a discussion). Cognitive engagement in tasks can be facilitated through training in specific strategies to support learning. While some learners may possess the underlying qualities and/or skills that allow them to make the most of potential learning opportunities with little or no assistance, the majority will likely require assistance over extended periods of time where the responsibility is gradually shifted from the teacher to the learners, and teacher guidance is slowly replaced with

more independent learner behaviour. Until learners have reached a point where they can act independently, emotional and social dimensions of task engagement can keep learners engaged in tasks for sustained periods of time (see Lewis, 2014, for a discussion of the role of the contribution of social aspects to the development of learner autonomy).

The role of technology to contribute to learner autonomy or long-term task engagement has also been somewhat overrated (Reinders & White, 2010), and this has also been the case with MALL (e.g., Kim & Kwon, 2012; Hsieh & Tsai, 2017). Technology in and of itself cannot lead to increased motivation in the long term (see Stockwell, 2013a), and it is somewhat naïve to believe that just providing technology itself is going to lead to learner autonomy, unless there is sufficient, planned, and ongoing training that gradually shifts the responsibility for learning away from the teacher and towards the learners themselves. While there may be some short-term motivational benefits, there are decades of research which have shown that it is highly unlikely that learners will maintain their interest in continually carrying out activities after the novelty effect has worn off (Karlström & Lundin, 2013). As with any task or activity, learners will not continue to engage in tasks and activities of their own volition unless they can see a link between them and their language learning goals. In other words, mobile technologies are useful tools for learners who are already autonomous. Rather, responsibility needs to be gradually handed over to the learner from the teacher through training learners to identify learning goals in the larger learning context. They need to be aware of what they currently know and what they need to know, and they need to acquire the skills to achieve through evaluating potential sources of language input and opportunities for output applying appropriate strategies to maximise the benefits from a particular encounter. The ways in which learners will engage with technology will depend on the individual skills, preferences, and experiences learners have regarding the technology (Levy & Stockwell, 2006). This includes, of course, their own mobile devices, which they may have limited or skewed familiarity with.

Learner training has been shown to have an enormous impact on sustainment of engagement in mobile-based language learning tasks and the exhibition of some early signs of autonomous behaviour (Stockwell & Hubbard, 2014; Stockwell, 2019). However, despite the enthusiasm shown by the learners during the activities, most were either unable or unwilling to transfer those skills to other activities that differed from the ones that they were trained in, and many learners even exhibited a lack of motivation after the cycle of learner training was completed (Stockwell & Hubbard, 2018). This outcome supports

the argument made by Schwienhorst (2008), who proposes that even if autonomy is developed in one area, there is no guarantee that autonomy will develop in others. A potential reason for the learners' inability to complete other tasks autonomously in Stockwell and Hubbard's (2018) study could be that the learner training focussed predominantly on providing learners with the skills that they needed to use the materials that were assigned to them, but the learners either could not or were not willing to transfer these skills to other tasks. Herein lies one of the largest challenges of learner training to develop autonomy. On the one hand, the training needs to be specific enough that learners can see precisely what they need to do to achieve certain task outcomes, but on the other hand, it must include sufficiently broad elements that enable learners to apply their skills to alternative contexts that may or may not be predictable. A second – perhaps equally difficult – challenge facing teachers is raising learners' own awareness of the need for autonomy. As has been pointed out previously, autonomy and motivation are in some sense two wheels of the same cart (e.g., Dornyei & Ushioda, 2011), where one without the other is unlikely to lead to any sustained learning outcomes. Providing learners with the skills that they need to become autonomous without simultaneously nurturing their motivation to learn is akin to putting them into a cart with one missing wheel and simply hoping that somehow they will still be able to reach their destination with one axle dragging in the dust.

Recent developments in technology have seen attempts to develop software that simulates the role of the teacher (e.g., Lai, Shum & Tian, 2016). Technology that has been designed for learning purposes has an instructional design embedded within it, and this instructional design is intended to resemble the presence of a teacher in some shape or form through providing feedback, guidance, or encouragement to make the most of potential learning opportunities. Unless learners are gradually pushed into making decisions about what and how to capitalise upon these opportunities without assistance, it is questionable whether it can be claimed that they have truly been able to achieve autonomy. In MALL, technology aims to enable learners to take advantage of real-world situations through “just-in-time” learning, prompting them to use the target language authentically (e.g., Conole & Paredes, 2018). While there are indeed possibilities in this area, care still needs to be taken to ensure that the dependence on the presence of the teacher hasn't simply been replaced by dependence on technology, where there is no training to wean learners off this dependency.

Systematic methods of providing learner training have been rare, with one of the most comprehensive models being proposed by Romeo

and Hubbard (2010), founded on the original ideas of Hubbard's (2004) learner training framework. The model is integrated into a course such that learners are familiarised with the technologies that are to be used, strategies for using these technologies for language learning, and a foundational knowledge of the theories underlying these strategies. The model is described in the following section.

7.5 A Model of Training in MALL

As Hubbard (2004) points out, training is often a forgotten aspect of using technology for teaching and learning. The “digital natives” myth often prompts teachers into believing that younger learners are brought up in an environment where they are surrounded by technologies and therefore do not need training to use them. Research has suggested that this is not necessarily the case (see Stockwell & Hubbard, 2013), and there is little correlation between use of technology for private uses and effective use of technology for educational uses. The learners themselves also may not share this confidence in their abilities, and there is evidence that learners indeed desire more training in how to use their mobile devices effectively for learning (e.g., Rashid, Cunningham, Watson & Howard, 2018; Stockwell, 2008; Yang, 2012), adding weight to the argument in favour of providing learner training to all learners, regardless of their apparent skills with technology.

The three steps that are included in Hubbard's model are *technical*, *strategic*, and *pedagogical* training (Hubbard, 2004, 2013; Romeo & Hubbard, 2010; Stockwell & Hubbard, 2014). Each of these steps has an important role in ensuring that learners are familiar with the using the technology and should be carried out in an ongoing and cyclical manner. A central part of the model is that it also includes debriefing sessions, where learners and the teacher have an opportunity to discuss what learners did with the technology to support their learning, to get ideas from other students about appropriate strategies, and to further their own understanding of their own strategies through explaining them to others. This accords with Lewis's (2014) view of the need for the social aspect of developing autonomy, and it also enables learners to have a sense of support. Each of the steps is described briefly in the following sections.

7.5.1 Technical Training

At the time that Hubbard's (2004) paper was written, he lamented the fact that there was very little information in published articles about

how, when, and where this training was carried out, leading to the assumption that either the training did not exist or it was not seen as having sufficient impact on the learning process to be included as an element of the research described. The training that was most commonly described in the literature tended to be of a technical nature, and the primary focus is to familiarise learners with the functions of the tools that they were using. These technologies were generic in nature – such as an online dictionary, a word processor, or social networking software – or were dedicated for language learning – such as in-house or commercially developed software. For example, Romeo and Hubbard (2010) trained learners in how to use functions of Windows Media Player that enabled them to change the playback speed, a feature that most of the learners were not aware of despite playing videos through this tool on countless occasions. In the same light, it should not be taken for granted that learners know how to use the generic – let alone the dedicated – apps or software on mobile devices, or even the various functions of learners' own devices. Learners who are used to only using search engines may not know how to do something as apparently straightforward as typing a URL into a mobile browser (from my own experience!). Introducing and explaining how to use apps such as audio players that allow for changing the playback speed or book readers that have built in dictionaries are other examples of mobile-based tools that can assist learners if only they know how to use them.

7.5.2 Strategic Training

Research has suggested that learners who apply various strategies to their language learning experiences are more likely to be successful (Chamot, 2005), but learners are often not provided sufficient instruction in learning strategies for learning specific language skills and areas. Despite being one of the largest predictors for success in learning a second language (Ellis, 2015), the strategies employed by learners are often skewed towards certain skills such as vocabulary, with little focus on areas that learners are specifically targeting to improve such as listening and speaking. Moreover, learners are unlikely to revise or evaluate their own learning strategies and will often continue to apply the strategies that they started using from previous language learning experiences. Strategy training through technology does not need to vary greatly from non-technological means, but the specific techniques for how to use a particular technology to employ these strategies need to be taught to learners. General learning principles in the form of strategies will likely be useful, but if learners do not know how to use

the technologies that they are expected to use to carry out these strategies, there is the danger that they will not be able to do this properly.

There have been several studies that explore the provision of strategies as a means of preparing learners to use various technologies (see Lai & Morrison, 2013) and have found these to have a positive impact on engagement in learning activities and on learning outcomes. It follows that teaching strategies to use mobile devices for language will also likely have an impact on the ways in which learners will use them for learning. Kim (2017), for instance, found that learners of English in South Korea who were given strategies for English listening to help them engage in a dedicated MALL application were able to outperform learners on post-treatment listening tests compared with learners who did not receive this strategy training. Given potential time constraints for the provision of strategies in face-to-face environments, other methods have also appeared; for example, Tran (2018) provided strategies for using *Quizlet* to her Japanese learners of English outside of class using Line, and Lai, Shum and Tian (2016) used an online training program to provide learners with strategies that would help them to select and use technologies to support their language learning. Both studies reported increased participation in available resources along with positive feedback, reinforcing the view that learners themselves came to understand the benefits of strategy training in language learning through mobile devices.

7.5.3 Pedagogical Training

The third step of the learner training model is pedagogical training, where learners not only employ strategies as a part of their teaching and learning, but also evaluate and teach others the strategies that they use. The ability to teach specific strategies suggests a higher level of processing and understanding of the strategies, entailing an understanding of the not only the “how” aspect but also the “why” of using them. Learning through teaching is not a new concept, and there is already research that suggests that learners can affirm their own learning through teaching others (e.g., Cortese, 2005), as well as reflect on their own learning processes (Travers, Morisano & Locke, 2015). Romeo and Hubbard (2010) refer to pedagogical training in CALL as giving learners teacher training; that is, they are not just trained in *how* do use strategies, but also *why*, so that “informed choices can be made by an individual learner in the same way as a professionally prepared teacher would be making them for a class” (p. 217). Pedagogical training has been seen in CALL research, such as

O'Bryan (2008), who taught her students to use a multimedia gloss tool; she taught her learners –overseas students enrolled in degree programs in US universities –not only what strategies would help them, but also why the glosses were designed to contribute to their learning. While pedagogical training is yet to be pervasive in MALL, there are some examples starting to emerge. Tran (2016), for example, attempted to provide strategic and pedagogical training for her Vietnamese learners of English using Facebook, instructing them on not only how to use Quizlet but also the pedagogy behind the different strategies that she asked them to perform. While learners responded positively to the training, there was little evidence that there was a significant change in behaviour as a result of it.

The three steps of training presented in the model developed by Hubbard depend on their being reinforced on an ongoing and recurrent manner due to the impact that they can have on one another. The foundational understanding of the technology is a prerequisite for the use of strategies, and understanding why these strategies can contribute to learning is likely to have a positive impact on their sustained usage. Through the debriefings with other learners, learners can reconsider their strategies and even decide to try out new ones that they hadn't considered previously, even if they knew about them beforehand. Employing new strategies may be dependent upon understanding the technical features of the technology that were not previously necessary for the strategies that they were using, meaning that they need further training with the technology. In this way, training needs to be an integrated part of a course, with time given on a planned and consistent basis. In the absence of longitudinal studies to explore this, whether the holistic training presented in this model will lead to autonomy or not remains to be seen, but preliminary evidence from Stockwell and Hubbard (2014) suggests that sustained interaction is indeed possible. As described earlier, sustained interaction might be considered as a forerunner to autonomy, providing them with at least some of the necessary skills to self-regulate their learning. This gives learners the *skill* that they require for agency, and as their needs change and the *will* arises, it is likely that autonomous behaviour will emerge.

From the Field: Incorporating Training in MALL

I remember when I first introduced systematic learner training into my classroom environment. Training had always been a part of my teaching, but training that took up a significant part of each of my classes on a

week-by-week basis was somewhat daunting from the outset. Following work by Romeo and Hubbard (2010), debriefings were included in the class for the learners to discuss what they had done with their mobile devices outside of class time and to evaluate and rethink their strategies for their learning and for me to provide further information about possible strategies that the learners might be able to use. My biggest concern was that these debriefing sessions took up about twenty to thirty minutes of a ninety-minute class, and that this might have a detrimental impact on the learners' performance. Ultimately, my fears proved to be unwarranted, and I found that there was significant improvement in the result of weekly quizzes and participation in class. Furthermore, when the semester ended, several learners took the time to ask me for more resources that they could apply the strategies that they had learned in class during the semester break. The environment in the classes with the strategy training was very different from previous years, and perhaps one of the deciding features was that learners could see that the teacher was concerned about individual learners, but at the same time, they realised that the ultimate responsibility for learning was on the learners themselves. They came to understand that mobile devices were only as effective as the strategies that they used with these devices. I've used learner training in each of my classes since that first attempt several years back, and although there is some variation between classes and cohorts, it is unmistakable that learner training makes an enormous difference to the classroom environment, the attitudes towards learning with mobile and non-mobile devices, and the relationship between the learners and the teacher.

7.6 Summary

The learners in MALL may take the form of those who are undertaking study of a language of their own volition for personal or vocational purposes, those who are in a course of formal study that requires them to learn a language, those who find themselves in an environment where they must learn the language in order to survive on a day-to-day basis, or indeed any combination of these. Different learners will possess varying degrees of agency, and some will be equipped to carry out behaviours that are associated with their learning without support, and others will require some kind of support at the individual or infrastructural level. The type of agency learners possess has a direct impact on the ways in they seek out and use the available resources and contributes to the ways in which they will engage with them. Agency also plays a role in the ways that learners

use mobile devices as a part of their learning. Learners with a higher degree of agency are likely to be more willing to proactively make use of the resources that their mobile devices give them access to, and as such, have more opportunities for exposure to and use of the target language. Agency can be supported by external factors, and with assistance learners can become more capable of making decisions and acting upon those decisions for learning.

A key factor leading to such behaviour is training, where learners can be shown specifically how mobile technologies can facilitate their learning, and over time learners become able to evaluate and even teach the strategies that they use. Sustained engagement in language learning tasks and activities over time can impact positively upon their beliefs and attitudes towards language learning, and learner engagement can evolve from being predominantly behavioural to emotional, with a higher degree of purposefulness (Philp & Duchesne, 2016). Sustained engagement in learning tasks can give learners a greater degree of familiarity with what is effective, and this can cause them to engage more cognitively and make more effective use of the time spent on task. Therefore, training can lead to sustained interaction, which can, in turn, lead learners to consider their own learning processes and practices. However, it should not be assumed that providing training targeting specific a language skill (i.e., reading, writing, listening, or speaking) or language area (i.e., vocabulary, pronunciation, grammar, etc.) will be sufficient to lead learners to become globally autonomous learners. There will likely be imbalance in what they are and are not able to do, which limits their ability to take responsibility for own learning in a holistic sense. However, part of training should be to help the learners to achieve direct personal agency, where even if they do not know what strategies are appropriate for targeting a particular language skill or area, they will at least be able to take the initiative to try to find out.

Learner training should be ongoing and cyclical, where the ultimate goal is to hand responsibility for learning completely over to the learners themselves to become globally autonomous with regards to language learning. How to provide this training continues to be elusive, but it needs to include a balance of supporting both skills and motivation. Skills training needs to teach learners about appropriate learning strategies, but also needs to teach learners to view these strategies with a critical eye, understanding the principles behind them. Teaching learners to take advantage of available resources through mobile devices must necessarily be a part of this training. Motivation is a complex construct of internal and external factors that are dependent upon short- and long-term goals, their skills with using

resources including mobile technologies, and the social dynamics of the learning context. The pedagogy behind learner training is still in a period of flux as teachers try to find the optimum combination of teaching content and skills for learners to be independent, but this should not happen at the expense of sufficient support for learners who need direction. A pertinent question is raised by Fisher, Evans and Esch (2004), who ask, “Where on the continuum between fully directed tasks and complete learner autonomy does good practice lie?” (p. 57). Giving too little or too much control to learners depending on their state of development can have a detrimental effect on the learning process, meaning that good practice in learner training will need to be fluid, adapting to the evolving needs of the learners. Whatever form learner training takes, it must be part of a larger picture of instruction that ultimately has an end to it, where the learners not only take responsibility for their own learning in the present, but do so with an eye on further developing their skills in the future.

7.7 Discussion Questions

1. Is it possible to determine whether or not a learner is achieved autonomy in their language learning? Why or why not? Is it possible to measure autonomy? If so, how?
2. What type of engagement do you think that teachers expect in their learners for language learning tasks? Is this being clearly conveyed to learners? Give reasons for your answers.
3. What barriers do you see in introducing the training model proposed by Romeo and Hubbard? How can these barriers be overcome?

8 *Designing MALL Environments*

8.1 Introduction

Design is so central to language teaching that it is easy to forget how much of our time is dedicated to it, ranging from designing syllabi, lesson plans, learning materials, and tasks, through to designing the layout of the classroom, the roles and interplay of in-class and out-of-class activities, homework, and assessment. In this way, teachers are very much designing an intricate learning environment, often within the constraints of a larger curriculum, technological resources, administrative workload, research objectives, and even funding. Designing a teaching and learning environment that includes technology needs to be founded on having a clear idea of the entire context, including understanding who the learners are and what their needs and skills are, but it also means knowing what technological resources and support infrastructure is in place, how to train both teachers and learners, and how to deal with technical problems as they arise. Design is not limited to language teachers, of course, and it can be viewed from the perspective of researchers, institutions, and even language learners, each of which has their own individual priorities and agendas. The scope of design in CALL is evident in the research which has been dedicated to it. Levy and Stockwell (2006), for example, point out that design in CALL can include *language learning tasks* that target one or more specific language skills and areas, but it can also include the design of CALL *artefacts* such as tutors and tools. In addition, design can refer to designing *research*, and this often entails learners engaging in activities to achieve research goals (see Chapter 4) or for the development of theories (see Chapter 5) (see Rodríguez, 2017, for an overview of design-based research and how it relates to theory).

Needless to say, design has also been a key issue in mobile learning. This may be from the most elemental levels that include software and

hardware (Pachler, Bachmair & Cook, 2010), but more recent perspectives of MALL consider design principles which coalesce the complex network of in-class and out-of-class social and cognitive interactions that make up learners' daily lives (Kukulska-Hulme, Lee & Norris, 2017; Pegrum, 2019). MALL can provide a rewarding learning experience that brings language learning opportunities closer to the learners – whether in class or as they go about their everyday business – but the difficulties lie in how to design learning environments that can create these opportunities. There have been a number of attempts to conceptualise what is necessary to make MALL effective (e.g., Sharples, 2013; Stockwell & Hubbard, 2013), and the points raised in these discussions have the potential to raise awareness of some of the key issues involved in using mobile devices in language teaching and learning. At the same time, there have also been criticisms of MALL for its lack of creativity in pedagogy (e.g., Burston, 2015), but the field is developing rapidly with innovations in design that help to make MALL a useful experience for learners, and even an integrated part of their lives (Gu, Gu & Laffey, 2011).

Design in MALL should be comprised of three distinct but inter-related elements: design of the *learning environment*, design of the *artefacts*, and design of the *tasks* that learners will engage in. The learning environment refers to the larger context in which learning takes place. It is extremely complex, both broad and yet localised at the same time, consisting of social, interactional, and institutional elements that are intertwined with both the *space* (the larger environment) and *place* (the immediately surroundings) of the learners (Luckin, 2010). Artefacts are, as described in Chapter 2, the objects that learners interact with, and may refer to both the hardware and the software that are used in learning. At the same time, interaction with artefacts is inseparably linked to the environment in which it occurs (Dourish, 2004), meaning that the design of artefacts must take place with a clear understanding of the context in which they are to be used. One of the foundations of artefact design is *human-computer interaction* (HCI). Approaches to HCI stipulate that designers need to know who the users are and when and how they will use the artefact to create an interface that can facilitate smooth interaction between the technology and the user (Tidwell, Brewer & Valencia, 2020). Finally, language learning tasks will dictate what learners will do with the technology to develop their second language skills. While the definition of tasks has evolved somewhat over the years, they should be meaning-centred, modifiable and flexible, include a gap which needs to be addressed, require learners to use their own language resources, and have a clearly-defined end to it (Ellis, Skehan, Li,

Shintani & Lambert, 2020). The role that mobile technologies has in facilitating language learning tasks can be defined – to a certain degree at least – by the design of the artefacts that are used, whether the interactions take place *with* the artefact (i.e., learners interacting with the technology) or *through* the artefact (i.e., learners interacting with other humans using the technology). The affordances of the devices that are used will also contribute to shaping the tasks that are carried out with them, which adds weight to the interdependency of the relationship artefacts, the devices, and ultimately, the overall learning environment, which includes the individual characteristics of the learners. Each of these three areas – learning environment, artefacts, and tasks – has a rich foundation in research and theory and have been applied widely to teaching and learning with technology. Their applicability to MALL is described in turn below.

8.2 Designing Learning Environments

Understanding the overall environment is the first step in deciding the role that technology will play and the tasks that should be selected and is dependent upon a clear idea of how the instruction will be designed and the skills that teachers and learners require. How the environment will be designed will depend upon the format that it takes, that is, how the learners interact with the content, the artefacts, the teacher, and/or other learners. Some possible environments may be *face-to-face*, *blended*, and *distance learning* (Stockwell & Tanaka-Ellis, 2012), and each may occur as a part of the others or in conjunction with them. For example, face-to-face environments may be a type of blended learning where technologies are used as a part of a course during class or for tasks and activities to be completed outside of class. They may also be a component of distance learning, where classes of students may be required to come together for a few days or weeks of the year enabling teachers and learners to meet one another in person.

It is possible to consider the applicability of mobile technologies in each of these environments. In face-to-face environments, learners may use their devices in class, either each using their own individual device or with multiple learners looking at a single device together. This type of environment can promote collaboration between learners, which is an aspect of MALL which has gained attention over the past several years (Kukulska-Hulme & Shield, 2008; Kukulska-Hulme & Lee, 2020). Blended learning has been plagued by unclear definitions (VanDerLinden, 2014), and a conclusion as to what blended learning actually consists of has been very much open to interpretation. A useful definition is “the thoughtful fusion of face-to-face and online learning

experiences ... [where] face-to-face oral communication and online written communication are optimally integrated such that the strengths of each are blended into a unique learning experience congruent with the context and intended educational purpose” (Garrison & Vaughan, 2008, p. 5). Despite the written focus, this definition serves to show that technology in blended learning should be used in a *meaningful* manner that takes advantage of the affordances of the technology and considers how it can be used as a part of the larger environment. This underlying perspective of blended learning agrees squarely with Laurillard’s (2012) observation that technology implementation needs to be based on understanding the principles of what good teaching are and assigning appropriate roles to technology based on this. It follows that regardless of what environment a technology is to be used in, the first and foremost consideration is how the technology can be used to achieve solid and identifiable learning goals. Distance learning has undergone a complete transformation over the past two decades or so as a result of technology, changing from posting of paper-based materials to electronic ones, and the position of mobile technologies as a way of keeping learners and teachers in communication with one another has become a central one (Sharples, 2014). Mobile technologies can be shaped to fit into most learning environments provided an appropriate design is implemented, as described below.

8.2.1 The ADDIE Approach

Designing of learning environments has largely been the domain of *instructional design* (ID). There have been several models of ID that have appeared over the years (Richey, Klein & Tracey, 2011), but one of the most influential instructional systems design frameworks is the ADDIE approach (Schegel, 1995; Tamez, 2016). ADDIE is based on the “Five-step approach” which was originally developed by the US military in the 1970s and as such does not have any identifiable starting point as an approach in its own right. ADDIE is made up of five phases (Analyse, Design, Development, Implementation, and Evaluation) each of which informs the next phase, and it is typically used cyclically where each iteration allows designers to systematically modify the environment based on empirical outcomes. It has been the foundation of several approaches to instructional design since it appeared, including ASSURE (Shelly, Cashman, Gunter & Gunter, 2006), the Kemp model (Kemp & Smellie, 1994; Morrison, Ross & Kemp, 2004), and the Dick and Carey model (Dick, Carey & Carey, 2005), each sharing very similar core concepts. Approaches like ADDIE and its various offshoots allow course designers to understand

the phases of creating a learning environment, and their broad nature makes them applicable to different contexts that need not follow the linear modes of design that has made them the target of criticism over the years (Branch, 2009). ADDIE is not specific to technology integration per se, but it is amenable to adaptation to include technology and can also be applied to language teaching and learning with technology such as courseware design (see Colpaert, 2006, for an overview). It is important to note that ADDIE is not a learning model (Woodill, 2015) but rather a reiterative framework for design. As such, it does not have a specific theory of learning or technology embedded in it, and designers approach their individual environments with their own understanding of how languages are acquired and the affordances of the various technologies that are available in that environment.

The ADDIE approach illustrates the complexities involved in understanding how the intricate elements of designing a learning environment fit together through the five interrelated phases. According to Branch (2009), in the *Analyse* phase, designers must determine what the instructional goals are to be, typically through identifying a performance gap, that is, an aspect of the learners' proficiency that needs improvement. In the analyse phase, it is also necessary to determine what technologies are available and specifically who the target audience will be. In MALL, this would mean ascertaining what mobile devices are owned and how they might fit together with other technological and non-technological resources that are available and understanding what support mechanisms are in place. Once the target aspect has been finalised, next is the *Design* phase. This entails conducting a task inventory, composing performance objectives, generate testing strategies, and calculate the return on investment. The design phase allows guidelines to be devised to stipulate precisely what tasks will be required to achieve the goals laid out in the previous phase, how much time and effort is to be dedicated to development, and how the effectiveness of these tasks will be evaluated. Where mobile technologies are to be used, this entails selecting the types of tasks to be used and deciding which of these will be available on mobile devices. It is during the *Development* phase that the actual content is generated along with the supporting media, which would include the creation of the learning artefacts. As part of the development phase, frameworks for learner and teacher guidance should be devised, and where possible the artefacts should be piloted to allow for revisions. According to the ADDIE approach, it is only after the artefacts have been successfully created and the learner and teacher training procedures have been established that the *Implementation* phase begins. As the name suggests, this is where the artefacts are used in the actual environment

for which they were designed, and it is the culmination of the previous phases to ensure that the tasks and artefacts are suited to the learning goals and the users are competent in carrying these out. The final phase, *Evaluation* is where the quality of the processes and products are confirmed. Appropriate evaluation criteria are essential in achieving this with the appropriate tools, and the outcomes of the evaluation should be applied to the next iteration of the ADDIE approach where any gaps that were not adequately filled or any newly identified performance gaps can be addressed.

8.2.2 The Ecology of Resources Model

A completely different perspective of the design of the learning environment stems from Vygotskian views of learning. The Ecology of Resources model was proposed by Luckin (2010) as a means of looking “beyond the single user, single system, single location paradigm” (p. 75). The motivation behind the Ecology of Resources model was to model a learner’s interactions both in and out of school and with various other participants in order to support the development of technology to link formal learning experiences to students’ lives outside of school. Luckin’s concept is logically compatible with MALL in that she views technology as being one of the resources that are a part of a learner’s ecology that can allow for provision of resources or facilitate interaction with others. At the heart of the Ecology of Resources model is a reinterpretation of Vygotsky’s Zone of Proximal Development (ZPD) where learners engage in scaffolded instructional interactions with others who possess a level of knowledge or skills that learners need in order to achieve a certain learning goal (“More Able Partners”). In Luckin’s model, the ZPD sits at the core of what she terms the Zone of Collaboration, which is “full of potential forms of assistance that might act as resources to facilitate learning” (p. 29). She also adds two new constructs, the Zone of Proximal Adjustment (ZPA) and the Zone of Available Assistance (ZAA), which form consecutively larger rings around the ZPD. The larger ZAA includes the variety of resources that learners can access for assistance when required, and the ZPA refers to those resources that are appropriate to the learner’s needs. As a design model, the Luckin’s model has three phases:

Phase 1. Identify and organise the forms of assistance around the learners that are potential resources for learning (the Ecology of Resources) in the following seven steps.

1. Brainstorm the potential resources in an Ecology of Resources
2. Specify the focus of attention of the learning environment

3. Categorise the resource elements
4. Identify barriers (filters) to access to resources
5. Identify the learners' own resources (skills, experiences, etc.)
6. Identify potential "More Able Partners"
7. Repeat the first six steps

Phase 2. Examine the interrelationships between the resource elements from the first phase and determine how they meet the learners' needs or can be optimised for individual learners. She identifies four types of relationships:

1. Influence relationships (how one element influences another)
2. Component relationships (whether one element is a part of another)
3. Typology relationships (whether one element is a type of another)
4. Social relationships (between family members, friends, or communities, etc.)

Phase 3. Develop the necessary scaffolds and adjustments for creating an appropriate ZPA for each learner. Luckin offers seven types of scaffolding and adjustment:

1. Interaction between the learner and resource elements
2. Interaction between the learner and More Able Partners
3. More Able Partner interaction for the learner's optimal use of resources
4. Interaction between resource elements
5. Interaction between the filters to resource usage
6. Interaction within a resource element
7. Interaction within a filter to resource usage

One of the primary distinguishing factors between the Ecology of Resources model and models of instructional design is that it focuses primarily on the resources that are already available to the learner and does not explicitly deal with design of artefacts or tasks themselves. These are tacitly embedded in the phases outlined in the design model (design of artefacts could be included as a part of Phase 1 and 2, and the design of tasks could be a part of Phase 3), but the main objective of the model is to limit the barriers and maximise the support from the "More Able Partners" so that the available resources around the learner can be used to maximum effect. The concept of fitting identifying all potential formal and informal learning resources – technological and human – and assisting the learner to gradually be able to use them independently has extremely important implications for MALL. In an era where the resources that

learners have access to extend beyond just those that are provided for them by teachers, it is essential that designers have a clear idea of what learners may potentially use and seek methods through which learners may receive guidance to using them appropriately.

8.2.3 Conversational Framework

While some of the existing instructional and educational models give a broad overview of any learning environment (such as the ADDIE approach), others stipulate how technology fits into the overall educational environment. One such example is Laurillard's *Conversational Framework* (2002, 2012). Laurillard explores how teaching itself may form the foundation of design, and has been one of the most influential frameworks of considering how technology may be used purposefully in education. The core emphasis of her framework is on learning in formal contexts, and she takes great care to ensure that all of the potential factors that may impact learning are given due consideration, including the workplace, educational theorists, the teacher, and even the educational establishment itself. Laurillard views teaching and learning as the complex intersection between teacher factors and learner factors, but these are seen within the learning environment itself, particularly in view of the course aims, teaching activities and assessment, intended learning outcomes, and logistical factors, and how these align with learner motivations, expectations, knowledge, skills, needs, and personal goals to produce the actual learning outcomes. Misalignments between intended and actual learning outcomes can then be investigated and resolved through reviewing the learning goals, the teaching activities, and assessment. Laurillard (2012) includes various other teaching principles in order to meet the expected learning goals, namely, monitoring alternative conceptions, using scaffold theory-based practice, fostering conceptual knowledge development, and encouraging meta-cognition. What is particularly interesting about Laurillard's view is how she explores the ways in which technology can be used in varied views towards teaching, including learning through acquisition, through inquiry, through discussion, through practice, and through collaboration. Applied to mobile learning, the Conversational Framework sees mobile devices as a means of communication between the teacher and learners and provision of site-specific practice are of particular relevance (Laurillard, 2007; Woodill, 2015). As a framework for learning environment design, the merit of the Conversational Framework is that it requires teachers to be aware of the complete range of factors in the context and to understand that there may be mismatches between

teacher and learner perspectives and outcomes, and suggests approaches in which to achieve the desired learning outcomes. At the same time, it also advocates the use of well-defined teaching principles to which mobile and other technologies should be applied purposefully.

8.2.4 *The iPAC Framework*

A framework which has been developed specifically for designing a mobile learning environment is the iPAC framework. It includes three main categories – *personalisation*, *authenticity*, and *collaboration* – that focus on the affordances of mobile devices in education and the use of *time-space* (Kearney, Schuck, Burden & Aubusson, 2012; Burden & Kearney, 2018). *Personalisation* includes what Kearney and colleagues term as “agency” and “customisation,” where agency is the extent to which learners can control when, where, and how they learn with mobile devices and the degree of autonomy that they have in doing this, and customisation is the degree to which learners can choose the tools and activities that they engage in to suit their individual needs. Agency is primarily an internal construct that develops as a result of individual factors and external support (see Chapter 7), but customisation will be heavily dependent upon the affordances of the tools and activities that are available. That is to say, even if a learner has agency, they may be limited in their capability to customise their learning experience due to a lack of appropriate resources. *Authenticity* links to the context in which mobile learning experiences are made real and meaningful through the environment in which they occur and is similar to the concept of situated learning (Lave & Wenger, 1991), and task refers to how realistic the tools and activities replicate those that would eventually be expected of learners in the real world. *Collaboration* includes conversation between peers, teachers, and “other experts” (Kearney, Burke & Schuck, 2019, p. 753) and co-creation, which refers to the creation of digital content collaboratively and sharing of information, data, and artefacts with others. Finally, the “time” in *time-space* relates to the freedom in time that mobile devices afford users (cf. Traxler, 2007), whereas “space” can mean both physical and non-physical or “virtual” spaces that learners may use to engage in learning or interact with others. The interrelationship of these factors gel together clearly to show how a learning environment may be designed in such a way as to make the most of the affordances of mobile devices.

8.2.5 Other Design Frameworks

There are numerous other frameworks or models of design that may be applicable. One instructional design model that has been applied especially to include technology (but not specifically mobile devices) is the *RASE learning design framework* (Churchill, 2013; Churchill, Fox & King, 2016). RASE (Resources-Activity-Support-Evaluation) is founded on the idea that resources themselves are insufficient for learners to achieve desired outcomes sufficiently, and there is a need to supplement them with suitable activities that lead learners to engage in using the available resources through active experience, provide the learners with support that enables them to work independently or with others to solve learning problems, and include an evaluation step where both teachers and learners can be informed of learner progress in achieving the learning outcomes. The RASE learning design framework is comprised of technology being used in each of the four elements, with *resources* including the various digital artefacts and learning objects, *activities* describing how these resources can be used, *support* exploring how technology may facilitate support from teachers, peers, and/or external parties, and *evaluation* considering how technology may be used to measure student learning and participation. Mobile technologies can also be applied to the RASE framework in terms of the interactivity of mobile devices to facilitate communication for ongoing support and their mobility to provide resources that are immediately relevant for activities carried out in authentic contexts. The RASE framework does not provide specific information about teaching or learning processes per se, but rather, it enables designers to see how technologies may be used in different aspects of the teaching and learning process, and as such has value in giving a global perspective of technology in the learning environment.

Other examples include Sharples, Taylor and Vavoula's (2007) *activity system of mobile learning* or Dennen and Hao's (2014) *M-COPE framework*. Both of these have a specific focus on mobile learning and may be applied to designing the learning environment but are related to other existing models. The activity system of mobile learning is based solidly on Laurillard's Conversational Framework (2012) but considers specifically the affordances of mobile devices and how they may be applied to learning within an activity theory framework. The M-COPE framework was designed to work in conjunction with the ADDIE approach considering the mobile affordances, conditions, outcomes, pedagogy, and ethics (which make up the acronym M-COPE) of mobile devices to inform the design process. The design of the overall environment is the first major step in determining what is

done by teachers, learners, and other supporters of the learning process, but the strength of this design is highly dependent upon the tools and resources which are used, which are described below.

8.3 Designing Artefacts

It would be impractical to try to recount all the current body of work that looks at design of resources for language teaching and learning in the limited space that is available here. There is some necessary overlap between theories and models that are used in the larger environment and those that are used for designing artefacts. One theory that is of particular relevance to mobile learning is *situated learning*. Used as the foundation for a number of the design frameworks described in the previous section, situated learning was conceived by Lave and Wenger (1995) and advocates that learning needs to take place through social interaction in the location in which skills and knowledge will be used, much as an apprentice might when learning a trade. If learners are connected to a real context and carrying out interactions with others who can provide them with the skills and knowledge that they need, they are more likely to have a practical sense of how these can be applied. Knowledge is situated within the environment in which it is acquired and is “in part a product of the activity, context, and culture in which it is developed and used” (Brown, Collins & Duguid, 1989, p. 32). It is not surprising that situated learning has been applied to mobile learning and, more specifically, to MALL. Mobile devices can be carried to the locations that they are required, and information that is needed to complete a communication act can be accessed easily. Furthermore, learning can be enhanced through communities of practice, which is another aspect that can be supported through mobile devices given the social and networking affordances that mobile devices not only make possible, but also enable to be accessible to learners in real situations. One criticism situated learning has faced is that learners in authentic environments may have difficulty in focussing their attention on the learning goals that designers may expect of them. However, as Little, Dam and Legenhausen (2017) contend, learners are constantly learning *something* when they are in such situations, but it might be simply something different from what the teacher intended them to learn. Acquiring skills and knowledge that learners need – even if it differs from teacher expectations of learner needs – is an important part of the learning process, provided that teachers can monitor learner progress to ensure that they are not deficient in any aspects that they are required to be able to function.

Studies based on situated learning as a design principle seek to take advantage of the mobility and communication aspects that mobile devices afford in artefact design. For example, Huang, Yang, Chiang and Su (2016) developed a mobile app that provided outdoor learning activities for young Taiwanese learners of English based on their location, which was determined through the GPS function on their devices. Learners were then able to engage in practice dialogues with their peers based on the contextualised activities and were reported to have shown improvement in their vocabulary as well as enhanced motivation to study English. In another study, Wong, King, Chai and Liu (2016) describe the development of an artefact called MyCLOUD, which was designed for learners of Chinese in Singapore. The system consisted of three main components: a mobile dictionary that learners could use to keep a record of new vocabulary that they encountered; a digitised version of their textbook, which enabled the text to be played audio using text-to-speech technology; and a social space for learners to post and interact with other learners' media. Learners could post pictures from their daily lives through the social space and comment on these and posts made by other learners, keeping a record of unfamiliar vocabulary that arose. Their results suggested that learners exhibited a broader range of vocabulary usage at the end of the thirteen-month study compared with the beginning, and this could at least be partially attributable to the MyCLOUD system that they used to contextualise new vocabulary that they encountered.

Other models and theories look at the way that the technology can facilitate learning through the specific affordances that technology provide. One such model is *dual coding theory*, which is a cognitive psychology theory that explores how access to information through both the auditory/verbal and visual/pictorial modes impacts a learner's cognitive processes to facilitate learning (Mayer, 2005; Sweller, van Merriënboer & Paas, 1998). An example of this is provided by Li and Yu (2017), who used dual coding in the design of their learning artefacts for Taiwanese learners of English – in this case, multimedia message service (MMS) messages that were sent to learners' mobile phones at 7 a.m. and 5 p.m., when the learners were likely at home. The messages consisted of text only; text and picture; text and sound; and text, picture, and sound in order to explore the effect of multiple modes on learner retention. Li and Yu found that there was enhanced retention if multiple modes were used in presenting new vocabulary, but there were not significant differences between the each of the modes. The purpose of using mobile devices in this study was to allow learners the flexibility to look at the messages at a time and place that suited them and to take advantage of the multimedia capabilities of the devices.

Another concept from cognitive psychology that has been applied to MALL design is *spaced learning*, which is founded on the notion that reviewing newly learned information within a relatively short time spans can enhance long-term retention of this knowledge (see Godwin-Jones, 2010, for a discussion). Spaced learning has formed the foundation of the design of commercial mobile-based artefacts such as Duolingo that have been used within a language learning environment (e.g., Loewen, Crowther, Isbell & Kim, 2019), but it can also be used to design a learning context that uses mobile devices, such as in the work of Cavus and Ibrahim (2009), who designed an SMS mail-out program called MOLT (MOBILE Learning Tool) for Turkish learners of English. The MOLT system was designed to send messages at timed intervals to cyclically remind them of new vocabulary on an ongoing basis. In order to bring in-class activities into learners' everyday lives, the timing of the messages was chosen so that the learners were outside of class, and there was evidence that learners had indeed acquired many of the vocabulary items that were taught through the system.

Psychometrics is another field that has contributed to design of MALL artefacts. *Item response theory* (IRT) was developed in the 1950s as a means of predicting the probability of getting an answer to a test item correct based on the difficulty of the item and the ability of the test taker, where test takers of higher ability will have a better chance of correctly answering more difficult test items (Baker & Kim, 2017). IRT has formed the foundation for computer-adaptive testing and has also been used in CALL (e.g., Browne & Culligan, 2008). For mobile devices, Chen and Chung (2008) developed a sophisticated vocabulary learning system based on IRT design principles for Taiwanese learners of English which learners used on PDAs provided by the researchers. The system enabled the designers to provide vocabulary activities that were tailored to the learners' needs and to develop a profile of each learner through which their progress could be monitored.

As these examples illustrate, design of artefacts is also an extremely complex process, and appropriate design is dependent upon the theoretical approaches that are selected (see Chapter 5). Needless to say, the design of language learning artefacts must take place with a clear view of the overall environment and an understanding of when, where, and how these artefacts will be used. There are countless artefacts on the market now for language learning on mobile devices, and the overwhelming majority of these take the form of apps. While I have intentionally avoided providing lists of available apps for language learning as they are inclined to become dated extremely

quickly, I will mention a few of the more enduring apps here. As described before, Duolingo (www.duolingo.com) is a commercial app for developing vocabulary, listening, and speaking through gradually introducing new content and testing the user as they progress (see Teske, 2017, for a review). Another commonly used app is Busuu (www.busuu.com), which includes an extra element on provision of content where learners can link up with others through the app, allowing them to interact through text or audio (Winans, 2020). Quizlet (www.quizlet.com, described later in the chapter) is a flashcard app that enables learners to produce their own lists of vocabulary. A useful list of mobile learning apps and an overview of learner attitudes towards them provided by Rosell-Aguilar (2018); he the list into self-study language apps, translation apps, news apps in the target language, flashcard apps, and other apps such as Facebook, Twitter, Skype, and YouTube; this is a useful guide when looking at possible alternatives.

8.4 Designing Tasks

The design of tasks is one of the most central aspects of language teaching and learning in formal environments. Any discussion of teaching practice should not be separated too much from the views of what language is and the ways in which it is taught, and the design of learning activities will be based on the view of language and the underlying assumptions about how languages are learned (cf. Richards & Rodgers, 2014). These views can be very broad in their focus, coming from as varied perspectives as cognitive-interactionalist, psycholinguistic, sociocultural, psychological, and educational (Ellis et al., 2020). Van Lier explores the relationship between underlying theories and the associated practices, as is described in Table 8.1. Examples from MALL research which have been designed based on these assumptions and/or practices are also provided.

An information processing view of model can be seen in a study investigating information gap tasks on mobile phones. In one example by Kiernan and Aizawa (2004), learners sent the information required to complete worksheets to one another using the text messaging function on their phones, and in another example, Razaee, Alavi and Razzaghifard (2019) used information gap activities as a way of facilitating interactions between learners in order to measure whether mobile-based interactions could lead to enhanced oral proficiency. The main goal in each case is that language is used as a tool in order to exchange information which facilitates interaction between the participants.

Table 8.1 Underlying views of learning and associated practices (adapted from van Lier, 2002, p. 142).

Assumptions	View of Language and/or Learning	Practices	Examples from MALL
Language use is information exchange, consisting of inputs and outputs	The conduit metaphor; sender-receiver model; information processing	Information gap tasks	Kiernan and Aizawa (2004), Rezaee, Alavi and Rhazzaghifard (2019)
Language learning means acquiring competence	Language as acquired habit; language as internal, mental competence	Memorising lists of words; sentence practice	Browne and Culligan (2008), Heo, Lim and Kim (2010), Nguyen and Pham (2011)
Language consists of form and meaning	Structural, generative linguistics; speech act theory, functionalism	Focus on form(s); consciousness raising; content-based teaching	Reinders and Cho (2010), Chen and Chang (2011)
Language consists of pronunciation, vocabulary, grammar, and meaning as building blocks	Descriptive linguistics	Skills-building exercises; practice in all skills areas, moving between smaller to larger items or vice versa	Cui and Bull (2005), Chen and Chung (2008)
Language use can be correct or incorrect, standard or non-standard	Normative linguistics	Error correction; formal essay structure; accent reduction; standard tests	Ghorbani and Ebadi (2019), Li and Hegelheimer (2013)
Early cognitive science; contrastive linguistics; behaviorism	Languages compete with one another in our brain for attention and storage	Avoidance of use of the first language; arguments against bilingual education	Tabatabaei and Goojani (2012), Lai (2016), Tran (2016)

With regards to a view of language learning as acquiring competence, vocabulary learning has long been a part of learning through mobile devices, and activities that focus on the use of memorising lists of words are quite common, such as flashcards (Browne & Culligan, 2008) or other more sophisticated systems that provide activities based on system estimates of learner abilities (e.g., Stockwell, 2010). Similarly, activities that target grammar are also relatively common, either through giving learners opportunities to write short passages such as blogs (Heo, Lim & Kim, 2010) or through intelligent systems that utilise adaptive modelling (Nguyen & Pham, 2011). Indeed, both vocabulary and grammar have been amongst the most commonly targeted language areas through technology since the early days of CALL (see Stockwell, 2007, for a discussion), and this trend has been repeated to some degree through mobile devices as well.

One study that looked at focus on form through mobile devices was carried out by Reinders and Cho (2010), in which learners used an audiobook that they could listen to on their mobile phones; the volume was raised slightly on specific targeted forms to increase their salience. In another study looking more at the retention of content, Chen and Chang (2011) compared learners who were provided access to content either through audio only or through a combination of audio and text. They found that learners who had both modes retained the content better than those who just had the audio and that lower-proficiency learners who had access to both audio and text were more likely to rely on the text to facilitate comprehension of the content.

Skills-building exercises, where practice typically targets easier items initially and gradually leads learners to more difficult items, have been greatly assisted by technologies that can adapt to individual learners. As has been seen in CALL more generally, there has been a strong trend towards vocabulary in MALL (Lin & Lin, 2019). This is obvious in numerous studies where various personalised adaptive systems based on algorithms such as item response theory (e.g., Chen & Chung, 2008) or through responses to activities (e.g., Cui & Bull, 2005) have been described in the literature. Other studies include pronunciation and grammar; for example, Wang (2017) explored the prospect of using WeChat as a tool to develop English pronunciation in Chinese learners, and Moghari and Marandi (2017) explored texting as means of developing Iranian learners' grammatical skills. Thus, descriptive linguistics views of language based on the assumption that incremental instruction assists learners in developing the building blocks of language can also be seen as the foundation of studies using mobile devices.

Regarding the correct use of language, the most typical focus of research in MALL has been on grammar. Ghorbani and Ebadi (2019), for example, explored the impact of feedback by instructors through the mobile-based messenger app Telegram on the grammatical accuracy of learners of English in Iran, finding that learners showed improvements in accuracy when instructors provided them with corrective feedback while chatting online. In another study, Li and Hegelheimer (2013) found that, when learners used a mobile app for grammar, they were more likely to engage in self-editing of texts in academic writing. The emphasis on the accuracy of language in both of these studies reflects a normative linguistics view of learning.

One of the more common ways of encouraging learners to communicate in the target language has been the use of CMC, which includes tools such as email or chat (e.g., Stockwell & Harrington, 2003). On mobile devices, text messaging has been the most commonly used method (e.g., Kim, 2011; Tabatabaei & Goojani, 2012), but more recent studies have used social networking tools such as Facebook (Tran, 2016) and Twitter (Kim & Lim, 2010), as well as communication tools such as WhatsApp (Lai, 2016) and Line (Tran, 2018). While CMC does not preclude the use of the L1, depending on the task design, it can be discouraged or even forbidden. Early cognitive studies which investigate interaction using CMC or social media have, however, tended to move away from strict enforcement of the L2, and other aspects such as participation and presence in communities of inquiry (CoI) have been the focus of an increasing body of research (e.g., Fornara & Lomicka, 2019; Tran, 2018).

The discussion here does not make judgements about which – if any – of these assumptions are more appropriate for language learning than any others, as there is sufficient evidence an extremely broad body of research completed over the years from that all have the potential to bring some positive (and some potentially negative) impacts on language learning. The reason for listing them here is to allow practitioners to reflect on what their own assumptions may be, and that these assumptions are unlikely to be the only possible assumptions that they may hold. The range of approaches applied and the respective results from the studies described here shows that the view of teaching and learning held by teachers is equally reflected in tasks and activities designed for mobile devices as they were likely to be carried out in more traditional non-technological environments. What this means is that the technology does not necessarily fundamentally alter practice, but, rather, the affordances of the mobile devices being used may be applied so that teachers may adapt their practices in innovative ways.

8.5 Designing Assessment in MALL

The appropriateness of using mobile devices for assessment purposes is a point that is understandably receiving more attention as they become more central in education. It makes sense that if learners are going to be required to use their devices for learning purposes, there should be some parallels with their assessment as well. As assessment procedures developed, there has been a move away from assessment that is based predominantly on memorisation of facts and figures to “open-book” methods where learners have access to reference materials and resources during testing as a more reliable way of determining deeper learner understanding (Eilertsen & Valdermo, 2000). Research has indicated that open-book testing is comparable to closed-book in terms of reliability and is not perceived as “easier” by test takers (Heijne-Penninga, Kuks, Schonrock-Adema, Snijders & Cohen-Schotanus, 2008). An extension of this has been the concept of “open-book–open-web” (OBOW) where examinees are able to refer to any resources – both offline and online – in order to complete their assessment (Cortese, Pupovac & Xu, 2019). Given Dikkers’s (2014) prediction of the need for change in assessment in mobile learning, how to make the most of devices that learners are carrying out a significant part of their learning will become increasingly relevant as mobile devices are used more widely in both formal and informal learning contexts.

In saying this, assessment through mobile devices is not without concern. As much as there are issues of reliability if test takers are able to freely access any information or interact with anyone as they undergo assessment, questions arise as to the sophistication of test responses if mobile devices are to be used as the *medium* for carrying out the assessment. An ongoing concern of assessment with mobile devices is, however, security concerns and making sure that the person who is taking the test is the one that should actually be taking it. One method that has been proposed includes the use of biometric authentication such as facial recognition (Kawahara, 2010; Kawahara, Nogimori & Matsumoto, 2015). As a completely online university, students enrolled at the researchers’ institution are required to take exams externally, meaning that there is a need to confirm the identity of the user. Using facial recognition, they are able to identify the user to ensure that the test taker is the person who is actually enrolled in the course. Assessment through mobile devices, and particularly in MALL, is still an emerging area at the time of writing, and challenges in designing effective and reliable assessment will likely be difficult to resolve.

8.6 Principles for MALL Design

The chapter provides a list of nine interrelated suggestions for mobile learning, supported by relevant findings from the recent literature, and can act as a practical checklist for those who are considering using mobile devices in their teaching and learning contexts.

8.6.1 *Understanding the Context*

Before engaging in any type of project using mobile devices, the point of departure must be having a sufficient understanding of the various intertwined elements that make up the context. This includes knowing who the learners are, their skills, preferences, learning styles, and habits (Levy & Stockwell, 2006). When mobile devices are included in the equation, it also means knowing what devices are owned and learner willingness to engage in activities using them, as there is a difference between knowing (i.e., being aware of) the idiosyncrasies of the context and understanding (i.e., realising the impact) that these may have on the selection of appropriate learning tasks and activities.

8.6.2 *Becoming Familiar with Available Resources*

There are numerous examples of resources that can be used on mobile devices and computers for language learning. I will not attempt to present a list of resources for learning here, as there is a constantly evolving and dynamic range of apps (see earlier in this chapter) and websites available to teachers and learners. In saying this, however, designers need to have some grasp of what is available and their applicability to their own context. As described earlier, MALL is far more than apps (see Chapter 1), but a look at any of the online app stores will quickly reveal that an enormous number of apps for language learning have been released over the past several years. Some of these are available completely free, but the majority provide either a free trial or free access to a limited number of materials with which learners can engage to determine whether or not it is worthwhile in paying for the full version. The quality of these apps is by no means uniform; some are produced by larger, well-known companies and/or organisations, and others are produced by smaller developers or individuals. However, the size of the organisation that developed the app is often not a very good indicator of the quality of the product itself, at least in terms of the content, despite a rather more impressive interface usually being associated with the more established companies.

There are also some apps which are completely commercial or offered as a part of a class or school or school licensing arrangement. Students may have limited access to the resources that are provided by these companies and may even be able to take out an individual subscription to be able to use the software, but the main targets are typically institutional sales. These materials often have both a computer-based interface and a mobile interface, adjusting to the device used by learners. Mobile activities are often designed in such a way that they can be completed easily using mobile devices in short chunks to facilitate atomised learning. One example of this is the learning site Newsela (www.newsela.com), which provides activities based on short videos of news segments. Students can use these in a limited capacity for free, but they will not have full access to the complete range of activities unless they have a subscription. Not only does the subscription give students full access to the materials, but it also gives teachers access to records of student activity on the site, which would allow it to be used for assessment purposes, should the teacher so desire. A similar model is the quiz and flashcard program Quizlet. Students can have access to all learning resources for free, and there are both computer-based and mobile interfaces; the content can be designed by teachers and/or students themselves, or they can access content that has been developed by others. A primary difference is that the content is designed on a voluntary basis by others with no guarantee of quality. If they pay for a yearly subscription, teachers can create trackable class groups, enabling them to have some record of student activity with Quizlet, which can be used for assessment or other purposes as necessary.

8.6.3 *Making the Most of the Affordances of the Device*

As described in Chapter 6, modern mobile devices have functionalities that make them very attractive tools for learning. It would be unrealistic to expect that mobile phones are the best choice to undertake all types of learning activities, and it is becoming apparent that there are some tasks and activities to which mobile devices are naturally more suited. A further consideration with affordances is that even if tools and apps are designed with mobile and PC versions, there is often a discrepancy between these two versions that can cause frustrations for learners, most notably a lack of functionality in the mobile app compared with web-based resources accessible by computers (Tran, 2019). While technological affordances may restrict some of the possibilities of what can be done with mobile devices, in many cases, good design can allow for the inclusion of multimedia, interactivity, and portability

appropriately. The affordances of mobile devices – such as their portability and their technical functions including cameras, music and video players, GPS, Internet browsing, and so forth (see Chapter 2) – do allow for an enormous range of possibilities that should persuade learners that there are clear benefits in using these devices for engaging in tasks and activities or language learning.

8.6.4 Setting Feasible Learning Goals Suitable for Mobile Devices

While the affordances of mobile devices make it possible to carry out learning in a way that promotes interactivity and multimodality, they also have limiting factors associated with the physical limitations of the device and the environment. As mentioned in Chapter 6, screen size, input methods, battery life, and network access can all have a restrictive impact on what can be done through mobile devices. On the one hand, mobility allows users to engage learning at times and in places that are suitable for them, but on the other hand, it means that these locations may not necessarily be ideal for concentrating on tasks. While MALL can, of course, be feasibly carried out in fixed locations like the classroom, the library, or at home, learning through mobile devices in noisier public spaces brings with it challenges of outside interferences that will likely cause starts and stops in activity. This has implications for design of both the tasks and the artefacts for MALL, and understanding the difficulties associated with the affordances of both the device and the environment is essential (see Stockwell & Hubbard, 2013; see also Chapter 9). Goal setting – both short and long term – needs to bear in mind the role that mobile devices fit in the larger picture with offline and other online learning.

8.6.5 Predicting Technical Problems and Preparing Support

Learning with mobile devices is something that can and does take place both inside and outside of the classroom. Technical difficulties are, to a certain extent, to be expected when using technology, and it is essential to ensure that there is either (a) a backup plan to deal with any problems that arise, or (b) means of immediately supporting problems that occur, rather than trying to deal with them after they actually occur. When problems occur outside of the classroom, it is natural that providing support is far more complex and can cause learners to have a very negative image of learning through mobile devices, leading to greatly reduced motivation, or even attrition. Ironically, mobile devices are

often cited as being an ideal *support* for learning (e.g., Järvelä, Näykki, Laru & Luokkanen, 2007; Ng & Nicholas, 2013), but descriptions of the support for mobile devices themselves are rarely made explicit. One observation is that teacher presence in online courses plays an important role in maintaining student engagement (Gunawardena et al., 2016; Stone & Springer, 2019). The key is not only to offer support but to prepare in advance exactly *how* this support can be given in a timely and accessible manner.

8.6.6 Preparing Learners Adequately

As has been alluded to at several points throughout this book, learners cannot be expected to engage in effective learning with technology without being sufficiently trained (Laurillard, 2002). Training is something that necessarily must be carried out in an ongoing and perhaps cyclical manner, where learners have multiple opportunities to determine what they do and do not know how to do. As is described in Chapter 7, training learners must go beyond just simply showing how to use a technology to ensure that learners know what to do to learn specific elements of language through the technology and, more importantly, what to do when they encounter difficulties. As has been found in a number of studies, learners are far more capable of engaging in self-directed learning using mobile devices outside of class when they are given sufficient training (Kondo et al., 2012; Stockwell & Hubbard, 2014). Thus, preparing for using mobile technologies goes beyond making sure learners understand the technology; they also need to know how to use the technology for learning in the context in which they find themselves with an understanding of their individual needs.

8.6.7 Allowing for Alternatives

There are two possible interpretations of *alternatives* here. The first means providing tasks and activities that will run suitably on a wide range of devices, and are not dependent on the sophistication of the device (i.e., older smart phone models vs. the latest models) or operating system (iOS, Android, Windows Phone, etc.). The second interpretation considers that, given there will be some learners who do not wish to engage in mobile learning, it is necessary to provide alternatives for learners (i.e., desktop computer, paper-based, etc.) that allow them to make choices about what is best for them depending on their individual learning preferences and their circumstances at that time. Some learners simply may not have access to the same level of sophistication of technologies of their peers (see Chapter 1), and this could

result in a reluctance to use outdated devices, while others may come in to learning environments with clear ideas on how they wish to learn, and this may entail not using mobile devices. Rather than forcing learners to use tools that they do not feel comfortable with, ideally, it is best to show them the benefits – and potential limitations – of the available tools (see Section 8.6.2), and let them choose for themselves at a time that is right for them. Provision of appropriate guidance may contribute to a change in their views, but the final decision of what tools to use should be left to the learners themselves.

8.6.8 Understanding the Impact of Non-learning Uses

The concept of knowing how learners use various technologies for non-learning purposes as a means of tapping into the skills that they have already acquired with the technologies is certainly not new, and there has been discussion of this for some years (see Levy & Stockwell, 2006). While it is true that learner engagement can be enhanced in this way (Steel, 2012), it is also helpful to note what learner expectations are with various apps (particularly relevant for app design), which can provide insights as to why and how learners behave using various tools that are available to them. An example of this is the apparent lack of participation in online learning communities, but when considering the high proportion of “lurkers” in Facebook, who are satisfied with just browsing the content of others without contributing, it is possible to understand why such similar trends are evident in learning contexts. It follows, then, that if learners are expected to participate in similar fora as they might in an SNS like Facebook, their instinctive behaviour will resemble their non-learning uses, meaning that active posting would be considered as an “unnatural” way of using a tool that they are familiar with, which could result in resistance to or even rejection of the tool. As a result, understanding learners’ regular non-learning uses may contribute to understanding learner usage patterns as well as provide insight about how to frame activities in a way that learners are more likely to follow behaviour expected by the teacher.

8.6.9 Dealing with Distractions

Distractions inside and outside mobile devices are described in Chapter 6, but both of these can have a severe negative impact on learning in that there is a great time loss caused by the shift in attention towards the new task and the time spent in trying to return attention to the original task. Part of this includes learners’ trying to find where they left off and reorganise ideas to be able to resume the task

(Stockwell & Hubbard, 2013). Some partially formed ideas may be easily forgotten, and time might be spent on trying to remember these ideas. If the impact of struggling to resume the original task after the distraction becomes too great, it may even result in the learners giving up on what they were doing and abandoning the task altogether.

Should learners be in an environment where distractions are inevitable, as is often the case with mobile learning (Traxler, 2007), it is possible to take some measures in order to minimise the potential waste of time and effort. Firstly, it is useful to leave some kind of markers to make it easier to know what has been done and how to continue. In a mobile context, this could be difficult, particularly if the distractions are unanticipated and there is little time beforehand to deal with them. This can be alleviated in part with the design of the learning resources themselves, either by providing activities that can be completed in microsegments and will automatically provide some indication of what the next step is (see Stockwell & Hubbard, 2013) or by enabling an automatic save function that can also give some hints to what has been done and how this can be continued, even if continuing on a different device. This type of function already exists in cloud-based products such as Dropbox and other tools that are designed for collaboration, providing sufficient information that someone can come in cold and still be able to determine what has been done with minimal effort.

8.6.10 Opening Channels of Communication

Reducing the barriers between teachers and learners has long been thought of as a means of promoting more active learning and of encouraging learners to take risks in the classroom. With the range of communication tools that are available, it may seem that greater communication between teachers and learners would be considered as natural but effective communication would be thought to depend on more than simply having the tools through which to carry out this communication. Showing a receptive attitude towards learner needs in face-to-face events, and then applying this to electronic communication tools, could be thought of as a good way to encourage learners to benefit from the teacher's experiences and knowledge to create a learning environment more conducive to active learning.

8.6.11 Experiencing MALL as a Learner

The point of experiencing learning through technology as a learner was initially raised by Hubbard (2004), who argued that the only way to really see what difficulties that learners experience is to actually sit

in the place of learners and undertake tasks as a learner. There is a lot to be said in favour of this behaviour, namely that it is easier to know exactly what types of problems may arise and what types of support are necessary to deal with them. Only when teachers and researchers have put themselves in the place of learners will they have a greater understanding of learner needs which can lead to a better learning experience.

8.7 Summary

As has been described in this chapter, design is an extremely complex and yet central part of any teaching and learning context, and this is perhaps even more the case when mobile technologies form a part of this context. The design of the learning environment means that teachers and course developers need to start from an overall perspective of the entire context and then see how this relates to actual practice that sees technologies as having a central role. Design needs to include both the artefacts themselves as well as the ways in which these artefacts are used in language learning tasks and activities that have been based on clear learning outcomes, and a process by which the intended and actual learning outcomes can be compared and any discrepancies can be resolved should be a part of the overall design. Mobile technologies should take advantage of the affordances of the devices that enable learners to interact with their surroundings, but should do this in a way that has sustainable pedagogical goals in mind.

8.8 Discussion Questions

1. Imagine you have been asked to advise colleagues who are introducing MALL into their learning context. What steps would you recommend they take to design their environment and tasks? What would you ask them to exercise caution with and why?
2. Try to find three MALL apps or resources and examine them for the underlying views of learning. What assumptions and practices are used? Do the assumptions and the practices match? Explain your answer.
3. Mobile-based communication means that it is possible for teachers and learners to be in constant contact with one another. What potential problems are there with this?

9 *Concepts, Contexts and Challenges in MALL*

9.1 Introduction

This chapter brings together the arguments covered in the previous eight chapters and looks specifically at the three main areas that are mentioned in the title of this book – *concepts*, *contexts*, and *challenges* in MALL. Although this is the final chapter in this book, I have intentionally decided not to call it the “conclusion,” as the term conclusion gives an image of completion. MALL has come a long way in the two decades since it emerged, but it is still very much in its infancy as an academic discipline, and as a result, much of what we will come to understand about MALL is in the future and not in the past. It is in this vein that I deal with the concepts, contexts, and challenges that need to be developed for the field to move ahead. Needless to say, the notions which have been dealt with in this book are complex, and many have not been given the attention that they could have been given, due to space limitations and the sheer amount of research which has been carried out. Part of the rationale behind this chapter is to offer alternative perspectives of MALL so that it may be seen from a broader perspective. For example, how could mobility be viewed to make it more applicable to a learner’s life? Could a different perspective lead us to design our learning environments differently? The same may be said of interactivity and agency, both of which have been shaped by a sociomaterial shift that continues to take place as mobile technologies evolve.

We have seen different viewpoints of MALL co-emerging in parallel in varying contexts, the most obvious of which being the distinction between learning in formal and informal contexts. There is, however, a definite blurring of the division between these, and this is also contributing to a shift in the way that we think of both, including the delivery of content, interaction between the players in the environment, and the products and processes of assessment. Undertaking

research in this increasingly complex environment is proving to be fraught with multifarious problems, and as learning becomes intertwined with learners' everyday lives, it becomes progressively difficult to disentangle learning episodes from non-learning ones, and, indeed, the question needs to be asked as to whether we really need to. Learning as a part of a person's life would be considered as a desirable outcome, and as such, separating learning experiences from their daily experiences would result in an artificial view of their engagement with the target language. It has also become more apparent that teaching is undergoing a transformation that is just as radical as learning. The teacher experience is a very different one compared to as little as a decade or two ago with the integration of technology into all aspects of teaching and learning (see Chapter 2), but this is undergoing yet another fundamental shift as mobile technologies move the range of educational opportunities into something that is more pervasive and yet more personal. These challenges are likely to affect all teachers, whether or not they themselves actively use MALL in their educational environments. These broader categories of concepts, contexts, and challenges are described in more depth in what follows, but it should be pointed out that each is interconnected with the other two. New concepts entail both new contexts and challenges, just as new contexts and challenges each give rise to new concepts, and new challenges force us to formulate new concepts and reconsider the contexts in which MALL takes place. This chapter aims to show the interrelatedness of these issues, prompt further discussion, provide a springboard for further deliberation and research, offers alternative perspectives on themes that have been covered previously, and relates them to practice in the present and foreseeable future.

9.2 Concepts in MALL

9.2.1 *Lifelong Mobility*

Given the centrality of the “mobility” aspect of mobile learning, it is natural to highlight it as the first concept to consider in MALL. As has been discussed in Chapter 1, mobility encompasses using personal mobile devices to learn across “multiple contexts, through social and content interactions” (Crompton, 2013, p. 4). Pegrum (2019) expands on this view of mobility in his *3 Mobilities Framework*, which encompasses three different levels of interaction: mobile devices, mobile learners, and mobile learning experiences. At the first level, the learners are using mobile devices, but they themselves are not mobile. At the second level, the learners are using these mobile devices outside

of the classroom to enter “centralised digital learning spaces to access materials and engage in tasks, perhaps together with their geographically scattered peers” (p. 105). Finally, at the third level, the learners engage in learning experiences through interacting with real-world objects and people through their mobile devices. This distinction is an extremely interesting one in that it sheds light on the complexity of what mobile learning actually is. Early research focussed almost exclusively on the first level, with experimental research taking place in classroom settings that was intended to determine the effectiveness of MALL (see Chapter 4). We are now seeing an increasing amount of research and practice that explores these next two levels of Pegrum’s framework; however, a persistent problem is how to assist learners to extend their learning experiences into these next levels and to successfully function within them.

The affordances of mobile devices no doubt make them suitable to the diversification of the mobility of learning spaces and experiences. Indeed, the almost-clichéd expression “anytime, anywhere learning” – which from the outset has served as the catchphrase of mobile learning – focusses on mobility. While it is true that mobile devices do indeed give learners enormous freedom to choose the “anytime” and the “anywhere” in their learning, how mobile can we really expect the learning experiences in the daily lives of average learners to be without some kind of intervention? As appealing as the catchphrase sounds, the reality is that learners need to have “sometime” and “someplace” in which to complete required tasks, and these need to be budgeted for within their busy lifestyles (Brown & Green, 2003). Learners who are immersed in environments with abundant opportunities for learning may well take advantage of them, but this depends on the predisposition and agency of the learner, access to tools and resources, and skills to use the tools effectively. Just as a hammer and saw will have a very different meaning to a master carpenter than to a novice, a mobile device in the hands of someone who is both aware of the almost overwhelming range of resources available and also skilled in using them will likely have a far greater chance of engaging in meaningful “mobile learning” than someone who is not. A fundamental problem is that many learners in both formal and informal language learning contexts see mobile devices as little more than a means of accessing information and carrying out activities through (usually freely available but all-too-often relatively uncreative) apps. Although this behaviour entails the use of mobile devices, it would be difficult to consider it as being beyond Pegrum’s first or second level of mobility.

As such, it is evident that mobility is a product of not only the affordances of learners' devices, but also the skills of the learners themselves and their teachers, peers, and mentors who are a part of the ecology of the educational environment in which learners find themselves. Some learners may possess the skills required to make the most of the existing opportunities in their context without intervention, but for the most part – and particularly learners who are in predominantly monolingual formal learning contexts – will have difficulty in implementing “mobility” beyond the most fundamental level. It is precisely these learners that we need to make aware of the tools and resources that are available. There is more than enough evidence to suggest that many teachers are not confident in providing learners with the training or support to use mobile technologies for learning purposes (see Chapter 3), but teachers should be competent in encouraging learners to make the most of any and all available resources through understanding and reflecting upon the learning process (Travers, Morisano & Locke, 2015), whether mobile or otherwise. This is where support from others in the larger learning environment can help learners to develop new skills, through using their networks of peers or mentors to deepen their command of effective use of mobile devices for language learning (Luckin, 2010). Through open communication between teachers and learners, the information can be passed back and forth between each other, and teachers can have access to new ideas that they had not considered themselves. Although learners may struggle to understand what resources are available and learn how to use them appropriately, the hurdle that will be even harder to overcome is to help them to understand how these resources are relevant to them not only in the short term but also after the period of formal study has finished.

This leads us to an important aspect that could be included in the definition of mobility – *lifelong mobility* – where learner interaction with resources throughout their lives can be supported through mobile devices. This does not refer only to the integration of mobile technologies into a learner's daily routine at a particular point in time but, rather, to the issue of how learners can use their mobile devices in a way that evolves with their long-term individual needs. This is an aspect of MALL which has received very little attention (Lyddon, 2016), although it is tacitly at the core of much of the discussion into the development of learner agency (see Chapter 7). Mobile devices (and particularly mobile phones) have become an indispensable part of a person's life, with many people owning their first mobile phone and/or tablet in their teens, after which time they will upgrade or update their devices constantly for the rest of their lives. Average

ownership of a mobile phone is two years (Dunn, 2017), but make it is relatively easy for people to transfer their data and apps from one smartphone or tablet to the next throughout their lives, providing them with a good degree of continuity. That means that users of mobile devices such as smartphones will often have access to these resources for decades unless they have a particular reason to delete it or if there are incompatibility issues.

Given this lifelong ownership, it is possible to see how learners' mobile devices can be enduring educational companions. As learners progress with their studies, they may advance from being predominantly *language learners* into *language users* (see Blaj-Ward, 2017), where their focus shifts from how to learn the language to how to use it effectively in real contexts. In the mix of changes that continually take place in their lives, one of the only constants may be their ownership of personal mobile devices. Resources that were essential as learning tools to language learners may be indispensable reference resources to them as language users, although their dependence on these resources will vary depending on the personal, educational, and career paths that they choose to follow. Even after extended breaks, learners may find that they return to resources that they used at an earlier point in their formal or informal language studies. This lifelong mobility of MALL is one that needs to be capitalised upon more fully if learners are to make the most of the potential of their devices and to keep ahead of the evolution of technologies. The technological developments from one technology to the next are typically manageable, and as a result, the learning curve between these ongoing hardware and software upgrades can be relatively mild, meaning that apps and resources can be compatible through multiple device or operating system changes. Ultimately, this indicates that learners have the potential to benefit from their devices as a part of their language learning for decades if they need to.

A stumbling block impeding lifelong mobility is, however, that the learner needs to be able to evolve along with the technologies and do so with little if any support. Some resources that were once central to a learner's needs may no longer be relevant several years later, but it is the learners themselves that need to be aware of this evolution and to keep abreast of it. This is where a predicament facing teachers and designers becomes clear – that is, providing a foundation for sustainability of mobile learning not only in the immediate learning context but also in learners' future lives. Designing learning for a known environment is, of course, a difficult undertaking (see Chapter 8), but designing for future, largely unknown environments is an obstacle that teachers and course administrators will need to bear in mind.

What is required is to lead learners to be evaluators of their own individual learning ecology, including the available resources and their immediate relevance and also their potential relevance in the future. This needs to be included in the design phase of any course of study that feature mobile devices. As has been lamented by Burston (2015) and others, true mobile learning needs to go beyond the here and now, and teachers and designers need to have an eye on the future needs and contexts of the learners to assist them in their lifelong learning that will likely be heavily supported by mobile devices.

9.2.2 *Interactivity*

Interactivity is considered a central element of successful mobile learning (Churchill, Fox & King 2016; Khamparia & Pandey, 2020; see also Chapter 5). Mobile devices have made it possible for learners to interact with one another, with teachers, with content, with other devices, with objects encountered in their daily environment, with learning spaces, and even with people that they will ever know only in a virtual capacity. This interactivity is extremely attractive, built on sociocultural principles that we learn from others through social encounters with those that possess the skills or knowledge that we require for the various tasks that we carry out in our lives (Bandura, 2001; Lantolf & Thorne, 2006). Mobile devices have changed the way in which we interact with people and information on a day-to-day basis, both in quantity and quality. A typical user will check their phone as many as sixty times a day, coming to a daily average of more than three hours, and 70 percent of the checks of their phones are shorter than two minutes (MacKay, 2019). These scattered exchanges of abbreviated snippets of intertwined information with numerous people through various apps and platforms form the foundations of the relationships that we have with people and comprise a large proportion of the interactions that we have with them. Similarly, we interact with news and information by scanning through headlines or key pieces of information (Xie, 2019), but we rarely engage with information in depth – we skip from one source of information to another until we ultimately forget what it was that we were looking for in the first place (Rosen, 2012). This rather bleak outlook on interactivity is intended to illustrate that while interaction is a key aspect of MALL, not all interactions can be considered as ideal, and simply having *more* interaction does not necessarily mean that this interaction is *better*, particularly if it is scattered and attention directed towards these interactions is limited. However, these brief encounters with people and information through mobile devices are becoming the “new

normal,” and as a result, it is likely that education – including language education – that features mobile devices as a prominent part of the learning process will face a fundamental change as our cognitive processing abilities adapt to this new model of interaction.

As previously alluded to, reading is one area where mobile devices have started to have a marked impact. There are the obvious fundamental differences compared to reading on paper, such as the fact that pages are scrolled through and not turned, and place marking requires digital bookmarks that are not the same as inserting a piece of paper or a finger between pages to quickly refer to something on a different page. Readers may feel less engaged with online texts (Carr, 2011), and distractions in the device such as push notifications can make reading more fragmented, adversely affecting their ability to process information in the text (Liu & Gu, 2019). Many people engage in reading – and, indeed, most activities during the day – with a mindset that Friedman (2006), citing former Microsoft executive Linda Stone, calls this *continuous partial attention* – a state where attention is focussed on multiple textual, visual, and audio sources of information around them, leading to a result where the ability to concentrate on any one of these sources is hindered. On mobile devices, information is often truncated to fit onto the smaller screens (Krug, 2014), with designers opting to include less text as a means of saving space. As reading takes place more frequently through mobile devices, many people are simply becoming less able to read longer texts, which is gradually leading to a decline in academic ability (Dewan, 2019). Thus, when it comes to reading, interaction with the content can be considered as engaging with a different quantity and quality of information as a result of mobile devices, and attention is prone to be split between multiple foci.

Visual interpersonal interaction is another area that mobile devices are being used, most particularly with one-to-one or group video conferencing. To some extent, this type of interaction most closely resembles face-to-face interaction, but there is increased evidence of differences emerging between video-conferencing-based and live face-to-face communication. Clearly, one difference is that interlocutors can choose to show as much or as little of themselves through the screen, but the lack of the physical presence of interlocutors at close proximity can make it easy to forget that fine details may be clearly visible to other participants in video conferencing. This may cause participants to not consider their attire when engaging in these interactions or to carry out subconscious actions such as nose blowing directly in front of the camera on their devices that will be seen clearly by the others and perceived as rude or inappropriate (Neustaedter,

Greenberg & Boyle, 2006). Eye contact is another part of interaction that is different, where participants who communicate through video are more likely to look at their interlocutors' faces when compared to face-to-face interaction (Kuhn, Teszka, Tenaw & Kingstone, 2016), most likely as a means of compensating for any visual cues that may be lost through using the video. While there are many multimodal tools that are emerging that are designed to simulate face-to-face communication, in many ways, it is still difficult to encapsulate the same experience. Screen sharing enables people communicating through video conferencing tools to see view documents, presentations, or even videos simultaneously, virtual "whiteboards" make it possible to handwrite on shared white spaces using a mouse or stylus, and participants can collaborate on documents in real time and see the revisions being made by others who are remotely distant. When video communication is carried out on mobile devices, however, there are other limitations that are linked closely to the affordances of the device. Smaller screens as always are an ongoing concern, but when screen sharing is employed, the amount of screen space that can be allocated to seeing the faces of the participants in the communication is drastically reduced, again changing the dimensions of the interaction between the participants. Despite these limitations, this is not to say that the communication is necessary inferior to face-to-face communication in terms of quantity and quality, but it is definitely different, and being aware of these differences can enable planning for interaction in a way that these potential difficulties can be circumvented to make a more rewarding interaction experience for all of the participants involved.

9.2.3 Agency and Sociomateriality

As discussed in Chapter 7, agency refers to the ability to make and act on decisions that can have an impact how one lives (Martin, 2004). Agency has typically been deemed as a trait that is applicable only to humans or sentient animals (Knappett & Malafouris, 2008), but the concept of "non-human agency" has, in fact, been a contentious issue for many years. Proponents of actor-network theory (ANT) advocate that even inanimate artefacts have agency as they act as mediators to shape cognitive or material engagement processes (e.g., Casper, 1994; Sayes, 2014; van der Leeuw, 2008). This concept has already been applied to education from an ANT standpoint, including the role of technology in education, where all of the elements of a learner's environment have the potential to transform the social networks they are a part of (e.g., Fenwick & Edwards, 2010). According

to Thumlert, de Castell and Jenson (2015), learner agency is “networked, co-constituted, and never a ‘standalone’ accomplishment” (p. 787), and technological artefacts can have an immense impact on the ecology of the learning environment in which learners find themselves. Obviously, how technology is used affects the impact that it has on the user – for example, as a resource repository, a means of interacting with others, or as a tutor with its own instructional design principles. As described in the previous section, while human–human interaction is necessarily altered by technology that takes place through it, proxy agency will typically depend on the actions taken by the interlocutor(s), which may be restricted by their own individual time and other constraints.

A technological artefact may not only support human agency in its ability to distribute cognition to support the cognitive aspects of information processing and storage (Sutton, 2008; see also Chapter 5), but it may also take on its own pseudo agency through being given the “illusion of life” (Suchman, 2007, p. 207). *Non-human agents*, Suchman argues, simulate human interaction through artificial intelligence, giving an image of agency. The use of artificial intelligence has been in existence in language teaching and learning for decades, particularly as the foundation for adaptive systems in testing (e.g., Chalhoub-Deville & Deville, 1999; Tseng, 2016). There is certainly an ongoing need for systems that can adapt to learner responses for evaluative purposes or to provide materials that are appropriate to learner needs such as placement tests, but the possibilities for software agency for mobile learning can extend beyond this. Suchman (2007) alludes to two different types of software agents: *support agents* and *autonomous agents*. A support agent provides guidance to help users overcome information overload that stems from a lack of familiarity with elements of a system. It may take the form of a *chatbot* (or chatterbot), which enables users to communicate with a user in much the same way they might with a live user. These have been used extensively with online customer service, although not without some difficulties with communication breakdown and misunderstanding (Følstad & Taylor, 2020). In language teaching and learning, chatbots have been as a means of practicing conversation – acting as a support for learners with anxiety about speaking in the target language (Kim, 2019) – but for the most part, the future intention has been to interact with a human conversation partner (Fryer, Nakao & Thompson, 2019). There is already an established body of research that has shown how support agents such as chatbots can facilitate learners with lower motivation (Fryer & Carpenter, 2006; Kim, 2017), but an area that is of particular interest to MALL is the potential of autonomous agents.

The nature of such autonomous agents is encapsulated by Suchman (2007), where she described ideal software agents as artefacts that will come to “know” their users and, through mobile technologies, will accompany them constantly. Through their interconnectedness with other devices, such agents will at times stay home in place of users and at other times “travel” to carry out services in their stead, providing “greater mobility without a loss of familiar ground” (p. 207). Their autonomy stems from their ability to not only follow the commands of users to fulfil predetermined goals but also to pre-empt user needs and to act in advance of a user giving it directions. This type of autonomous agent can work as a personal assistant by understanding learner needs and responding to them. While the technical and ethical aspects of AI software that maintains data on users have been come into question (e.g., Hois, 2019), a mobile-based autonomous agent needs to acquire a certain amount of information about users to function, which may include location data and other information from apps. A support agent for language teaching and learning would need to have similar functionality, enabling parsing of information that the learner has access to pertaining to their use of the target language and include functions to curate new information into an accessible format for the user, presented at the right time and place. Quite obviously, there is a need to address the privacy concerns of an autonomous software agent that could provide proxy agency for learners (see Chapter 7, for a discussion of agency), but coupled with the notion of lifelong mobility, the possibilities of tools that can keep the learner in touch with relevant language input through mobile devices are worth exploring.

While ANT was also originally conceived as a social theory, it is a related concept, *sociomateriality*, that further explores the social interrelationship between artefacts and humans. According to sociomaterial principles, the social and the material are inextricably intertwined to the extent that one has become indistinguishable from the other in many respects. The user is seen as part of the “sociomaterial assemblage that comprises a functioning machine” (Suchman, 2007, p. 190), unwittingly “configured” by the technology despite the illusion of the reverse. User behaviour is shaped by the way in which a technology is designed; for example, abbreviations in text messaging emerged from the affordances of early hardware and software, and decisions about when to download large files or videos may be dependent on the amount of available data remaining or access to Wi-Fi. Orlikowski (2007) provides a stark example of the potential impact of technology on human behaviour with the results of a

search on Google. A search using the Google search engine will provide different results depending on when the search is conducted, based on a complex interplay of “servers, databases, indexes, and algorithms, and reflects considerable choices made about its design, construction, and operation” (p. 1439). The results that feature highly in the PageRank algorithm from this combination of factors at the time the search is conducted are “temporally emergent” (Pickering, 1995, p. 14), yet this technologically supported presentation of information may have a direct impact on the choices a person may make. From the opposite perspective, Google itself has not remained unchanged over time, and the way in which information is curated is shaped by the voices of the users who use it to conduct searches (Gabe, 2019), and mobile phones are constantly being redesigned as a result of customer feedback (Seker, 2020). Thus, sociomaterialism may be viewed from a *technocentric perspective*, which examines how technology modifies human action, and a *human-centred perspective*, focussing on how humans “make sense of and interact with technology in various circumstances” Orlikowski (2007, p. 1437). Each perspective needs to be considered in terms of the other, and separating them can give a skewed view of the reality of the relationship between them.

The connotations of sociomateriality also have a direct bearing on language learning, particularly as we see mobile devices being used more widely in the overall learning ecology. The technological artefacts are shaping what, when, how, and even why a language is learned, with choices being made about which apps or resources to use as a result of the rankings in the relevant app stores, calculated by sophisticated algorithms that reflect those in Google’s search engine. The resources that learners choose to view on their mobile devices will depend upon the technical affordances of the devices that they own, with decisions made about what tasks and activities to engage in being influenced by factors such as screen size, storage, processor speed, and Internet access. Design of language learning resources for mobile devices also needs to be carried out cognizant of these sociomaterial aspects; that is, what impact will the software and hardware have on the learner, and, in turn, what procedures are there in place to gather feedback from learners that can be used for improved design? Keeping the symbiotic relationship of artefact and user in mind can contribute to understanding learner behaviour while learning with mobile devices and can inform how to continue to evolve with technological developments as well as the changing needs and skills of the learner.

9.3 Contexts in MALL

9.3.1 *Pluralistic Development of MALL Contexts*

Although it has been argued that the distinction between formal and informal learning are becoming increasingly blurred over time (e.g., Hubbard, 2020) – particularly when mobile devices are involved – there remains, to a certain degree, a gap between the research and practice that takes place in formal settings compared to the research and practice that takes place in informal ones. A lot of research surrounding informal learning focusses on identity development as a user of the target language (e.g., Chik, 2020), but to many learners who are in compulsory classes where learning a foreign language is little more than memorising syntax and vocabulary in order to solve convoluted questions in order to attain a high score on an achievement test or to gain entry into a higher-level educational institution where the language is often taught at a level that fails to even reach that of the exit point of the previous one. The rather gloomy picture painted here of formal learning contexts obviously does not reflect all environments, and there are many examples of learners engaging actively with the target language in authentic language learning events as a part of their formal studies (e.g., Fazzi & Lasagabaster, 2020; Wigglesworth & Harvor, 2018). As Merriam and Bierema (2013) suggest, learning successfully in informal contexts can be quite subtle, being self-directed, incidental, subconscious, and integrated with conscious learning activities. Each of these points is dependent upon learner agency, an attribute that is more likely to emerge when learners have clear goals in mind for learning the target language, except the last point, *integration*, which alludes to a bridge between formal and informal learning contexts.

Learners who are in situations where the onus for learning falls on their own shoulders may struggle to find ways to relate their everyday self-directed experiences to conscious learning activities, preferring to seek out opportunities to gain exposure to the language instead. At the same time, learners in formal learning environments simply may not see how the study that they are undertaking relates to any meaningful target language usage. A critical look at the practice of language teaching and learning reveals that for the most part, pedagogies for formal contexts have developed quite independently of pedagogies for informal ones, and there is little cross-pollination between the two. As discussed in Chapter 7, learners in formal learning contexts are rarely provided with sufficient training in order to achieve a level of autonomy that will allow them to continue with the studies in a

self-directed manner unless there is sufficient motivation for learners to do so. At the same time, learners who are informal learning environments find that they lack the skills to engage in useful language learning activities, a problem which is particularly evident in new migrants (Cummings, Sturm & Avram, 2020).

It is here that mobile technologies provide the means of bridging the gap, where learning activities and tasks can be taken out into the real world, and the world can be brought into the classroom. However, the only way that this gap can be bridged – even with the existence of the tools to do so – is to ensure that training can be made available to learners in each of these situations.

The difficulty of the rift between formal and informal learning is also evident in research as well. Research that takes place in formal learning contexts is seldom relevant to informal ones and vice versa. Indeed, the divergence in the ways in which learners are interacting with mobile devices means that it is becoming increasingly difficult to conduct worthwhile research that can capture the myriad ways that learners can exploit their devices to engage with the target language to boost their acquisition of the language. These learning experiences are happening as a part of both formal and informal environments, very much prompted by the skills and motivation of the learners themselves. Research in formal contexts is easier to conduct because the variables are easier to control for when compared to learners who are in naturalistic settings, but there is still a great deal that is not understood about what learners do once they step outside of the classroom. One method that can be used is to explore individual learners' use of technology through the systematic use of case studies (Grin, Rotmans & Schot, 2010). Case studies can be written up as historical narratives without an explicit theory, but parts of this narrative can be explained by hypotheses and theoretical mechanisms, which in turn allows the narrative to be converted into an analytical explanation through identifying patterns embedded in the theoretical perspectives that can be used to develop more generalisable theoretical arguments (George & Bennett, 2004). In this way, it is possible to go beyond speculation and anecdotal evidence of what learners are doing when they are not immediately visible to teachers or researchers and create a solid foundation on which to build pedagogies that would be appropriate for learners who are in formal contexts, informal contexts, or a combination of the two.

9.3.2 *Mobile Learning for Crisis Management*

Teaching with technology is something that is considered as a natural part of the educational environment to some, but to many – if not

most – others, the thought of using technology as the *primary* means of instruction is almost unimaginable. As discussed in Chapter 5, it is not uncommon for in-service (and indeed pre-service) teachers to have never had the opportunity to experience technology as a key element of their own education, and while they see the potential, the time and energy needed to acquire the skills to include technology in their teaching often has a low priority. The COVID-19 coronavirus that spread around the world in early 2020 turned essentially every walk of life upside down, with self-isolation being enforced on a global scale. Needless to say, education was one of the areas that was also heavily affected, with teachers suddenly finding themselves in situations where they were expected to teach their students from a distance. Most institutions were ill-prepared and scrambled to put together materials that could be taught using online methods in what came to be termed as “pandemic pedagogy” (Milman, 2020). With just weeks or even days to shift curriculum into online modes, a patchwork of resources was combined into a teaching regime to try to avoid falling behind in their teaching goals.

For teachers with experience with teaching with technology, getting through the crisis largely entailed some adjustment of the tools that they had already prepared for their regular teaching. The COVID-19 outbreak showed, however, that for those teachers without such experience, the logistics of how to conduct online teaching typically needed to be decided quickly, with little or no time for pre-planning, and implementation was usually done “on the fly” as teachers prepared for online delivery on an ongoing basis throughout the courses they were teaching. While the dangers in this are obvious, in many – if not most – cases, teachers found themselves with little alternative, as they attempted to apply the limited skills that they had with tools they were largely unfamiliar with to prepare for classes that attempted to simulate what they were used to in their existing face-to-face environments. At the time of writing this book, the situation was not concluded, and speculations regarding the prospects of online teaching and learning as a viable method of instruction were widely circulated. Expectations appeared through various media outlets, with predictions for both the success and the failure of online learning, and some saw this crisis as a chance to prove the (in)effectiveness of learning through technology at a distance (Zimmerman, 2020).

Some have argued that the term “online teaching” was not appropriate to use in the crisis, advocating expressions such as “emergency remote teaching” (Hodges, Moore, Lockee, Trust & Bond, 2020), which emphasised the difficulties in trying to carry out effective online teaching without having the time or resources to prepare properly.

The reality is, however, that it actually *was* online teaching, and to many, the first experience with it that they had as either a teacher or a learner. It is here that the expression “first impressions last” takes on a particularly poignant meaning. As Norman (2013) argues, the experiences that users have are critical in that they determine how positively users remember their interactions with an artefact or event. Negative emotions are long-lasting and can have a detrimental impact on a user’s willingness to engage with an artefact again in the future. This is a point of particular relevance for both teachers and learners in crisis situations. It is unlikely that it is just the learners who felt the inadequacies of many of the support framework put in place under hurried conditions, and teachers found themselves stressed and simultaneously both proud of yet dissatisfied with the finished products that they created and used in their teaching.

The crisis highlighted the fact that it sometimes takes emergency situations such as the pandemic to heighten the awareness of what can and cannot be done through teaching online. As Milman (2020) points out, online teaching needs to be planned in advance with an understanding of how the different elements of the course fit together with one another and with the available tools. As has been mentioned at a number of points throughout this book, learners will likely be using a combination of mobile and non-mobile devices as a part of their learning, and designers will need to consider how these will work together with one another. Mobile devices are very much a central part of the online teaching that takes place, generally as this is the primary technology owned by many, particularly if there are multiple users in a single household that are required to engage in online learning simultaneously. However, the question of how to engage in MALL during such crises is a difficult one. In crisis situations like the COVID-19 outbreak, self-isolation meant that learners could not be physically mobile, often forced to self-isolate to avoid spreading the virus. However, learners *were* able to use their mobile devices to make their learning experiences mobile, even if virtually. Resources were not only stored locally, but teachers could screen-share their own resources to make them available to learners, in one sense moving themselves into the learners’ individual spaces in a way that they had not done before, with teachers and learners exposing elements of their private lives to one another, such as sharing videos showing the inside of their own homes.

A potential outcome of the COVID-19 crisis is that it demonstrated a need for further consideration of *crisis management of education*. Crisis management in itself has been covered as an academic field of study and research in organisational management for many years

(e.g., Mintzberg, 1994), but there has been little discussion of crisis management that occurs within education itself. One of the conclusions of the COVID-19 outbreak was that it confirmed Slater's (2017) view that dealing with unexpected emergencies in education is dependent upon human resource and structural leadership frames, and both of these were stretched to their limits during the pandemic. The crisis emphasised how ill-prepared most institutions were to deal with emergencies, making the need for new infrastructure and new policies clear. The problem is, however, that both require time, money, and know-how to be put in place appropriately. An issue which has been raised with regards to crisis management is the danger in over-engineering as a result of crisis, where restructuring without sufficient planning and keeping an eye on the needs of both employees (teachers) and customers (students) can be a recipe for disaster (Bolman & Deal, 2017). Plans need to be reasonable, and not a simple knee-jerk reaction that can have a damaging impact on other areas of education.

As was discussed in Chapter 1, technology has often been considered as a potential means of saving money, despite the fact that the evidence invariably points to the opposite. During the crisis, costs were, to a certain extent, kept down by many service providers making their services available for free or very low costs as an interim measure, and as a result, various providers rose from anonymity to fame virtually overnight with huge demand for the free or extended services they offered for education during this time. However, if introduced on a large scale, these measures will, of course, have ongoing costs associated with them, and there is a risk that ineffective, cost-inefficient strategies will be put in place that may not fulfil the desired purpose, should another crisis arise. As has been seen all too often with the outcomes of calamities such as natural disasters, designing for crisis situations rarely matches the needs of the moment when they actually occur (e.g., Farazmand, 2016). The key point to bear in mind here is that designing online learning is no different from other forms of learning with technology, requiring a thoughtful design process (as described in Chapter 8). Sometimes crises can be the catalyst needed for changes in perception, and new systems introduced as a result of crisis have the potential to provide dramatically improved results, despite the chaos and breakdown that occurs in the short term (Bolman & Deal, 2017). If change is to take place, then it needs to be done in an informed manner. Models for teaching language skills online certainly exist, and it is where teachers and researchers who have extensive experience with teaching languages at a distance can provide invaluable insight into how to employ best practice (e.g.,

Stickler, Hampel & Emke, 2020). In regions where computers are prohibitively expensive, mobile devices have very often been used as the primary device for learning. In other cases, as schools reopen, students may be expected to use their mobile devices as a means to maintain social distancing rather than using shared computers. It is possible that the COVID-19 outbreak may eventually lead to a shift in views towards mobile learning – including MALL – as a viable option to support language teaching and learning, but only time will tell regarding how it is viewed by teachers, learners, and administrators in the long term.

9.4 Challenges in MALL

9.4.1 *The Nexus of Learner and Teacher Training*

The range of challenges that could be listed in this section are far too great to give any of them suitable coverage in the space here, so the discussion will be limited to two main areas: the complexities of the relationship between teacher and learner education and the challenge of how to deal with evolving technologies. As is so often the case, there is a degree of overlap between them, but they also have their own individual idiosyncrasies. The importance of teacher education – and most notably with regards to teaching with technology – has until recently been somewhat undervalued, in part due to the greater focus on *learning* (as opposed to teaching) that has dominated education over the past several decades (Laurillard, 2002, p. 5). A look through available teacher education titles reveals a paucity of coverage of technology in books on teacher education unless they specifically focus on technology, implying that technology is still in many regards as much on the edges of mainstream language teaching pedagogy as was bemoaned by Coleman (2005) nearly two decades ago.

As was discussed in Chapter 8, there are several frameworks for designing learning environments – some explicitly outlining the role for technology, some applying this to language teaching, and others relating pedagogy to learning through mobile devices. Guidelines for instructing teachers to use technology in their individual learning environments have been less widely employed, but frameworks such as TPACK (Mishra & Koehler, 2006) have been used as a model of teacher education, allowing teachers to see the position of technological knowledge as a part of teaching practice. The main components of TPACK – *Technical, Pedagogical, [and] Content Knowledge* – are designed to enable teachers to see the balance between each of

these elements when they are learning how to use technology as a part of their teaching and learning. While TPACK provides a basic framework to show the importance of understanding how to avoid being overly (or under-) biased towards technology amongst other skills, it is not without criticism, particularly with regards to having too many domains of knowledge that are difficult to distinguish from one another (Brantley-Dias & Ertmer, 2013). From a contrasting perspective, Stickler and Hampel (2015) argue for the development of teaching skills as a pyramid with basic ICT competence at its base, and the need to develop specific technical competence to deal with particular artefacts, the ability to facilitate communicative competence and online socialisation, and – finally, at the top – the importance of one's own creativity. Foundational technological skills have gradually become the norm with the penetration of mobile devices across all age ranges (Statistica Research Department, 2019), bringing with it a marked decrease in anxiety towards using technology and even active usage in higher age ranges for seeking information pertaining to health and support (Halmdienst, Radhuber & Winter-Ebmer, 2019). Thus, the hurdle facing the use of technology lies not in familiarity with technology per se but, rather, with pedagogies that can capitalise upon it.

There is evidence that suggests learner training in using technology to achieve specific learning goals can have a positive on learner attitudes and outcomes (Stockwell & Hubbard, 2014; see also Chapter 7), but learner training hinges on teachers being able to conduct this training. Herein lies the challenge for using mobile devices for learning. Both teachers and learners possess basic skills with using mobile technologies but lack the knowledge to apply these skills to effective teaching and learning. Once the psychological barrier of ignoring mobile devices as potentially powerful learning tools is broken down, both teachers and learners can start to explore the possibilities. Collaborative debriefings regarding technology use (Romeo & Hubbard, 2010) can help both groups to understand ways in which technologies can be used, and not only can learners acquire the necessary strategies and skills to use technology for focussed language learning through such debriefings, but teachers can equally learn techniques from learners that they can apply back to their educational environments. In other words, teacher training does not have to be formalised in all cases, and every interaction between teachers and learners has the potential to be a rich source of information about how to improve practice, which is worthy of consideration.

9.4.2 *Dealing with Evolving Technologies*

As the final section of this book, it may seem prudent to consider the directions that mobile learning will take in the future. However, any predictions of the future directions of technology – of which mobile technologies are now a major part – are fraught with errors, tending to either overstate or understate the possibilities. For example, cognitive scientist and an expert in AI at the time, Marvin Minsky, is quoted as saying “in from three to eight years we will have a machine with the general intelligence of an average human being” (Darrach, 1970, p. 58D). Fifty years on, and although artificial intelligence has progressed remarkably, it would be difficult to claim that it would possess intelligence of a human on any general level. On the other extreme, co-founder of YouTube Steve Chen is reported as saying that YouTube would fail due to a lack of videos that he thought would be of interest. According to Statistica Research Department (2020), YouTube now has two billion users worldwide, making it the second-most popular social network in the world.

Predictions of MALL are equally difficult in many regards. As Pegrum (2016) points out, “in any discussion of m-learning, we must take into consideration the larger cultural, political, economic, and educational ecologies in which mobile technologies, like all technologies, are embedded” (p. 413). Each of these ecologies is both fragile and dynamic, and the interplay of these factors can lead to very different outcomes regarding what devices are developed, when and where they are available, and how they are used and viewed, including in educational settings (Glover & Rodger, 2018). And as described earlier, from a sociomaterial perspective, the uses that emerge from technological advances ultimately affect the next generation of development.

New devices are appearing at a phenomenal rate, with mobile, wearable, and implanted technologies flooding the market (see Chapter 1). Meanwhile, the rate of flow of these new technologies into widespread use is dependent upon the financial resources of individuals and institutions, timing of contracts or leases, and larger infrastructure shifts such as the introduction of 5G networks (Rao & Prasad, 2018). One of the consequences of the multifarious rate of penetration of devices into the wider market is that there will be technologies of varying levels of sophistication being used by different users at any one point in time. The development of a new technology does not mean that older technologies rotate out of common use in the short term, and older devices will be used in parallel with newer ones for several years if not even decades. It is largely for this reason that

the future of MALL is not simply limited to the technological changes that are taking place, but, rather, MALL needs to be viewed from a psychosocial perspective, one that includes the perceptions of the ultimate players in the overall equation that makes up the educational ecology – the learners, teachers, administrators, designers, and even family.

Future mobile learning will likely see two main areas that will expand the use of mobile technologies as viable tools in language learning: *augmented reality* and *context awareness*. Both of these are enormous fields that have seen exponential developments over the past decade and have also extended into educational uses. Augmented reality enables learners to overlay reality with digitally enhanced multimedia in order to add information that may assist them in learning – including turning still images on a page to video, providing extra information about vocabulary, or adding imagery to real life places (e.g., Godwin-Jones, 2016). The possibilities are indeed very encouraging, but research has been plagued by technical difficulties and problems in providing correct overlays (Yang & Mei, 2018). Context awareness enables mobile devices to tailor the learning experience to individual locational requirements (Crompton, 2016; Koenraad, 2019). Through personalisation, learners can have access to information that can assist them with the needs of the moment, which can support situated learning (e.g., Lave & Wenger, 1991; see also Chapter 8). The ways in which these tools will be used are very much up to the imagination of teachers and designers as well as the future technological developments, but both make it possible for learners to capitalise upon the full potential of the mobility of MALL.

9.5 Final Comments

Mobile devices have already firmly secured their place in language teaching and learning environments, with their uses being shaped not only by the evolving physical characteristics of the devices themselves, but also by the constantly shifting landscapes of second language pedagogies, research agendas, socioeconomic change, and even world-wide disasters and crises. The contexts in which MALL can take place are as varied as there are learners, and each individual ecology will vary – sometimes imperceptibly – from another. These variances can lead to differences in the processes, the products, the hurdles, and the successes that learners will experience. It is the personal nature of the learning ecologies that empowers mobile learning, and learners have freedom in the choices that they make about their learning, not only in short-term goals but also lifelong ones. However, the fact that the

spread of mobile learning brings about not only possibilities but also challenges is becoming increasingly evident as research and practice into the use of mobile devices in education continues – bringing to the forefront such issues as equity, sustainability, and dealing with changes, and problems of scale (Crompton & Traxler, 2018). Some challenges are easier to deal with, and others more difficult, but as long as there are teachers and learners that dedicate themselves to the difficult task of broadening language learning experiences, there is no doubt that mobile devices will feature as an indispensable tool to enrich and enhance these experiences as an integral part of their lives.

Glossary

ADDIE – Short for *analyse, design, development, implementation* and *evaluation*, ADDIE has been the foundation of several models of instructional design since it was originally developed by the US military as the “five-step approach” in the 1970s. It later became known as ADDIE in the 1990s (see Chapter 8).

actor–network theory – A social theory proposed by primarily by Callon (1986), Latour (1987) and Law (1994) that human and both living and inanimate non-human entities exist in constantly shifting networks that impact on one another affording mutual change.

affordances – Proposed by Gibson in 1979, the theory of affordances explains what an artefact makes possible. It has been widely used in CALL to refer to the technological functions of a device, but it is also used to refer to the possibilities that can arise from the properties of a device that may or may not have been intended by the developers. Affordances are not only inherent to a device but are also dependent on the skills, knowledge, experience and imagination of the users of a device (see Chapter 2).

agency – See learner agency.

artefact – An object that is typically not naturally occurring but rather made by humans with the purpose of subsequent use. In CALL and MALL, it is frequently used to refer to either hardware (such as a computer or mobile phone) or software (such as learning courseware or an app).

atomised learning – The provision of modules of content for learners that can be accessed as fragments of a larger whole. It has been claimed to provide learners with more freedom as learners can choose small “micro-content” in any order that suits their own preferences or understanding, but it has also been criticised, as it may lead to gaps in understanding the overall course and the lack of compatibility with current models of assessment.

attribution theory – The view that an individual has on the causes of behaviour and events based on the information that they have

available to them. These causes may be internal (i.e., a direct result of the individual themselves) or external (i.e., factors that are beyond the individual's control such as circumstances or other people).

augmented reality – The ability to add information to the real world through technology, such as adding visual overlays to enhance the user's perception of reality. It is most commonly used with a mobile device or wearable technology where the original scene is displayed with some extra element digitally drawn over the top of this scene. A common example of this was the Pokémon Go game that was launched in 2016.

autonomy – See learner autonomy.

app – A term coined by Apple with the release of its first iPhone in 2007 to refer to an application. These were initially limited to telephone and messaging apps, calendars, music players, memos and cameras but rapidly spread to include video players, games, office and weather apps, as well as a range of niche apps such as online banking and shopping. Apps have also been developed specifically for language learning.

blended learning – The mixture of technological and non-technological means as part of a course. In its initial definition, blended learning expected a balance between these elements where both were an integral part of the whole learning experience, but there has been a shift in recent years to include basically any combination of technology with traditional teaching approaches (see Chapter 8).

complexity theory – See dynamic systems theory.

complex dynamic systems theory – Initially based on work by Larsen-Freeman and Cameron (2008), complex dynamic system theory explains the lack of linearity in language learning, suggesting that there are multiple factors at play that may or may not be visible or consistent. One of the tenets behind complex dynamic systems theory is that learning outcomes are difficult to predict due to the unknown influences of various interconnected factors (see Chapter 5).

context awareness – The ability of a system to determine its environment based on information it gathers through various sensors such as GPS, NFC, QR codes or other types of sensors. A context-aware system is able to adapt to the environment that it detects or provide contextually relevant information to users.

Conversational Framework – Proposed by Laurillard (2002), the role of technologies in facilitating teaching and learning through interaction between the participants in the learning environment (see Chapter 8).

courseware – Combined from the words *course* and *software*. It is used to refer to educational software that has specific learning goals or targets embedded in the instructional design, usually incorporating some kind of multimedia. Courseware is most commonly designed for running on a computer, but courseware apps recently have also been created to be used on mobile devices.

cyberslacking – Spending time focussed on tasks through digital technologies that are unrelated to the required tasks at hand, such as texting, web browsing, online shopping and checking social media.

directed motivational currents – A proposed theory of sustained motivation by Dörnyei, Muir and Henry (2016) that learners may be pulled along towards completion of their goals without a sense of burden, much like being pulled in a strong current of water. Directed motivational currents are dependent upon clear goal setting that can focus learners' motivation (see Chapter 5).

digital divide – The gap in technology access as a result of socio-economic factors. People in less affluent regions or situations find it difficult to afford the latest technologies, sometimes resulting in a gap of the “haves” and the “have-nots,” which can have an impact not only on pedagogy but also on how learners feel about themselves placed in the larger context they find themselves in.

digital literacy – The ability to seek out, evaluate and synthesise information through the use of various digital platforms. This may include the use of search engines, word processors, social media and other digital resources which may be used to perform desired tasks. Digital literacy does not replace traditional literacies but, rather, is largely dependent upon them as the foundation for how digital tools are used.

digital native – A term initially coined by Prensky (2001) which sees the younger generation as being more comfortable with using technologies on the basis of being born into a world where technology surrounds them. It has faced criticism based on its lack of empirical validity and due to its potentially discriminatory implications.

dual systems theory – Developed from work by Evans (2003), dual systems theory suggests that the human mind processes information through two systems, one conscious and the other unconscious.

Habitual behaviour can be attributed to the unconscious system and requires the conscious system to inhibit these habits (see Chapter 5).

dynamic systems theory – See complex dynamic systems theory.

emergency remote teaching – A response to a crisis (such as the 2020 COVID-19 pandemic) where educators around the world are forced to commence online teaching in the midst of school closures with minimal skills training and preparation time.

engagement – The way in which a learner undertakes a task or activity. It has most typically referred to the amount of time that a learner spends on the task or activity, but it may also refer to the cognitive, affective and social impact that it may have.

expectancy-value theory – The concept that the expectancies and values that an individual places on a given activity will impact upon subsequent behaviour. The greater the expectation that is held by individuals regarding their success through the activity, the more likely that they will be to expend time and effort on it. It is also dependent upon their beliefs regarding their own competences and abilities to complete the activity.

flipped classroom – A model of learning that seeks to maximise effective use of class time by requiring learners to preview content before class through readings or videos so that they can discuss it in class. It is possible to use this model without technology, but it has been popularised with the penetration of digital devices. It has faced some criticism in that it requires learners to be sufficiently disciplined to review materials in advance of class.

GSM phone – One of the many terms which have been used to refer to phones prior to smartphones. Also sometimes called “dumb phones,” these phones typically had considerably smaller screens and fewer functions than smartphones but still provided SMS, email and Internet access through a GSM (initially 2G and later 3G) network. Most GSM phones also allowed for a limited number of applications, usually restricted to the particular service provider.

hype cycle – Originally coined in 1995 by a research organisation called Gartner, it is used to describe the stages that a technology is likely to go through from its initial inception. It is broken down into five stages: innovation (technology) trigger, peak of inflated expectations, trough of disillusionment, slope of enlightenment and plateau of productivity.

learner agency – The degree to which learners proactively involve themselves in the learning process. It is based on the tenet that learners will have a more fulfilling and successful learning experience if they take charge of their own learning.

language socialisation – The modification of language by users to meet the conventions of a particular social context (see Chapter 5).

learner autonomy – The concept of learners taking responsibility for their own learning, popularised by Holec (1981) (see Chapter 7).

learner training – The provision of instruction in how to use technologies for learning purposes with the goal of enabling learners to consider their own strategy usage (see Chapter 6).

LMS – A system that allows teachers to provide content and carry out assessment online. Also known as a virtual learning environment (VLE), an LMS typically makes it possible to maintain communication with learners through various text, audio and video-based tools and to track learner access to the online resources. There are both open-source (e.g., Moodle) and proprietary (e.g., Blackboard) LMSs, and many institutions will use an LMS to support courses they offer, even if the course is predominantly face to face.

MOOC – An abbreviation for *massive open online course*, it is type of free online course offering unlimited enrolments that was started as a part of the open educational resources (OER) movement in 2008. More traditional MOOCs have been referred to as xMOOCs, whereas more interactive models that allow for more freedom of selection of content have been called cMOOCs. Attrition rates of MOOCs have remained extremely high, with the vast majority of students who enrol failing to complete them.

MP3 player – A type of portable digital audio player. Most MP3 players are capable of playing audio files of several different formats including but not limited to MP3 and also include a recording function and allow the downloading of audio recordings called podcasts. MP3 players have been largely replaced by music-playing and sound-recording apps of smartphones since the mid-2010s.

multitasking – The ability to perform two or more tasks or routines at the same time. The term multitasking has been applied to operating systems where routines can run in the background at the same time as primary routines can be accessed by users. It has been used by humans in their ability to do multiple things at the same time, but there is evidence to suggest that most people are simply shifting their

attention between tasks rather than actually performing multiple tasks simultaneously.

normalisation – A concept proposed by Bax (2003) where technology will become such a natural part of the learning context that it will no longer be noticed by the users, but rather it will be picked up as easily as pen and paper.

novelty effect – The initial enthusiasm associated with using a technology in the early stages.

PDA – Often thought of as a forerunner to the smartphone, a PDA is a handheld mobile device (also known as a palmtop) that allows for making telephone calls; sending of text messages, faxes and emails; and accessing the Internet. Early PDAs were typically stylus based with character recognition technology, but later PDAs incorporated a small physical keyboard. They have now been largely replaced by smartphones and tablets.

selective attention shifting – Diverting attention away from a primary task in order to do other tasks like checking messages for some period of time before returning to the original task again (see Chapter 6).

self-determination theory – Made up of three basic human needs – autonomy, competence and relatedness – self-determination theory explores the motivations behind a learner's actions to understand the degree to which they are self-determined.

serious games – Games that do not have entertainment or enjoyment as their primary function but, rather, may be used for training or promotion of critical thinking. Serious games can also include educational games, where users may be trained in a specific skill or theme including a focus on a particular language skill or practice in behaving in a certain situation.

smartphone – A mobile device that was created in 2007 that includes telephoning; Internet access; sending and receiving of messages; and various apps such as calendars, memos, music players, games and cameras. They differ from GSM phones that preceded them in that they have a far larger screen with an onscreen keyboard, and a range of apps are made available through dedicated online stores rather than the limited range installed by the network provider.

social networking – The use of online websites or apps to find and interact with other people. Many people use social networking as a way of catching up with friends and family members that they know

offline, but others use it as a way of meeting new people with similar interests to their own. Common social networking service (SNS) sites include Facebook, Twitter and Instagram, but there are many localised social networking services that are used for a particular purpose or in specific regions.

sociomateriality – The complex relationship between artefacts and people who use them. All the view of any object is social, and its functions and value are dependent upon social contexts. Conversely, any social action is made possible through the use of artefacts, and actions are shaped in some way by these artefacts.

SNS – See social networking.

tablet – A roughly book-sized portable electronic device with touch-screen technology and onscreen inputting, initially with a stylus but later with an onscreen keyboard, that preceded the smartphone. Although there were several forerunners, tablets (originally “tablet computers”) that resemble modern devices were developed in 2000 by Microsoft. Many current tablets now use software that is typically associated with laptop computers.

teacher training – A field of education that looks at how both pre-service and in-service teachers can receive training that will enhance their usage of technology in their teaching and learning contexts.

TPACK – Devised by Mishra and Koehler (2006), TPACK (technological, pedagogical and content knowledge) is a teaching model that encourages teachers to focus on more than just the technology but also to have an understanding of appropriate pedagogy and sufficient knowledge of the content.

training – See learner training *or* teacher training.

translanguaging – Strategic use of available linguistic resources in order to convey meaning. Speakers of multiple languages may rely on any or all of their language skills to communicate to speakers who may share some knowledge of these languages.

ubiquitous learning – Learning that is supported by mobile, fixed and/or wearable technologies that are networked to allow a seamless flow of information between them. Learners may choose the tool or tools most appropriate to the environment that they are in at a given point in time.

virtual reality – The simulation of real or imagined environments through the use of technology that makes users feel as though they

are immersed in it. Commonly achieved through a wearable head-mounted display that may provide a three-dimensional effect, users may interact with the objects that are digitally created within the virtual world that they are in.

wearable technology – Technologies that are worn by the users at very close physical proximity rather than carried in a pocket or bag. The most common examples of wearable technologies are smart watches and smart glasses, which like their traditional namesakes, are worn by the users as a part of their daily routine.

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